

1 *Introduction*

1.1 Motivation

Most tasks require a person or an automated system to *reason*: to take the available information and reach conclusions, both about what might be true in the world and about how to act. For example, a doctor needs to take information about a patient — his symptoms, test results, personal characteristics (gender, weight) — and reach conclusions about what diseases he may have and what course of treatment to undertake. A mobile robot needs to synthesize data from its sonars, cameras, and other sensors to conclude where in the environment it is and how to move so as to reach its goal without hitting anything. A speech-recognition system needs to take a noisy acoustic signal and infer the words spoken that gave rise to it.

In this book, we describe a general framework that can be used to allow a computer system to answer questions of this type. In principle, one could write a special-purpose computer program for every domain one encounters and every type of question that one may wish to answer. The resulting system, although possibly quite successful at its particular task, is often very brittle: If our application changes, significant changes may be required to the program. Moreover, this general approach is quite limiting, in that it is hard to extract lessons from one successful solution and apply it to one which is very different.

We focus on a different approach, based on the concept of a *declarative representation*. In this approach, we construct, within the computer, a *model* of the system about which we would like to reason. This model encodes our knowledge of how the system works in a computer-readable form. This representation can be manipulated by various algorithms that can answer questions based on the model. For example, a model for medical diagnosis might represent our knowledge about different diseases and how they relate to a variety of symptoms and test results. A reasoning algorithm can take this model, as well as observations relating to a particular patient, and answer questions relating to the patient's diagnosis. **The key property of a declarative representation is the separation of knowledge and reasoning. The representation has its own clear semantics, separate from the algorithms that one can apply to it. Thus, we can develop a general suite of algorithms that apply any model within a broad class, whether in the domain of medical diagnosis or speech recognition. Conversely, we can improve our model for a specific application domain without having to modify our reasoning algorithms constantly.**



declarative
representation
model

Declarative representations, or model-based methods, are a fundamental component in many fields, and models come in many flavors. Our focus in this book is on models for complex sys-

uncertainty

tems that involve a significant amount of *uncertainty*. Uncertainty appears to be an inescapable aspect of most real-world applications. It is a consequence of several factors. We are often uncertain about the true state of the system because our observations about it are partial: only some aspects of the world are observed; for example, the patient's true disease is often not directly observable, and his future prognosis is never observed. Our observations are also noisy — even those aspects that are observed are often observed with some error. The true state of the world is rarely determined with certainty by our limited observations, as most relationships are simply not deterministic, at least relative to our ability to model them. For example, there are few (if any) diseases where we have a clear, universally true relationship between the disease and its symptoms, and even fewer such relationships between the disease and its prognosis. Indeed, while it is not clear whether the universe (quantum mechanics aside) is deterministic when modeled at a sufficiently fine level of granularity, it is quite clear that it is not deterministic relative to our current understanding of it. To summarize, uncertainty arises because of limitations in our ability to observe the world, limitations in our ability to model it, and possibly even because of innate nondeterminism.

Because of this ubiquitous and fundamental uncertainty about the true state of world, we need to allow our reasoning system to consider different possibilities. One approach is simply to consider any state of the world that is possible. Unfortunately, it is only rarely the case that we can completely eliminate a state as being impossible given our observations. In our medical diagnosis example, there is usually a huge number of diseases that are *possible* given a particular set of observations. Most of them, however, are highly unlikely. If we simply list all of the possibilities, our answers will often be vacuous of meaningful content (e.g., “the patient can have any of the following 573 diseases”). **Thus, to obtain meaningful conclusions, we need to reason not just about what is possible, but also about what is *probable*.**



probability theory

The calculus of *probability theory* (see section 2.1) provides us with a formal framework for considering multiple possible outcomes and their likelihood. It defines a set of mutually exclusive and exhaustive possibilities, and associates each of them with a *probability* — a number between 0 and 1, so that the total probability of all possibilities is 1. This framework allows us to consider options that are unlikely, yet not impossible, without reducing our conclusions to content-free lists of every possibility.



Furthermore, one finds that probabilistic models are very liberating. Where in a more rigid formalism we might find it necessary to enumerate every possibility, here we can often sweep a multitude of annoying exceptions and special cases under the “probabilistic rug,” by introducing outcomes that roughly correspond to “something unusual happens.” In fact, as we discussed, this type of approximation is often inevitable, as we can only rarely (if ever) provide a deterministic specification of the behavior of a complex system. Probabilistic models allow us to make this fact explicit, and therefore often provide a model which is more faithful to reality.

1.2 Structured Probabilistic Models

This book describes a general-purpose framework for constructing and using probabilistic models of complex systems. We begin by providing some intuition for the principles underlying this framework, and for the models it encompasses. This section requires some knowledge of

basic concepts in probability theory; a reader unfamiliar with these concepts might wish to read section 2.1 first.

random variable

Complex systems are characterized by the presence of multiple interrelated aspects, many of which relate to the reasoning task. For example, in our medical diagnosis application, there are multiple possible diseases that the patient might have, dozens or hundreds of symptoms and diagnostic tests, personal characteristics that often form predisposing factors for disease, and many more matters to consider. These domains can be characterized in terms of a set of *random variables*, where the value of each variable defines an important property of the world. For example, a particular disease, such as *Flu*, may be one variable in our domain, which takes on two values, for example, *present* or *absent*; a symptom, such as *Fever*, may be a variable in our domain, one that perhaps takes on continuous values. The set of possible variables and their values is an important design decision, and it depends strongly on the questions we may wish to answer about the domain.

joint probability distribution

Our task is to reason probabilistically about the values of one or more of the variables, possibly given observations about some others. In order to do so using principled probabilistic reasoning, we need to construct a *joint distribution* over the space of possible assignments to some set of random variables $\mathcal{X}$. This type of model allows us to answer a broad range of interesting queries. For example, we can make the observation that a variable X_i takes on the specific value x_i , and ask, in the resulting *posterior distribution*, what the probability distribution is over values of another variable X_j .

posterior distribution

Example 1.1

Consider a very simple medical diagnosis setting, where we focus on two diseases — flu and hayfever; these are not mutually exclusive, as a patient can have either, both, or none. Thus, we might have two binary-valued random variables, Flu and Hayfever. We also have a 4-valued random variable Season, which is correlated both with flu and hayfever. We may also have two symptoms, Congestion and Muscle Pain, each of which is also binary-valued. Overall, our probability space has $2 \times 2 \times 4 \times 2 \times 2 = 64$ values, corresponding to the possible assignments to these five variables. Given a joint distribution over this space, we can, for example, ask questions such as how likely the patient is to have the flu given that it is fall, and that she has sinus congestion but no muscle pain; as a probability expression, this query would be denoted

$$P(\text{Flu} = \text{true} \mid \text{Season} = \text{fall}, \text{Congestion} = \text{true}, \text{Muscle Pain} = \text{false}).$$

■

1.2.1 Probabilistic Graphical Models

Specifying a joint distribution over 64 possible values, as in example 1.1, already seems fairly daunting. When we consider the fact that a typical medical- diagnosis problem has dozens or even hundreds of relevant attributes, the problem appears completely intractable. This book describes the framework of probabilistic graphical models, which provides a mechanism for exploiting structure in complex distributions to describe them compactly, and in a way that allows them to be constructed and utilized effectively.

Probabilistic graphical models use a graph-based representation as the basis for compactly encoding a complex distribution over a high-dimensional space. In this graphical representation, illustrated in figure 1.1, the nodes (or ovals) correspond to the variables in our domain, and the edges correspond to direct probabilistic interactions between them. For example, figure 1.1a (top)

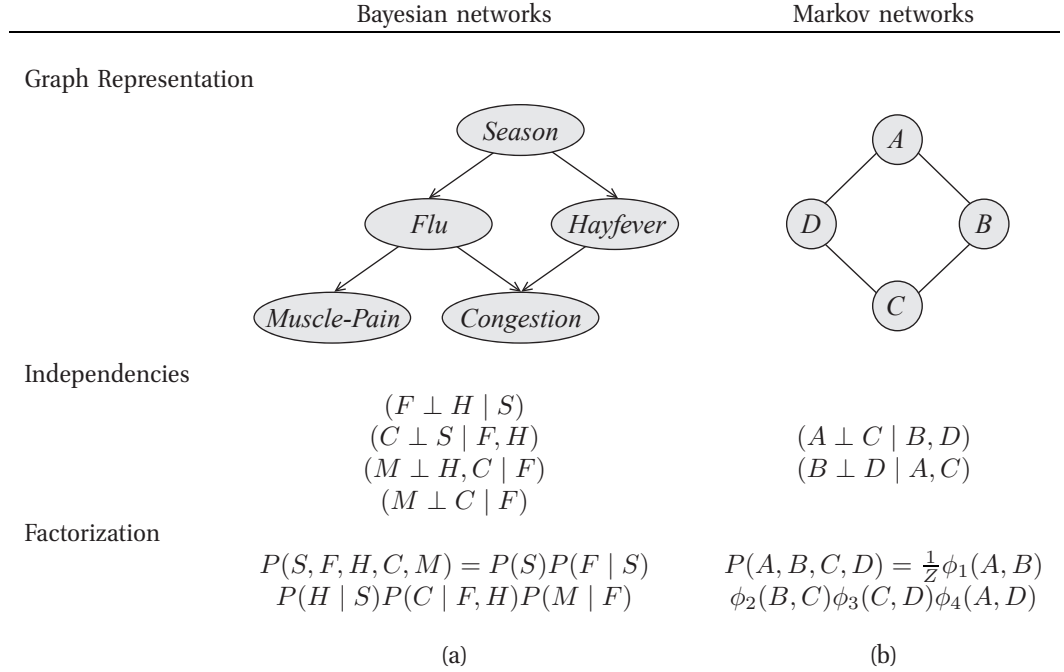


Figure 1.1 Different perspectives on probabilistic graphical models: top — the graphical representation; middle — the independencies induced by the graph structure; bottom — the factorization induced by the graph structure. (a) A sample Bayesian network. (b) A sample Markov network.

illustrates one possible graph structure for our flu example. In this graph, we see that there is no direct interaction between *Muscle Pain* and *Season*, but both interact directly with *Flu*.

There is a dual perspective that one can use to interpret the structure of this graph. From one perspective, the graph is a compact representation of a set of *independencies* that hold in the distribution; these properties take the form $\mathbf{X}$ is independent of $\mathbf{Y}$ given $\mathbf{Z}$, denoted $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$, for some subsets of variables $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$. For example, our “target” distribution P for the preceding example — the distribution encoding our beliefs about this particular situation — may satisfy the conditional independence $(\text{Congestion} \perp \text{Season} \mid \text{Flu}, \text{Hayfever})$. This statement asserts that

$$P(\text{Congestion} \mid \text{Flu}, \text{Hayfever}, \text{Season}) = P(\text{Congestion} \mid \text{Flu}, \text{Hayfever});$$

that is, if we are interested in the distribution over the patient having congestion, and we know whether he has the flu and whether he has hayfever, the season is no longer informative. Note that this assertion does *not* imply that *Season* is independent of *Congestion*; only that all of the information we may obtain from the season on the chances of having congestion we already obtain by knowing whether the patient has the flu and has hayfever. Figure 1.1a (middle) shows the set of independence assumptions associated with the graph in figure 1.1a (top).

factor

The other perspective is that the graph defines a skeleton for compactly representing a high-dimensional distribution: Rather than encode the probability of every possible assignment to all of the variables in our domain, we can “break up” the distribution into smaller *factors*, each over a much smaller space of possibilities. We can then define the overall joint distribution as a product of these factors. For example, figure 1.1(a-bottom) shows the factorization of the distribution associated with the graph in figure 1.1 (top). It asserts, for example, that the probability of the event “spring, no flu, hayfever, sinus congestion, muscle pain” can be obtained by multiplying five numbers: $P(\text{Season} = \text{spring})$, $P(\text{Flu} = \text{false} \mid \text{Season} = \text{spring})$, $P(\text{Hayfever} = \text{true} \mid \text{Season} = \text{spring})$, $P(\text{Congestion} = \text{true} \mid \text{Hayfever} = \text{true}, \text{Flu} = \text{false})$, and $P(\text{Muscle Pain} = \text{true} \mid \text{Flu} = \text{false})$. This parameterization is significantly more compact, requiring only $3 + 4 + 4 + 4 + 2 = 17$ nonredundant parameters, as opposed to 63 nonredundant parameters for the original joint distribution (the 64th parameter is fully determined by the others, as the sum over all entries in the joint distribution must sum to 1). The graph structure defines the factorization of a distribution P associated with it — the set of factors and the variables that they encompass.



It turns out that these two perspectives — the graph as a representation of a set of independencies, and the graph as a skeleton for factorizing a distribution — are, in a deep sense, equivalent. The independence properties of the distribution are precisely what allow it to be represented compactly in a factorized form. Conversely, a particular factorization of the distribution guarantees that certain independencies hold.

Bayesian network

Markov network

We describe two families of graphical representations of distributions. One, called *Bayesian networks*, uses a directed graph (where the edges have a source and a target), as shown in figure 1.1a (top). The second, called *Markov networks*, uses an undirected graph, as illustrated in figure 1.1b (top). It too can be viewed as defining a set of independence assertions (figure 1.1b [middle] or as encoding a compact factorization of the distribution (figure 1.1b [bottom])). Both representations provide the duality of independencies and factorization, but they differ in the set of independencies they can encode and in the factorization of the distribution that they induce.

1.2.2 Representation, Inference, Learning

The graphical language exploits structure that appears present in many distributions that we want to encode in practice: the property that variables tend to interact *directly* only with very few others. Distributions that exhibit this type of structure can generally be encoded naturally and compactly using a graphical model.

This framework has many advantages. First, it often allows the distribution to be written down tractably, even in cases where the explicit representation of the joint distribution is astronomically large. Importantly, the type of representation provided by this framework is *transparent*, in that a human expert can understand and evaluate its semantics and properties. This property is important for constructing models that provide an accurate reflection of our understanding of a domain. Models that are opaque can easily give rise to unexplained, and even undesirable, answers.

inference

Second, as we show, the same structure often also allows the distribution to be used effectively for *inference* — answering queries using the distribution as our model of the world. In particular, we provide algorithms for computing the posterior probability of some variables given evidence

on others. For example, we might observe that it is spring and the patient has muscle pain, and we wish to know how likely he is to have the flu, a query that can formally be written as $P(\text{Flu} = \text{true} \mid \text{Season} = \text{spring}, \text{Muscle Pain} = \text{true})$. These inference algorithms work directly on the graph structure and are generally orders of magnitude faster than manipulating the joint distribution explicitly.

Third, this framework facilitates the effective construction of these models, whether by a human expert or automatically, by *learning* from data a model that provides a good approximation to our past experience. For example, we may have a set of patient records from a doctor's office and wish to learn a probabilistic model encoding a distribution consistent with our aggregate experience. Probabilistic graphical models support a *data-driven approach* to model construction that is very effective in practice. In this approach, a human expert provides some rough guidelines on how to model a given domain. For example, the human usually specifies the attributes that the model should contain, often some of the main dependencies that it should encode, and perhaps other aspects. The details, however, are usually filled in automatically, by fitting the model to data. The models produced by this process are usually much better reflections of the domain than models that are purely hand-constructed. Moreover, they can sometimes reveal surprising connections between variables and provide novel insights about a domain.

data-driven
approach



These three components — representation, inference, and learning — are critical components in constructing an intelligent system. We need a declarative representation that is a reasonable encoding of our world model. We need to be able to use this representation effectively to answer a broad range of questions that are of interest. And we need to be able to acquire this distribution, combining expert knowledge and accumulated data. Probabilistic graphical models are one of a small handful of frameworks that support all three capabilities for a broad range of problems.

1.3 Overview and Roadmap

1.3.1 Overview of Chapters

The framework of probabilistic graphical models is quite broad, and it encompasses both a variety of different types of models and a range of methods relating to them. This book describes several types of models. For each one, we describe the three fundamental cornerstones: representation, inference, and learning.

We begin in part I, by describing the most basic type of graphical models, which are the focus of most of the book. These models encode distributions over a fixed set $\mathcal{X}$ of random variables. We describe how graphs can be used to encode distributions over such spaces, and what the properties of such distributions are.

Specifically, in chapter 3, we describe the Bayesian network representation, based on directed graphs. We describe how a Bayesian network can encode a probability distribution. We also analyze the independence properties induced by the graph structure.

In chapter 4, we move to Markov networks, the other main category of probabilistic graphical models. Here also we describe the independencies defined by the graph and the induced factorization of the distribution. We also discuss the relationship between Markov networks and Bayesian networks, and briefly describe a framework that unifies both.

In chapter 5, we delve a little deeper into the representation of the parameters in probabilistic

models, focusing mostly on Bayesian networks, whose parameterization is more constrained. We describe representations that capture some of the finer-grained structure of the distribution, and show that, here also, capturing structure can provide significant gains.

In chapter 6, we turn to formalisms that extend the basic framework of probabilistic graphical models to settings where the set of variables is no longer rigidly circumscribed in advance. One such setting is a *temporal* one, where we wish to model a system whose state evolves over time, requiring us to consider distributions over entire trajectories. We describe a compact representation — a *dynamic Bayesian network* — that allows us to represent structured systems that evolve over time. We then describe a family of extensions that introduce various forms of higher level structure into the framework of probabilistic graphical models. Specifically, we focus on domains containing *objects* (whether concrete or abstract), characterized by attributes, and related to each other in various ways. Such domains can include repeated structure, since different objects of the same type share the same probabilistic model. These languages provide a significant extension to the expressive power of the standard graphical models.

In chapter 7, we take a deeper look at models that include continuous variables. Specifically, we explore the properties of the multivariate Gaussian distribution and the representation of such distributions as both directed and undirected graphical models. Although the class of Gaussian distributions is a limited one and not suitable for all applications, it turns out to play a critical role even when dealing with distributions that are not Gaussian.

In chapter 8, we take a deeper, more technical look at probabilistic models, defining a general framework called the *exponential family*, that encompasses a broad range of distributions. This chapter provides some basic concepts and tools that will turn out to play an important role in later development.

We then turn, in part II, to a discussion of the inference task. In chapter 9, we describe the basic ideas underlying exact inference in probabilistic graphical models. We first analyze the fundamental difficulty of the exact inference task, separately from any particular inference algorithm we might develop. We then present two basic algorithms for exact inference — variable elimination and conditioning — both of which are equally applicable to both directed and undirected models. Both of these algorithms can be viewed as operating over the graph structure defined by the probabilistic model. They build on basic concepts, such as graph properties and dynamic programming algorithms, to provide efficient solutions to the inference task. We also provide an analysis of their computational cost in terms of the graph structure, and we discuss where exact inference is feasible.

In chapter 10, we describe an alternative view of exact inference, leading to a somewhat different algorithm. The benefit of this alternative algorithm is twofold. First, it uses dynamic programming to avoid repeated computations in settings where we wish to answer more than a single query using the same network. Second, it defines a natural algorithm that uses message passing on a graph structure; this algorithm forms the basis for approximate inference algorithms developed in later chapters.

Because exact inference is computationally intractable for many models of interest, we then proceed to describe approximate inference algorithms, which trade off accuracy with computational cost. We present two main classes of such algorithms. In chapter 11, we describe a class of methods that can be viewed from two very different perspectives: On one hand, they are direct generalizations of the graph-based message-passing approach developed for the case of exact inference in chapter 10. On the other hand, they can be viewed as solving an optimization

problem: one where we approximate the distribution of interest using a simpler representation that allows for feasible inference. The equivalence of these views provides important insights and suggests a broad family of algorithms that one can apply to approximate inference.

In chapter 12, we describe a very different class of methods: *particle-based methods*, which approximate a complex joint distribution by considering samples from it (also known as particles). We describe several methods from this general family. These methods are generally based on core techniques from statistics, such as importance sampling and Markov-chain Monte Carlo methods. Once again, the connection to this general class of methods suggests multiple opportunities for new algorithms.

While the representation of probabilistic graphical models applies, to a great extent, to models including both discrete and continuous-valued random variables, inference in models involving continuous variables is significantly more challenging than the purely discrete case. In chapter 14, we consider the task of inference in continuous and *hybrid* (continuous/discrete) networks, and we discuss whether and how the exact and approximate inference methods developed in earlier chapters can be applied in this setting.

The representation that we discussed in chapter 6 allows a compact encoding of networks whose size can be unboundedly large. Such networks pose particular challenges to inference algorithms. In this chapter, we discuss some special-purpose methods that have been developed for the particular settings of networks that model dynamical systems.

We then turn, in part III, to the third of our main topics — learning probabilistic models from data. We begin in chapter 16 by reviewing some of the fundamental concepts underlying the general task of learning models from data. We then present the spectrum of learning problems that we address in this part of the book. These problems vary along two main axes: the extent to which we are given prior knowledge specifying the model, and whether the data from which we learn contain complete observations of all of the relevant variables. In contrast to the inference task, where the same algorithms apply equally to Bayesian networks and Markov networks, the learning task is quite different for these two classes of models. We begin with studying the learning task for Bayesian networks.

In chapter 17, we focus on the most basic learning task: learning parameters for a Bayesian network with a given structure, from fully observable data. Although this setting may appear somewhat restrictive, it turns out to form the basis for our entire development of Bayesian network learning. As we show, the factorization of the distribution, which was central both to representation and to inference, also plays a key role in making inference feasible.

We then move, in chapter 18, to the harder problem of learning both Bayesian network structure and the parameters, still from fully observed data. The learning algorithms we present trade off the accuracy with which the learned network represents the empirical distribution for the complexity of the resulting structure. As we show, the type of independence assumptions underlying the Bayesian network representation often hold, at least approximately, in real-world distributions. Thus, these learning algorithms often result in reasonably compact structures that capture much of the signal in the distribution.

In chapter 19, we address the Bayesian network learning task in a setting where we have access only to partial observations of the relevant variables (for example, when the available patient records have missing entries). This type of situation occurs often in real-world settings. Unfortunately, the resulting learning task is considerably harder, and the resulting algorithms are both more complex and less satisfactory in terms of their performance.

We conclude the discussion of learning in chapter 20 by considering the problem of learning Markov networks from data. It turns out that the learning tasks for Markov networks are significantly harder than the corresponding problem for Bayesian networks. We explain the difficulties and discuss the existing solutions.

Finally, in part IV, we turn to a different type of extension, where we consider the use of this framework for other forms of reasoning. Specifically, we consider cases where we can act, or intervene, in the world.

In chapter 21, we focus on the semantics of intervention and its relation to causality. We present the notion of a *causal model*, which allows us to answer not only queries of the form “if I observe X, what do I learn about Y,” but also *intervention queries*, of the form “if I manipulate X, what effect does it have on Y.”

We then turn to the task of *decision making* under uncertainty. Here, we must consider not only the distribution over different states of the world, but also the preferences of the agent regarding these outcomes. In chapter 22, we discuss the notion of *utility functions* and how they can encode an agent’s preferences about complex situations involving multiple variables. As we show, the same ideas that we used to provide compact representations of probability distribution can also be used for utility functions.

In chapter 23, we describe a unified representation for decision making, called *influence diagrams*. Influence diagrams extend Bayesian networks by introducing actions and utilities. We present algorithms that use influence diagrams for making decisions that optimize the agent’s expected utility. These algorithms utilize many of the same ideas that formed the basis for exact inference in Bayesian networks.

We conclude with a high-level synthesis of the techniques covered in this book, and with some guidance on how to use them in tackling a new problem.

1.3.2 Reader’s Guide

As we mentioned, the topics described in this book relate to multiple fields, and techniques from other disciplines — probability theory, computer science, information theory, optimization, statistics, and more — are used in various places throughout it. While it is impossible to present all of the relevant material within the scope of this book, we have attempted to make the book somewhat self-contained by providing a very brief review of the key concepts from these related disciplines in chapter 2.

Some of this material, specifically the review of probability theory and of graph-related concepts, is very basic yet central to most of the development in this book. Readers who are less familiar with these topics may wish to read these sections carefully, and even knowledgeable readers may wish to briefly review them to gain familiarity with the notations used. Other background material, covering such topics as information theory, optimization, and algorithmic concepts, can be found in the appendix.

The chapters in the book are structured as follows. The main text in each chapter provides the detailed technical development of the key ideas. Beyond the main text, most chapters contain boxes that contain interesting material that augments these ideas. These boxes come in three types: *Skill boxes* describe “hands-on” tricks and techniques, which, while often heuristic in nature, are important for getting the basic algorithms described in the text to work in practice. *Case study boxes* describe empirical case studies relating to the techniques described in the text.

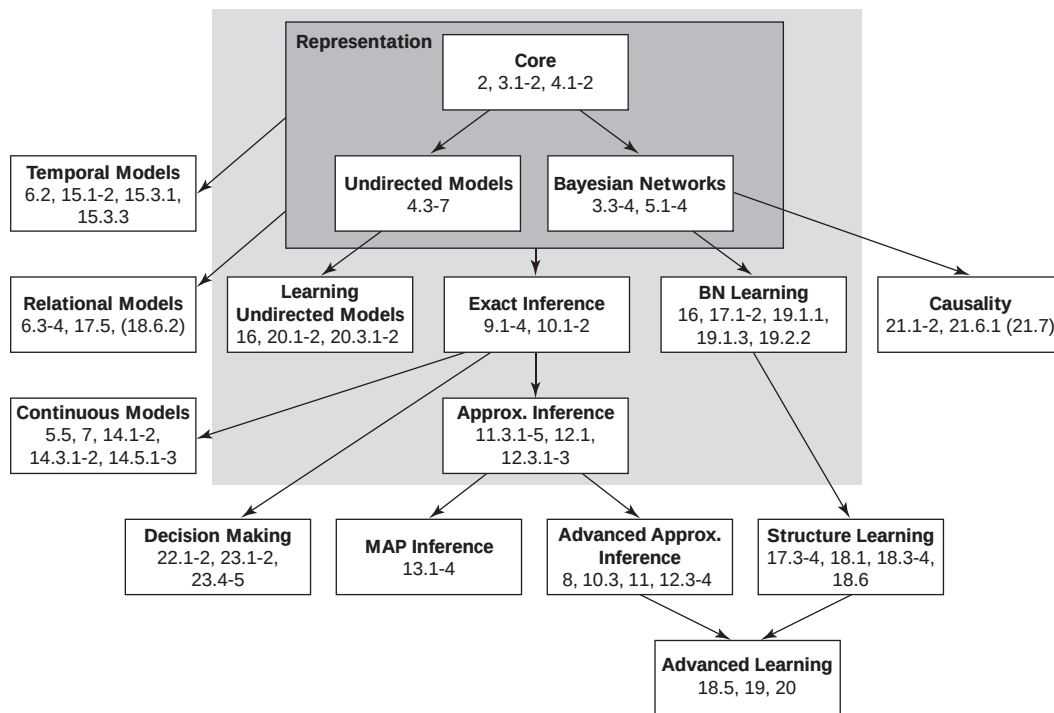


Figure 1.2 A reader's guide to the structure and dependencies in this book

These case studies include both empirical results on how the algorithms perform in practice and descriptions of applications of these algorithms to interesting domains, illustrating some of the issues encountered in practice. Finally, *concept boxes* present particular instantiations of the material described in the text, which have had significant impact in their own right.

This textbook is clearly too long to be used in its entirety in a one-semester class. Figure 1.2 tries to delineate some coherent subsets of the book that can be used for teaching and other purposes. The small, labeled boxes represent “units” of material on particular topics. Arrows between the boxes represent dependencies between these units. The first enclosing box (solid line) represents material that is fundamental to everything else, and that should be read by anyone using this book. One can then use the dependencies between the boxes to expand or reduce the depth of the coverage on any given topic. The material in the larger box (dashed line) forms a good basis for a one-semester (or even one-quarter) overview class. Some of the sections in the book are marked with an asterisk, denoting the fact that they contain more technically advanced material. In most cases, these sections are self-contained, and they can be skipped without harming the reader’s ability to understand the rest of the text.

We have attempted in this book to present a synthesis of ideas, most of which have been developed over many years by multiple researchers. To avoid futile attempts to divide up the credit precisely, we have omitted all bibliographical references from the technical presentation

in the chapters. Rather, each chapter ends with a section called “Relevant Literature,” which describes the historical evolution of the material in the chapter, acknowledges the papers and books that developed the key concepts, and provides some additional readings on material relevant to the chapter. We encourage the reader who is interested in a topic to follow up on some of these additional readings, since there are many interesting developments that we could not cover in this book.

Finally, each chapter includes a set of exercises that explore in additional depth some of the material described in the text and present some extensions to it. The exercises are annotated with an asterisk for exercises that are somewhat more difficult, and with two asterisks for ones that are truly challenging.

Additional material related to this book, including slides and figures, solutions to some of the exercises, and errata, can be found online at <http://pgm.stanford.edu>.

1.3.3 Connection to Other Disciplines

The ideas we describe in this book are connected to many fields. From probability theory, we inherit the basic concept of a probability distribution, as well as many of the operations we can use to manipulate it. From computer science, we exploit the key idea of using a graph as a data structure, as well as a variety of algorithms for manipulating graphs and other data structures. These algorithmic ideas and the ability to manipulate probability distributions using discrete data structures are some of the key elements that make the probabilistic manipulations tractable. Decision theory extends these basic ideas to the task of decision making under uncertainty and provides the formal foundation for this task.

From computer science, and specifically from artificial intelligence, these models inherit the idea of using a declarative representation of the world to separate procedural reasoning from our domain knowledge. This idea is of key importance to the generality of this framework and its applicability to such a broad range of tasks.

Various ideas from other disciplines also arise in this field. Statistics plays an important role both in certain aspects of the representation and in some of the work on learning models from data. Optimization plays a role in providing algorithms both for approximate inference and for learning models from data. Bayesian networks first arose, albeit in a restricted way, in the setting of modeling genetic inheritance in human family trees; in fact, restricted version of some of the exact inference algorithms we discuss were first developed in this context. Similarly, undirected graphical models first arose in physics as a model for systems of electrons, and some of the basic concepts that underlie recent work on approximate inference developed from that setting.

Information theory plays a dual role in its interaction with this field. Information-theoretic concepts such as entropy and information arise naturally in various settings in this framework, such as evaluating the quality of a learned model. Thus, tools from this discipline are a key component in our analytic toolkit. On the other side, the recent successes in coding theory, based on the relationship between inference in probabilistic models and the task of decoding messages sent over a noisy channel, have led to a resurgence of work on approximate inference in graphical models. The resulting developments have revolutionized both the development of error-correcting codes and the theory and practice of approximate message-passing algorithms in graphical models.

1.3.3.1 What Have We Gained?

Although the framework we describe here shares common elements with a broad range of other topics, it has a coherent common core: the use of structure to allow a compact representation, effective reasoning, and feasible learning of general-purpose, factored, probabilistic models. These elements provide us with a general infrastructure for reasoning and learning about complex domains.

As we discussed earlier, by using a declarative representation, we essentially separate out the description of the model for the particular application, and the general-purpose algorithms used for inference and learning. Thus, this framework provides a general algorithmic toolkit that can be applied to many different domains.

Indeed, probabilistic graphical models have made a significant impact on a broad spectrum of real-world applications. For example, these models have been used for medical and fault diagnosis, for modeling human genetic inheritance of disease, for segmenting and denoising images, for decoding messages sent over a noisy channel, for revealing genetic regulatory processes, for robot localization and mapping, and more. Throughout this book, we will describe how probabilistic graphical models were used to address these applications and what issues arise in the application of these models in practice.

In addition to practical applications, these models provide a formal framework for a variety of fundamental problems. For example, the notion of conditional independence and its explicit graph-based representation provide a clear formal semantics for irrelevance of information. This framework also provides a general methodology for handling data fusion — we can introduce *sensor variables* that are noisy versions of the true measured quantity, and use Bayesian conditioning to combine the different measurements. The use of a probabilistic model allows us to provide a formal measure for model quality, in terms of a numerical fit of the model to observed data; this measure underlies much of our work on learning models from data. The temporal models we define provide a formal framework for defining a general trend toward persistence of state over time, in a way that does not raise inconsistencies when change does occur.

In general, part of the rich development in this field is due to the close and continuous interaction between theory and practice. In this field, unlike many others, the distance between theory and practice is quite small, and there is a constant flow of ideas and problems between them. Problems or ideas arise in practical applications and are analyzed and subsequently developed in more theoretical papers. Algorithms for which no theoretical analysis exists are tried out in practice, and the profile of where they succeed and fail often provides the basis for subsequent analysis. This rich synergy leads to a continuous and vibrant development, and it is a key factor in the success of this area.

1.4 Historical Notes

The foundations of probability theory go back to the sixteenth century, when Gerolamo Cardano began a formal analysis of games of chance, followed by additional key developments by Pierre de Fermat and Blaise Pascal in the seventeenth century. The initial development involved only discrete probability spaces, and the analysis methods were purely combinatorial. The foundations of modern probability theory, with its measure-theoretic underpinnings, were laid by Andrey Kolmogorov in the 1930s.

2 *Foundations*

In this chapter, we review some important background material regarding key concepts from probability theory, information theory, and graph theory. This material is included in a separate introductory chapter, since it forms the basis for much of the development in the remainder of the book. Other background material — such as discrete and continuous optimization, algorithmic complexity analysis, and basic algorithmic concepts — is more localized to particular topics in the book. Many of these concepts are presented in the appendix; others are presented in concept boxes in the appropriate places in the text. All of this material is intended to focus only on the minimal subset of ideas required to understand most of the discussion in the remainder of the book, rather than to provide a comprehensive overview of the field it surveys. We encourage the reader to explore additional sources for more details about these areas.

2.1 Probability Theory

The main focus of this book is on complex probability distributions. In this section we briefly review basic concepts from probability theory.

2.1.1 Probability Distributions

When we use the word “probability” in day-to-day life, we refer to a degree of confidence that an event of an uncertain nature will occur. For example, the weather report might say “there is a low probability of light rain in the afternoon.” Probability theory deals with the formal foundations for discussing such estimates and the rules they should obey.

Before we discuss the representation of probability, we need to define what the events are to which we want to assign a probability. These events might be different outcomes of throwing a die, the outcome of a horse race, the weather configurations in California, or the possible failures of a piece of machinery.

2.1.1.1 Event Spaces

event
outcome space

Formally, we define *events* by assuming that there is an agreed upon *space* of possible outcomes, which we denote by Ω . For example, if we consider dice, we might set $\Omega = \{1, 2, 3, 4, 5, 6\}$. In the case of a horse race, the space might be all possible orders of arrivals at the finish line, a much larger space.

measurable event

In addition, we assume that there is a set of *measurable events* $\mathcal{S}$ to which we are willing to assign probabilities. Formally, each event $\alpha \in \mathcal{S}$ is a subset of Ω . In our die example, the event $\{6\}$ represents the case where the die shows 6, and the event $\{1, 3, 5\}$ represents the case of an odd outcome. In the horse-race example, we might consider the event “Lucky Strike wins,” which contains all the outcomes in which the horse Lucky Strike is first.

Probability theory requires that the event space satisfy three basic properties:

- It contains the *empty event* $\emptyset$, and the *trivial event* Ω .
- It is closed under union. That is, if $\alpha, \beta \in \mathcal{S}$, then so is $\alpha \cup \beta$.
- It is closed under complementation. That is, if $\alpha \in \mathcal{S}$, then so is $\Omega - \alpha$.

The requirement that the event space is closed under union and complementation implies that it is also closed under other Boolean operations, such as intersection and set difference.

2.1.1.2 Probability Distributions

Definition 2.1probability
distribution

A probability distribution P over $(\Omega, \mathcal{S})$ is a mapping from events in $\mathcal{S}$ to real values that satisfies the following conditions:

- $P(\alpha) \geq 0$ for all $\alpha \in \mathcal{S}$.
- $P(\Omega) = 1$.
- If $\alpha, \beta \in \mathcal{S}$ and $\alpha \cap \beta = \emptyset$, then $P(\alpha \cup \beta) = P(\alpha) + P(\beta)$. ■

The first condition states that probabilities are not negative. The second states that the “trivial event,” which allows all possible outcomes, has the maximal possible probability of 1. The third condition states that the probability that one of two mutually disjoint events will occur is the sum of the probabilities of each event. These two conditions imply many other conditions. Of particular interest are $P(\emptyset) = 0$, and $P(\alpha \cup \beta) = P(\alpha) + P(\beta) - P(\alpha \cap \beta)$.

2.1.1.3 Interpretations of Probability

Before we continue to discuss probability distributions, we need to consider the interpretations that we might assign to them. Intuitively, the probability $P(\alpha)$ of an event α quantifies the degree of confidence that α will occur. If $P(\alpha) = 1$, we are certain that one of the outcomes in α occurs, and if $P(\alpha) = 0$, we consider all of them impossible. Other probability values represent options that lie between these two extremes.

This description, however, does not provide an answer to what the numbers mean. There are two common interpretations for probabilities.

frequentist
interpretation

The *frequentist* interpretation views probabilities as frequencies of events. More precisely, the probability of an event is the fraction of times the event occurs if we repeat the experiment indefinitely. For example, suppose we consider the outcome of a particular die roll. In this case, the statement $P(\alpha) = 0.3$, for $\alpha = \{1, 3, 5\}$, states that if we repeatedly roll this die and record the outcome, then the fraction of times the outcomes in α will occur is 0.3. More precisely, the limit of the sequence of fractions of outcomes in α in the first roll, the first two rolls, the first three rolls, $\dots$, the first n rolls, $\dots$ is 0.3.

The frequentist interpretation gives probabilities a tangible semantics. When we discuss concrete physical systems (for example, dice, coin flips, and card games) we can envision how these frequencies are defined. It is also relatively straightforward to check that frequencies must satisfy the requirements of proper distributions.

The frequentist interpretation fails, however, when we consider events such as “It will rain tomorrow afternoon.” Although the time span of “Tomorrow afternoon” is somewhat ill defined, we expect it to occur exactly once. It is not clear how we define the frequencies of such events. Several attempts have been made to define the probability for such an event by finding a *reference class* of similar events for which frequencies are well defined; however, none of them has proved entirely satisfactory. Thus, the frequentist approach does not provide a satisfactory interpretation for a statement such as “the probability of rain tomorrow afternoon is 0.3.”

reference class



subjective interpretation

An alternative interpretation views probabilities as *subjective degrees of belief*. Under this interpretation, the statement $P(\alpha) = 0.3$ represents a subjective statement about one's own degree of belief that the event α will come about. Thus, the statement “the probability of rain tomorrow afternoon is 50 percent” tells us that in the opinion of the speaker, the chances of rain and no rain tomorrow afternoon are the same. Although tomorrow afternoon will occur only once, we can still have uncertainty about its outcome, and represent it using numbers (that is, probabilities).

This description still does not resolve what exactly it means to hold a particular degree of belief. What stops a person from stating that the probability that Bush will win the election is 0.6 and the probability that he will lose is 0.8? The source of the problem is that we need to explain how subjective degrees of beliefs (something that is internal to each one of us) are reflected in our actions.

This issue is a major concern in subjective probabilities. One possible way of attributing degrees of beliefs is by a betting game. Suppose you believe that $P(\alpha) = 0.8$. Then you would be willing to place a bet of \$1 against \$3. To see this, note that with probability 0.8 you gain a dollar, and with probability 0.2 you lose \$3, so on average this bet is a good deal with expected gain of 20 cents. In fact, you might be even tempted to place a bet of \$1 against \$4. Under this bet the average gain is 0, so you should not mind. However, you would not consider it worthwhile to place a bet \$1 against \$4 and 10 cents, since that would have negative expected gain. Thus, by finding which bets you are willing to place, we can assess your degrees of beliefs.

The key point of this mental game is the following. If you hold degrees of belief that do not satisfy the rule of probability, then by a clever construction we can find a series of bets that would result in a sure negative outcome for you. Thus, the argument goes, a rational person must hold degrees of belief that satisfy the rules of probability.¹

In the remainder of the book we discuss probabilities, but we usually do not explicitly state their interpretation. Since both interpretations lead to the same mathematical rules, the technical definitions hold for both interpretations.

1. As stated, this argument assumes as that people's preferences are directly proportional to their expected earnings. For small amounts of money, this assumption is quite reasonable. We return to this topic in chapter 22.

2.1.2 Basic Concepts in Probability

2.1.2.1 Conditional Probability

To use a concrete example, suppose we consider a distribution over a population of students taking a certain course. The space of outcomes is simply the set of all students in the population. Now, suppose that we want to reason about the students' intelligence and their final grade. We can define the event α to denote "all students with grade A," and the event β to denote "all students with high intelligence." Using our distribution, we can consider the probability of these events, as well as the probability of $\alpha \cap \beta$ (the set of intelligent students who got grade A). This, however, does not directly tell us how to update our beliefs given new evidence. Suppose we learn that a student has received the grade A; what does that tell us about her intelligence?

This kind of question arises every time we want to use distributions to reason about the real world. More precisely, after learning that an event α is true, how do we change our probability about β occurring? The answer is via the notion of *conditional probability*. Formally, the conditional probability of β given α is defined as

conditional
probability

$$P(\beta \mid \alpha) = \frac{P(\alpha \cap \beta)}{P(\alpha)} \quad (2.1)$$

That is, the probability that β is true given that we know α is the relative proportion of outcomes satisfying β among these that satisfy α . (Note that the conditional probability is not defined when $P(\alpha) = 0$.)

The conditional probability given an event (say α) satisfies the properties of definition 2.1 (see exercise 2.4), and thus it is a probability distribution by its own right. Hence, we can think of the conditioning operation as taking one distribution and returning another over the same probability space.

2.1.2.2 Chain Rule and Bayes Rule

From the definition of the conditional distribution, we immediately see that

$$P(\alpha \cap \beta) = P(\alpha)P(\beta \mid \alpha). \quad (2.2)$$

chain rule

This equality is known as the *chain rule* of conditional probabilities. More generally, if $\alpha_1, \dots, \alpha_k$ are events, then we can write

$$P(\alpha_1 \cap \dots \cap \alpha_k) = P(\alpha_1)P(\alpha_2 \mid \alpha_1) \cdots P(\alpha_k \mid \alpha_1 \cap \dots \cap \alpha_{k-1}). \quad (2.3)$$

In other words, we can express the probability of a combination of several events in terms of the probability of the first, the probability of the second given the first, and so on. It is important to notice that we can expand this expression using any order of events — the result will remain the same.

Bayes' rule

Another immediate consequence of the definition of conditional probability is *Bayes' rule*

$$P(\alpha \mid \beta) = \frac{P(\beta \mid \alpha)P(\alpha)}{P(\beta)}. \quad (2.4)$$

A more general conditional version of Bayes' rule, where all our probabilities are conditioned on some background event γ , also holds:

$$P(\alpha \mid \beta \cap \gamma) = \frac{P(\beta \mid \alpha \cap \gamma)P(\alpha \mid \gamma)}{P(\beta \mid \gamma)}.$$

Bayes' rule is important in that it allows us to compute the conditional probability $P(\alpha \mid \beta)$ from the “inverse” conditional probability $P(\beta \mid \alpha)$.

Example 2.1

prior

*Consider the student population, and let Smart denote smart students and GradeA denote students who got grade A. Assume we believe (perhaps based on estimates from past statistics) that $P(\text{GradeA} \mid \text{Smart}) = 0.6$, and now we learn that a particular student received grade A. Can we estimate the probability that the student is smart? According to Bayes' rule, this depends on our prior probability for students being smart (before we learn anything about them) and the prior probability of students receiving high grades. For example, suppose that $P(\text{Smart}) = 0.3$ and $P(\text{GradeA}) = 0.2$, then we have that $P(\text{Smart} \mid \text{GradeA}) = 0.6 * 0.3 / 0.2 = 0.9$. That is, an A grade strongly suggests that the student is smart. On the other hand, if the test was easier and high grades were more common, say, $P(\text{GradeA}) = 0.4$ then we would get that $P(\text{Smart} \mid \text{GradeA}) = 0.6 * 0.3 / 0.4 = 0.45$, which is much less conclusive about the student. ■*

Another classic example that shows the importance of this reasoning is in disease screening. To see this, consider the following hypothetical example (none of the mentioned figures are related to real statistics).

Example 2.2

Suppose that a tuberculosis (TB) skin test is 95 percent accurate. That is, if the patient is TB-infected, then the test will be positive with probability 0.95, and if the patient is not infected, then the test will be negative with probability 0.95. Now suppose that a person gets a positive test result. What is the probability that he is infected? Naïve reasoning suggests that if the test result is wrong 5 percent of the time, then the probability that the subject is infected is 0.95. That is, 95 percent of subjects with positive results have TB.

If we consider the problem by applying Bayes' rule, we see that we need to consider the prior probability of TB infection, and the probability of getting positive test result. Suppose that 1 in 1000 of the subjects who get tested is infected. That is, $P(\text{TB}) = 0.001$. What is the probability of getting a positive test result? From our description, we see that $0.001 \cdot 0.95$ infected subjects get a positive result, and $0.999 \cdot 0.05$ uninfected subjects get a positive result. Thus, $P(\text{Positive}) = 0.0509$. Applying Bayes' rule, we get that $P(\text{TB} \mid \text{Positive}) = 0.001 \cdot 0.95 / 0.0509 \approx 0.0187$. Thus, although a subject with a positive test is much more probable to be TB-infected than is a random subject, fewer than 2 percent of these subjects are TB-infected. ■

2.1.3 Random Variables and Joint Distributions

2.1.3.1 Motivation

Our discussion of probability distributions deals with events. Formally, we can consider any event from the set of measurable events. The description of events is in terms of sets of outcomes. In many cases, however, it would be more natural to consider *attributes* of the outcome. For example, if we consider a patient, we might consider attributes such as “age,”

“gender,” and “smoking history” that are relevant for assigning probability over possible diseases and symptoms. We would like then consider events such as “age > 55, heavy smoking history, and suffers from repeated cough.”

To use a concrete example, consider again a distribution over a population of students in a course. Suppose that we want to reason about the intelligence of students, their final grades, and so forth. We can use an event such as *GradeA* to denote the subset of students that received the grade A and use it in our formulation. However, this discussion becomes rather cumbersome if we also want to consider students with grade B, students with grade C, and so on. Instead, we would like to consider a way of directly referring to a student’s grade in a clean, mathematical way.

random variable The formal machinery for discussing attributes and their values in different outcomes are *random variables*. A random variable is a way of reporting an attribute of the outcome. For example, suppose we have a random variable *Grade* that reports the final grade of a student, then the statement $P(\text{Grade} = A)$ is another notation for $P(\text{Grade}A)$.

2.1.3.2 What Is a Random Variable?

Formally, a random variable, such as *Grade*, is defined by a function that associates with each outcome in Ω a value. For example, *Grade* is defined by a function f_{Grade} that maps each person in Ω to his or her grade (say, one of A, B, or C). The event $\text{Grade} = A$ is a shorthand for the event $\{\omega \in \Omega : f_{\text{Grade}}(\omega) = A\}$. In our example, we might also have a random variable *Intelligence* that (for simplicity) takes as values either “high” or “low.” In this case, the event “*Intelligence* = *high*” refers, as can be expected, to the set of smart (high intelligence) students.

Random variables can take different sets of values. We can think of *categorical* (or *discrete*) random variables that take one of a few values, as in our two preceding examples. We can also talk about random variables that can take infinitely many values (for example, integer or real values), such as *Height* that denotes a student’s height. We use $\text{Val}(X)$ to denote the set of values that a random variable X can take.

In most of the discussion in this book we examine either categorical random variables or random variables that take real values. We will usually use uppercase roman letters X, Y, Z to denote random variables. In discussing generic random variables, we often use a lowercase letter to refer to a value of a random variable. Thus, we use x to refer to a generic value of X . For example, in statements such as “ $P(X = x) \geq 0$ for all $x \in \text{Val}(X)$.” When we discuss categorical random variables, we use the notation $x^1, \dots, x^k$, for $k = |\text{Val}(X)|$ (the number of elements in $\text{Val}(X)$), when we need to enumerate the specific values of X , for example, in statements such as

$$\sum_{i=1}^k P(X = x^i) = 1.$$

multinomial
distribution

The distribution over such a variable is called a *multinomial*. In the case of a binary-valued random variable X , where $\text{Val}(X) = \{\text{false}, \text{true}\}$, we often use x^1 to denote the value *true* for X , and x^0 to denote the value *false*. The distribution of such a random variable is called a *Bernoulli distribution*.

Bernoulli
distribution

We also use boldface type to denote sets of random variables. Thus, $\mathbf{X}, \mathbf{Y}$, or $\mathbf{Z}$ are typically used to denote a set of random variables, while $\mathbf{x}, \mathbf{y}, \mathbf{z}$ denote assignments of values to the

variables in these sets. We extend the definition of $Val(\mathbf{X})$ to refer to sets of variables in the obvious way. Thus, $\mathbf{x}$ is always a member of $Val(\mathbf{X})$. For $\mathbf{Y} \subseteq \mathbf{X}$, we use $\mathbf{x}\langle\mathbf{Y}\rangle$ to refer to the assignment within $\mathbf{x}$ to the variables in $\mathbf{Y}$. For two assignments $\mathbf{x}$ (to $\mathbf{X}$) and $\mathbf{y}$ (to $\mathbf{Y}$), we say that $\mathbf{x} \sim \mathbf{y}$ if they agree on the variables in their intersection, that is, $\mathbf{x}\langle\mathbf{X} \cap \mathbf{Y}\rangle = \mathbf{y}\langle\mathbf{X} \cap \mathbf{Y}\rangle$.

In many cases, the notation $P(X = x)$ is redundant, since the fact that x is a value of X is already reported by our choice of letter. Thus, in many texts on probability, the identity of a random variable is not explicitly mentioned, but can be inferred through the notation used for its value. Thus, we use $P(x)$ as a shorthand for $P(X = x)$ when the identity of the random variable is clear from the context. Another shorthand notation is that $\sum_x$ refers to a sum over all possible values that X can take. Thus, the preceding statement will often appear as $\sum_x P(x) = 1$. Finally, another standard notation has to do with conjunction. Rather than write $P((X = x) \cap (Y = y))$, we write $P(X = x, Y = y)$, or just $P(x, y)$.

2.1.3.3 Marginal and Joint Distributions

marginal
distribution

Once we define a random variable X , we can consider the distribution over events that can be described using X . This distribution is often referred to as the *marginal distribution* over the random variable X . We denote this distribution by $P(X)$.

Returning to our population example, consider the random variable *Intelligence*. The marginal distribution over *Intelligence* assigns probability to specific events such as $P(\text{Intelligence} = \text{high})$ and $P(\text{Intelligence} = \text{low})$, as well as to the trivial event $P(\text{Intelligence} \in \{\text{high}, \text{low}\})$. Note that these probabilities are defined by the probability distribution over the original space. For concreteness, suppose that $P(\text{Intelligence} = \text{high}) = 0.3$, $P(\text{Intelligence} = \text{low}) = 0.7$.

If we consider the random variable *Grade*, we can also define a marginal distribution. This is a distribution over all events that can be described in terms of the *Grade* variable. In our example, we have that $P(\text{Grade} = A) = 0.25$, $P(\text{Grade} = B) = 0.37$, and $P(\text{Grade} = C) = 0.38$.

It should be fairly obvious that the marginal distribution is a probability distribution satisfying the properties of definition 2.1. In fact, the only change is that we restrict our attention to the subsets of $\mathcal{S}$ that can be described with the random variable X .

joint distribution

In many situations, we are interested in questions that involve the values of several random variables. For example, we might be interested in the event “*Intelligence* = high and *Grade* = A.” To discuss such events, we need to consider the *joint distribution* over these two random variables. In general, the joint distribution over a set $\mathcal{X} = \{X_1, \dots, X_n\}$ of random variables is denoted by $P(X_1, \dots, X_n)$ and is the distribution that assigns probabilities to events that are specified in terms of these random variables. We use ξ to refer to a full assignment to the variables in $\mathcal{X}$, that is, $\xi \in Val(\mathcal{X})$.

The joint distribution of two random variables has to be consistent with the marginal distribution, in that $P(x) = \sum_y P(x, y)$. This relationship is shown in figure 2.1, where we compute the marginal distribution over *Grade* by summing the probabilities along each row. Similarly, we find the marginal distribution over *Intelligence* by summing out along each column. The resulting sums are typically written in the row or column margins, whence the term “marginal distribution.”

Suppose we have a joint distribution over the variables $\mathcal{X} = \{X_1, \dots, X_n\}$. The most fine-grained events we can discuss using these variables are ones of the form “ $X_1 = x_1$ and $X_2 = x_2, \dots$, and $X_n = x_n$ ” for a choice of values $x_1, \dots, x_n$ for all the variables. Moreover,

		<i>Intelligence</i>		
		<i>low</i>	<i>high</i>	
<i>Grade</i>	<i>A</i>	0.07	0.18	0.25
	<i>B</i>	0.28	0.09	0.37
	<i>C</i>	0.35	0.03	0.38
		0.7	0.3	1

Figure 2.1 Example of a joint distribution $P(\textit{Intelligence}, \textit{Grade})$: Values of *Intelligence* (columns) and *Grade* (rows) with the associated marginal distribution on each variable.

canonical
outcome space

atomic outcome

any two such events must be either identical or disjoint, since they both assign values to all the variables in $\mathcal{X}$. In addition, any event defined using variables in $\mathcal{X}$ must be a union of a set of such events. Thus, we are effectively working in a *canonical outcome space*: a space where each outcome corresponds to a joint assignment to $X_1, \dots, X_n$. More precisely, all our probability computations remain the same whether we consider the original outcome space (for example, all students), or the canonical space (for example, all combinations of intelligence and grade). We use ξ to denote these *atomic outcomes*: those assigning a value to each variable in $\mathcal{X}$. For example, if we let $\mathcal{X} = \{\textit{Intelligence}, \textit{Grade}\}$, there are six atomic outcomes, shown in figure 2.1. The figure also shows one possible joint distribution over these six outcomes.

Based on this discussion, from now on we will not explicitly specify the set of outcomes and measurable events, and instead implicitly assume the canonical outcome space.

2.1.3.4 Conditional Probability

conditional
distribution

The notion of conditional probability extends to induced distributions over random variables. For example, we use the notation $P(\textit{Intelligence} \mid \textit{Grade} = A)$ to denote the *conditional distribution* over the events describable by *Intelligence* given the knowledge that the student's grade is A. Note that the conditional distribution over a random variable given an observation of the value of another one is not the same as the marginal distribution. In our example, $P(\textit{Intelligence} = \textit{high}) = 0.3$, and $P(\textit{Intelligence} = \textit{high} \mid \textit{Grade} = A) = 0.18/0.25 = 0.72$. Thus, clearly $P(\textit{Intelligence} \mid \textit{Grade} = A)$ is different from the marginal distribution $P(\textit{Intelligence})$. The latter distribution represents our *prior* knowledge about students before learning anything else about a particular student, while the conditional distribution represents our more informed distribution after learning her grade.

We will often use the notation $P(X \mid Y)$ to represent a set of conditional probability distributions. Intuitively, for each value of Y , this object assigns a probability over values of X using the conditional probability. This notation allows us to write the shorthand version of the chain rule: $P(X, Y) = P(X)P(Y \mid X)$, which can be extended to multiple variables as

$$P(X_1, \dots, X_k) = P(X_1)P(X_2 \mid X_1) \cdots P(X_k \mid X_1, \dots, X_{k-1}). \quad (2.5)$$

Similarly, we can state Bayes' rule in terms of conditional probability distributions:

$$P(X \mid Y) = \frac{P(X)P(Y \mid X)}{P(Y)}. \quad (2.6)$$

2.1.4 Independence and Conditional Independence

2.1.4.1 Independence

As we mentioned, we usually expect $P(\alpha \mid \beta)$ to be different from $P(\alpha)$. That is, learning that β is true changes our probability over α . However, in some situations equality can occur, so that $P(\alpha \mid \beta) = P(\alpha)$. That is, learning that β occurs did not change our probability of α .

Definition 2.2
independent
events

We say that an event α is independent of event β in P , denoted $P \models (\alpha \perp \beta)$, if $P(\alpha \mid \beta) = P(\alpha)$ or if $P(\beta) = 0$. ■

We can also provide an alternative definition for the concept of independence:

Proposition 2.1

A distribution P satisfies $(\alpha \perp \beta)$ if and only if $P(\alpha \cap \beta) = P(\alpha)P(\beta)$.

PROOF Consider first the case where $P(\beta) = 0$; here, we also have $P(\alpha \cap \beta) = 0$, and so the equivalence immediately holds. When $P(\beta) \neq 0$, we can use the chain rule; we write $P(\alpha \cap \beta) = P(\alpha \mid \beta)P(\beta)$. Since α is independent of β , we have that $P(\alpha \mid \beta) = P(\alpha)$. Thus, $P(\alpha \cap \beta) = P(\alpha)P(\beta)$. Conversely, suppose that $P(\alpha \cap \beta) = P(\alpha)P(\beta)$. Then, by definition, we have that

$$P(\alpha \mid \beta) = \frac{P(\alpha \cap \beta)}{P(\beta)} = \frac{P(\alpha)P(\beta)}{P(\beta)} = P(\alpha). \quad \blacksquare$$

As an immediate consequence of this alternative definition, we see that independence is a symmetric notion. That is, $(\alpha \perp \beta)$ implies $(\beta \perp \alpha)$.

Example 2.3

For example, suppose that we toss two coins, and let α be the event “the first toss results in a head” and β the event “the second toss results in a head.” It is not hard to convince ourselves that we expect that these two events to be independent. Learning that β is true would not change our probability of α . In this case, we see two different physical processes (that is, coin tosses) leading to the events, which makes it intuitive that the probabilities of the two are independent. In certain cases, the same process can lead to independent events. For example, consider the event α denoting “the die outcome is even” and the event β denoting “the die outcome is 1 or 2.” It is easy to check that if the die is fair (each of the six possible outcomes has probability $\frac{1}{6}$), then these two events are independent. ■

2.1.4.2 Conditional Independence



While independence is a useful property, it is not often that we encounter two independent events. A more common situation is when two events are independent given an additional event. For example, suppose we want to reason about the chance that our student is accepted to graduate studies at Stanford or MIT. Denote by *Stanford* the event “admitted to Stanford” and by *MIT* the event “admitted to MIT.” In most reasonable distributions, these two events are not independent. If we learn that a student was admitted to Stanford, then our estimate of her probability of being accepted at MIT is now higher, since it is a sign that she is a promising student.

Now, suppose that both universities base their decisions only on the student's grade point average (GPA), and we know that our student has a GPA of A. In this case, we might argue that learning that the student was admitted to Stanford should not change the probability that she will be admitted to MIT: Her GPA already tells us the information relevant to her chances of admission to MIT, and finding out about her admission to Stanford does not change that. Formally, the statement is

$$P(\text{MIT} \mid \text{Stanford}, \text{GradeA}) = P(\text{MIT} \mid \text{GradeA}).$$

In this case, we say that *MIT* is *conditionally independent* of *Stanford* given *GradeA*.

Definition 2.3

conditional
independence

We say that an event α is conditionally independent of event β given event γ in P , denoted $P \models (\alpha \perp \beta \mid \gamma)$, if $P(\alpha \mid \beta \cap \gamma) = P(\alpha \mid \gamma)$ or if $P(\beta \cap \gamma) = 0$. ■

It is easy to extend the arguments we have seen in the case of (unconditional) independencies to give an alternative definition.

Proposition 2.2

P satisfies $(\alpha \perp \beta \mid \gamma)$ if and only if $P(\alpha \cap \beta \mid \gamma) = P(\alpha \mid \gamma)P(\beta \mid \gamma)$.

2.1.4.3 Independence of Random Variables

Until now, we have focused on independence between events. Thus, we can say that two events, such as one toss landing heads and a second also landing heads, are independent. However, we would like to say that any pair of outcomes of the coin tosses is independent. To capture such statements, we can examine the generalization of independence to sets of random variables.

Definition 2.4

conditional
independence

observed variable

marginal
independence

Let X, Y, Z be sets of random variables. We say that X is conditionally independent of Y given Z in a distribution P if P satisfies $(X = x \perp Y = y \mid Z = z)$ for all values $x \in \text{Val}(X)$, $y \in \text{Val}(Y)$, and $z \in \text{Val}(Z)$. The variables in the set Z are often said to be observed. If the set Z is empty, then instead of writing $(X \perp Y \mid \emptyset)$, we write $(X \perp Y)$ and say that X and Y are marginally independent. ■

Thus, an independence statement over random variables is a universal quantification over all possible values of the random variables.

The alternative characterization of conditional independence follows immediately:

Proposition 2.3

The distribution P satisfies $(X \perp Y \mid Z)$ if and only if $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$.

Suppose we learn about a conditional independence. Can we conclude other independence properties that must hold in the distribution? We have already seen one such example:

symmetry

• **Symmetry:**

$$(X \perp Y \mid Z) \implies (Y \perp X \mid Z). \quad (2.7)$$

There are several other properties that hold for conditional independence, and that often provide a very clean method for proving important properties about distributions. Some key properties are:

decomposition

- **Decomposition:**

$$(X \perp Y, W \mid Z) \implies (X \perp Y \mid Z). \quad (2.8)$$

weak union

- **Weak union:**

$$(X \perp Y, W \mid Z) \implies (X \perp Y \mid Z, W). \quad (2.9)$$

contraction

- **Contraction:**

$$(X \perp W \mid Z, Y) \& (X \perp Y \mid Z) \implies (X \perp Y, W \mid Z). \quad (2.10)$$

An additional important property does not hold in general, but it does hold in an important subclass of distributions.

Definition 2.5positive
distribution

A distribution P is said to be positive if for all events $\alpha \in \mathcal{S}$ such that $\alpha \neq \emptyset$, we have that $P(\alpha) > 0$. ■

For positive distributions, we also have the following property:

intersection

- **Intersection:** For positive distributions, and for mutually disjoint sets X, Y, Z, W :

$$(X \perp Y \mid Z, W) \& (X \perp W \mid Z, Y) \implies (X \perp Y, W \mid Z). \quad (2.11)$$

The proof of these properties is not difficult. For example, to prove Decomposition, assume that $(X \perp Y, W \mid Z)$ holds. Then, from the definition of conditional independence, we have that $P(X, Y, W \mid Z) = P(X \mid Z)P(Y, W \mid Z)$. Now, using basic rules of probability and arithmetic, we can show

$$\begin{aligned} P(X, Y \mid Z) &= \sum_w P(X, Y, w \mid Z) \\ &= \sum_w P(X \mid Z)P(Y, w \mid Z) \\ &= P(X \mid Z) \sum_w P(Y, w \mid Z) \\ &= P(X \mid Z)P(Y \mid Z). \end{aligned}$$

The only property we used here is called “reasoning by cases” (see exercise 2.6). We conclude that $(X \perp Y \mid Z)$.

2.1.5 Querying a Distribution

Our focus throughout this book is on using a joint probability distribution over multiple random variables to answer queries of interest.

2.1.5.1 Probability Queries

probability query	Perhaps the most common query type is the <i>probability query</i> . Such a query consists of two parts:
evidence	The <i>evidence</i>: a subset $\mathbf{E}$ of random variables in the model, and an instantiation e to these variables;
query variables	the <i>query variables</i>: a subset $\mathbf{Y}$ of random variables in the network. Our task is to compute $P(\mathbf{Y} \mid \mathbf{E} = e),$ that is, the <i>posterior probability distribution</i> over the values $\mathbf{y}$ of $\mathbf{Y}$, conditioned on the fact that $\mathbf{E} = e$. This expression can also be viewed as the marginal over $\mathbf{Y}$, in the distribution we obtain by conditioning on e.

2.1.5.2 MAP Queries

MAP assignment	A second important type of task is that of finding a high-probability joint assignment to some subset of variables. The simplest variant of this type of task is the <i>MAP</i> query (also called <i>most probable explanation (MPE)</i>), whose aim is to find the <i>MAP assignment</i> — the most likely assignment to all of the (non-evidence) variables. More precisely, if we let $\mathbf{W} = \mathcal{X} - \mathbf{E}$, our task is to find the most likely assignment to the variables in $\mathbf{W}$ given the evidence $\mathbf{E} = e$:
----------------	---

$$\text{MAP}(\mathbf{W} \mid e) = \arg \max_{\mathbf{w}} P(\mathbf{w}, e), \quad (2.12)$$

where, in general, $\arg \max_x f(x)$ represents the value of x for which $f(x)$ is maximal. Note that there might be more than one assignment that has the highest posterior probability. In this case, we can either decide that the MAP task is to return the set of possible assignments, or to return an arbitrary member of that set.



It is important to understand the difference between MAP queries and probability queries. In a MAP query, we are finding the most likely *joint* assignment to $\mathbf{W}$. To find the most likely assignment to a single variable A , we could simply compute $P(A \mid e)$ and then pick the most likely value. **However, the assignment where each variable individually picks its most likely value can be quite different from the most likely joint assignment to all variables simultaneously.** This phenomenon can occur even in the simplest case, where we have no evidence.

Example 2.4

Consider a two node chain $A \rightarrow B$ where A and B are both binary-valued. Assume that:

$$\begin{array}{cc|cc} a^0 & a^1 & A & b^0 & b^1 \\ \hline 0.4 & 0.6 & a^0 & 0.1 & 0.9 \\ & & a^1 & 0.5 & 0.5 \end{array} \quad (2.13)$$

We can see that $P(a^1) > P(a^0)$, so that $\text{MAP}(A) = a^1$. However, $\text{MAP}(A, B) = (a^0, b^1)$: Both values of B have the same probability given a^1 . Thus, the most likely assignment containing a^1 has probability $0.6 \times 0.5 = 0.3$. On the other hand, the distribution over values of B is more skewed given a^0 , and the most likely assignment (a^0, b^1) has the probability $0.4 \times 0.9 = 0.36$. Thus, we have that $\arg \max_{a,b} P(a, b) \neq (\arg \max_a P(a), \arg \max_b P(b))$. ■

2.1.5.3 Marginal MAP Queries

marginal MAP

To motivate our second query type, let us return to the phenomenon demonstrated in example 2.4. Now, consider a medical diagnosis problem, where the most likely disease has multiple possible symptoms, each of which occurs with some probability, but not an overwhelming probability. On the other hand, a somewhat rarer disease might have only a few symptoms, each of which is very likely given the disease. As in our simple example, the MAP assignment to the data and the symptoms might be higher for the second disease than for the first one. The solution here is to look for the most likely assignment to the disease variable(s) only, rather than the most likely assignment to both the disease and symptom variables. This approach suggests the use of a more general query type. In the *marginal MAP* query, we have a subset of variables $\mathbf{Y}$ that forms our query. The task is to find the most likely assignment to the variables in $\mathbf{Y}$ given the evidence $\mathbf{E} = \mathbf{e}$:

$$\text{MAP}(\mathbf{Y} \mid \mathbf{e}) = \arg \max_{\mathbf{y}} P(\mathbf{y} \mid \mathbf{e}).$$

If we let $\mathbf{Z} = \mathcal{X} - \mathbf{Y} - \mathbf{E}$, the marginal MAP task is to compute:

$$\text{MAP}(\mathbf{Y} \mid \mathbf{e}) = \arg \max_{\mathbf{Y}} \sum_{\mathbf{Z}} P(\mathbf{Y}, \mathbf{Z} \mid \mathbf{e}).$$

Thus, marginal MAP queries contain both summations and maximizations; in a way, it contains elements of both a conditional probability query and a MAP query.

Note that example 2.4 shows that marginal MAP assignments are not monotonic: the most likely assignment $\text{MAP}(Y_1 \mid \mathbf{e})$ might be completely different from the assignment to Y_1 in $\text{MAP}(\{Y_1, Y_2\} \mid \mathbf{e})$. Thus, in particular, we cannot use a MAP query to give us the correct answer to a marginal MAP query.

2.1.6 Continuous Spaces

In the previous section, we focused on random variables that have a finite set of possible values. In many situations, we also want to reason about continuous quantities such as weight, height, duration, or cost that take real numbers in $\mathbb{R}$.

When dealing with probabilities over continuous random variables, we have to deal with some technical issues. For example, suppose that we want to reason about a random variable X that can take values in the range between 0 and 1. That is, $\text{Val}(X)$ is the interval $[0, 1]$. Moreover, assume that we want to assign each number in this range equal probability. What would be the probability of a number x ? Clearly, since each x has the same probability, and there are infinite number of values, we must have that $P(X = x) = 0$. This problem appears even if we do not require uniform probability.

2.1.6.1 Probability Density Functions

density function

How do we define probability over a continuous random variable? We say that a function $p : \mathbb{R} \mapsto \mathbb{R}$ is a *probability density function* or (*PDF*) for X if it is a nonnegative integrable

function such that

$$\int_{\text{Val}(X)} p(x)dx = 1.$$

That is, the integral over the set of possible values of X is 1. The PDF defines a distribution for X as follows: for any x in our event space:

$$P(X \leq a) = \int_{-\infty}^a p(x)dx.$$

cumulative
distribution

The function P is the *cumulative distribution* for X . We can easily employ the rules of probability to see that by using the density function we can evaluate the probability of other events. For example,

$$P(a \leq X \leq b) = \int_a^b p(x)dx.$$

Intuitively, the value of a PDF $p(x)$ at a point x is the incremental amount that x adds to the cumulative distribution in the integration process. The higher the value of p at and around x , the more mass is added to the cumulative distribution as it passes x .

The simplest PDF is the uniform distribution.

Definition 2.6

uniform
distribution

A variable X has a uniform distribution over $[a, b]$, denoted $X \sim \text{Unif}[a, b]$ if it has the PDF

$$p(x) = \begin{cases} \frac{1}{b-a} & b \geq x \geq a \\ 0 & \text{otherwise.} \end{cases}$$

■

Thus, the probability of any subinterval of $[a, b]$ is proportional its size relative to the size of $[a, b]$. Note that, if $b - a < 1$, then the density can be greater than 1. Although this looks unintuitive, this situation can occur even in a legal PDF, if the interval over which the value is greater than 1 is not too large. We have only to satisfy the constraint that the total area under the PDF is 1.

As a more complex example, consider the Gaussian distribution.

Definition 2.7

Gaussian
distribution

A random variable X has a Gaussian distribution with mean μ and variance σ^2 , denoted $X \sim \mathcal{N}(\mu, \sigma^2)$, if it has the PDF

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

standard
Gaussian

A standard Gaussian is one with mean 0 and variance 1.

■

A Gaussian distribution has a bell-like curve, where the mean parameter μ controls the location of the peak, that is, the value for which the Gaussian gets its maximum value. The variance parameter σ^2 determines how peaked the Gaussian is: the smaller the variance, the

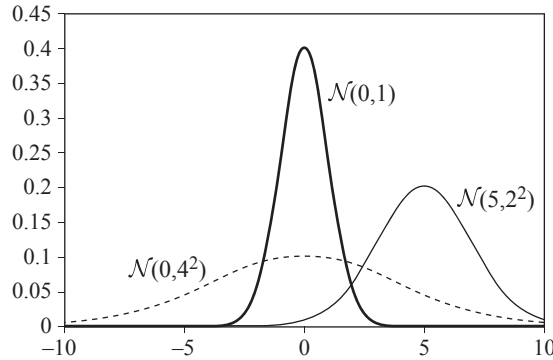


Figure 2.2 Example PDF of three Gaussian distributions

more peaked the Gaussian. Figure 2.2 shows the probability density function of a few different Gaussian distributions.

More technically, the probability density function is specified as an exponential, where the expression in the exponent corresponds to the square of the number of standard deviations σ that x is away from the mean μ . The probability of x decreases exponentially with the square of its deviation from the mean, as measured in units of its standard deviation.

2.1.6.2 Joint Density Functions

The discussion of density functions for a single variable naturally extends for joint distributions of continuous random variables.

Definition 2.8

joint density

Let P be a joint distribution over continuous random variables $X_1, \dots, X_n$. A function $p(x_1, \dots, x_n)$ is a joint density function of $X_1, \dots, X_n$ if

- $p(x_1, \dots, x_n) \geq 0$ for all values $x_1, \dots, x_n$ of $X_1, \dots, X_n$.
- p is an integrable function.
- For any choice of $a_1, \dots, a_n$, and $b_1, \dots, b_n$,

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} p(x_1, \dots, x_n) dx_1 \dots dx_n. \quad \blacksquare$$

Thus, a joint density specifies the probability of any joint event over the variables of interest.

Both the uniform distribution and the Gaussian distribution have natural extensions to the multivariate case. The definition of a multivariate uniform distribution is straightforward. We defer the definition of the multivariate Gaussian to section 7.1.

From the joint density we can derive the marginal density of any random variable by integrating out the other variables. Thus, for example, if $p(x, y)$ is the joint density of X and Y ,

then

$$p(x) = \int_{-\infty}^{\infty} p(x, y) dy.$$

To see why this equality holds, note that the event $a \leq X \leq b$ is, by definition, equal to the event “ $a \leq X \leq b$ and $-\infty \leq Y \leq \infty$.” This rule is the direct analogue of marginalization for discrete variables. Note that, as with discrete probability distributions, we abuse notation a bit and use p to denote both the joint density of X and Y and the marginal density of X . In cases where the distinction is not clear, we use subscripts, so that p_X will be the marginal density, of X , and $p_{X,Y}$ the joint density.

2.1.6.3 Conditional Density Functions

As with discrete random variables, we want to be able to describe conditional distributions of continuous variables. Suppose, for example, we want to define $P(Y | X = x)$. Applying the definition of conditional distribution (equation (2.1)), we run into a problem, since $P(X = x) = 0$. Thus, the ratio of $P(Y, X = x)$ and $P(X = x)$ is undefined.

To avoid this problem, we might consider conditioning on the event $x - \epsilon \leq X \leq x + \epsilon$, which can have a positive probability. Now, the conditional probability is well defined. Thus, we might consider the limit of this quantity when $\epsilon \rightarrow 0$. We define

$$P(Y | x) = \lim_{\epsilon \rightarrow 0} P(Y | x - \epsilon \leq X \leq x + \epsilon).$$

When does this limit exist? If there is a continuous joint density function $p(x, y)$, then we can derive the form for this term. To do so, consider some event on Y , say $a \leq Y \leq b$. Recall that

$$\begin{aligned} P(a \leq Y \leq b | x - \epsilon \leq X \leq x + \epsilon) &= \frac{P(a \leq Y \leq b, x - \epsilon \leq X \leq x + \epsilon)}{P(x - \epsilon \leq X \leq x + \epsilon)} \\ &= \frac{\int_a^b \int_{x-\epsilon}^{x+\epsilon} p(x', y) dy dx'}{\int_{x-\epsilon}^{x+\epsilon} p(x') dx'}. \end{aligned}$$

When ϵ is sufficiently small, we can approximate

$$\int_{x-\epsilon}^{x+\epsilon} p(x') dx' \approx 2\epsilon p(x).$$

Using a similar approximation for $p(x', y)$, we get

$$\begin{aligned} P(a \leq Y \leq b | x - \epsilon \leq X \leq x + \epsilon) &\approx \frac{\int_a^b 2\epsilon p(x, y) dy}{2\epsilon p(x)} \\ &= \int_a^b \frac{p(x, y)}{p(x)} dy. \end{aligned}$$

We conclude that $\frac{p(x, y)}{p(x)}$ is the density of $P(Y | X = x)$.

Definition 2.9conditional
density function

Let $p(x, y)$ be the joint density of X and Y . The conditional density function of Y given X is defined as

$$p(y | x) = \frac{p(x, y)}{p(x)}$$

When $p(x) = 0$, the conditional density is undefined. ■

The conditional density $p(y | x)$ characterizes the conditional distribution $P(Y | X = x)$ we defined earlier.

The properties of joint distributions and conditional distributions carry over to joint and conditional density functions. In particular, we have the chain rule

$$p(x, y) = p(x)p(y | x) \tag{2.14}$$

and Bayes' rule

$$p(x | y) = \frac{p(x)p(y | x)}{p(y)}. \tag{2.15}$$

As a general statement, whenever we discuss joint distributions of continuous random variables, we discuss properties with respect to the joint density function instead of the joint distribution, as we do in the case of discrete variables. Of particular interest is the notion of (conditional) independence of continuous random variables.

Definition 2.10conditional
independence

Let $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$ be sets of continuous random variables with joint density $p(\mathbf{X}, \mathbf{Y}, \mathbf{Z})$. We say that $\mathbf{X}$ is conditionally independent of $\mathbf{Y}$ given $\mathbf{Z}$ if

$$p(\mathbf{x} | \mathbf{z}) = p(\mathbf{x} | \mathbf{y}, \mathbf{z}) \text{ for all } \mathbf{x}, \mathbf{y}, \mathbf{z} \text{ such that } p(\mathbf{z}) > 0. \quad \blacksquare$$

2.1.7 Expectation and Variance

2.1.7.1 Expectation

expectation

Let X be a discrete random variable that takes numerical values; then the *expectation* of X under the distribution P is

$$E_P[X] = \sum_x x \cdot P(x).$$

If X is a continuous variable, then we use the density function

$$E_P[X] = \int x \cdot p(x) dx.$$

For example, if we consider X to be the outcome of rolling a fair die with probability $1/6$ for each outcome, then $E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = 3.5$. On the other hand, if we consider a biased die where $P(X = 6) = 0.5$ and $P(X = x) = 0.1$ for $x < 6$, then $E[X] = 1 \cdot 0.1 + \cdots + 5 \cdot 0.1 + \cdots + 6 \cdot 0.5 = 4.5$.

Often we are interested in expectations of a function of a random variable (or several random variables). Thus, we might consider extending the definition to consider the expectation of a functional term such as $X^2 + 0.5X$. Note, however, that any function g of a set of random variables $X_1, \dots, X_k$ is essentially defining a new random variable Y : For any outcome $\omega \in \Omega$, we define the value of Y as $g(f_{X_1}(\omega), \dots, f_{X_k}(\omega))$.

indicator function

Based on this discussion, we often define new random variables by a functional term. For example $Y = X^2$, or $Y = e^X$. We can also consider functions that map values of one or more categorical random variables to numerical values. One such function that we use quite often is the *indicator function*, which we denote $\mathbf{I}\{X = x\}$. This function takes value 1 when $X = x$, and 0 otherwise.

In addition, we often consider expectations of functions of random variables without bothering to name the random variables they define. For example $E_P[X + Y]$. Nonetheless, we should keep in mind that such a term does refer to an expectation of a random variable.

We now turn to examine properties of the expectation of a random variable.

First, as can be easily seen, the expectation of a random variable is a linear function in that random variable. Thus,

$$E[a \cdot X + b] = aE[X] + b.$$

A more complex situation is when we consider the expectation of a function of several random variables that have some joint behavior. An important property of expectation is that the expectation of a sum of two random variables is the sum of the expectations.

Proposition 2.4

$$E[X + Y] = E[X] + E[Y].$$

linearity of
expectation

This property is called *linearity of expectation*. It is important to stress that this identity is true even when the variables are not independent. As we will see, this property is key in simplifying many seemingly complex problems.

Finally, what can we say about the expectation of a product of two random variables? In general, very little:

Example 2.5

Consider two random variables X and Y , each of which takes the value $+1$ with probability $1/2$, and the value -1 with probability $1/2$. If X and Y are independent, then $E[X \cdot Y] = 0$. On the other hand, if X and Y are correlated in that they always take the same value, then $E[X \cdot Y] = 1$. ■

However, when X and Y are independent, then, as in our example, we can compute the expectation simply as a product of their individual expectations:

Proposition 2.5

If X and Y are independent, then

$$E[X \cdot Y] = E[X] \cdot E[Y].$$

conditional
expectation

We often also use the expectation given some evidence. The *conditional expectation* of X given y is

$$E_P[X \mid y] = \sum_x x \cdot P(x \mid y).$$

2.1.7.2 Variance

variance

The expectation of X tells us the mean value of X . However, It does not indicate how far X deviates from this value. A measure of this deviation is the *variance* of X .

$$\mathbf{Var}_P[X] = \mathbf{E}_P \left[(X - \mathbf{E}_P[X])^2 \right].$$

Thus, the variance is the expectation of the squared difference between X and its expected value. It gives us an indication of the spread of values of X around the expected value.

An alternative formulation of the variance is

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2. \quad (2.16)$$

(see exercise 2.11).

Similar to the expectation, we can consider the expectation of a functions of random variables.

Proposition 2.6

If X and Y are independent, then

$$\mathbf{Var}[X + Y] = \mathbf{Var}[X] + \mathbf{Var}[Y].$$

It is straightforward to show that the variance scales as a quadratic function of X . In particular, we have:

$$\mathbf{Var}[a \cdot X + b] = a^2 \mathbf{Var}[X].$$

standard
deviation

For this reason, we are often interested in the square root of the variance, which is called the *standard deviation* of the random variable. We define

$$\sigma_X = \sqrt{\mathbf{Var}[X]}.$$

The intuition is that it is improbable to encounter values of X that are farther than several standard deviations from the expected value of X . Thus, σ_X is a normalized measure of “distance” from the expected value of X .

As an example consider the Gaussian distribution of definition 2.7.

Proposition 2.7

Let X be a random variable with Gaussian distribution $N(\mu, \sigma^2)$, then $\mathbf{E}[X] = \mu$ and $\mathbf{Var}[X] = \sigma^2$.

Thus, the parameters of the Gaussian distribution specify the expectation and the variance of the distribution. As we can see from the form of the distribution, the density of values of X drops exponentially fast in the distance $\frac{x-\mu}{\sigma}$.

Not all distributions show such a rapid decline in the probability of outcomes that are distant from the expectation. However, even for arbitrary distributions, one can show that there is a decline.

Theorem 2.1

Chebyshev's
inequality

(Chebyshev inequality):

$$P(|X - \mathbf{E}_P[X]| \geq t) \leq \frac{\mathbf{Var}_P[X]}{t^2}.$$

We can restate this inequality in terms of standard deviations: We write $t = k\sigma_X$ to get

$$P(|X - E_P[X]| \geq k\sigma_X) \leq \frac{1}{k^2}.$$

Thus, for example, the probability of X being more than two standard deviations away from $E[X]$ is less than $1/4$.

2.2 Graphs

Perhaps the most pervasive concept in this book is the representation of a probability distribution using a graph as a data structure. In this section, we survey some of the basic concepts in graph theory used in the book.

2.2.1 Nodes and Edges

A graph is a data structure $\mathcal{K}$ consisting of a set of nodes and a set of edges. Throughout most this book, we will assume that the set of nodes is $\mathcal{X} = \{X_1, \dots, X_n\}$. A pair of nodes X_i, X_j can be connected by a *directed edge* $X_i \rightarrow X_j$ or an *undirected edge* $X_i - X_j$. Thus, the set of edges $\mathcal{E}$ is a set of pairs, where each pair is one of $X_i \rightarrow X_j$, $X_j \rightarrow X_i$, or $X_i - X_j$, for $X_i, X_j \in \mathcal{X}$, $i < j$. We assume throughout the book that, for each pair of nodes X_i, X_j , at most one type of edge exists; thus, we cannot have both $X_i \rightarrow X_j$ and $X_j \rightarrow X_i$, nor can we have $X_i \rightarrow X_j$ and $X_i - X_j$.² The notation $X_i \leftarrow X_j$ is equivalent to $X_j \rightarrow X_i$, and the notation $X_j - X_i$ is equivalent to $X_i - X_j$. We use $X_i \rightleftharpoons X_j$ to represent the case where X_i and X_j are connected via some edge, whether directed (in any direction) or undirected.

In many cases, we want to restrict attention to graphs that contain only edges of one kind or another. We say that a graph is *directed* if all edges are either $X_i \rightarrow X_j$ or $X_j \rightarrow X_i$. We usually denote directed graphs as $\mathcal{G}$. We say that a graph is *undirected* if all edges are $X_i - X_j$. We denote undirected graphs as $\mathcal{H}$. We sometimes convert a general graph to an undirected graph by ignoring the directions on the edges.

Definition 2.11

graph's
undirected
version
child
parent
neighbor
boundary

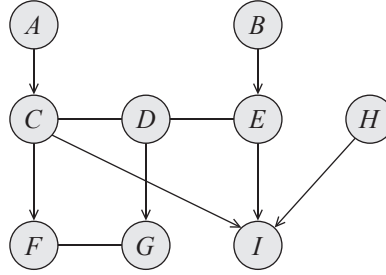
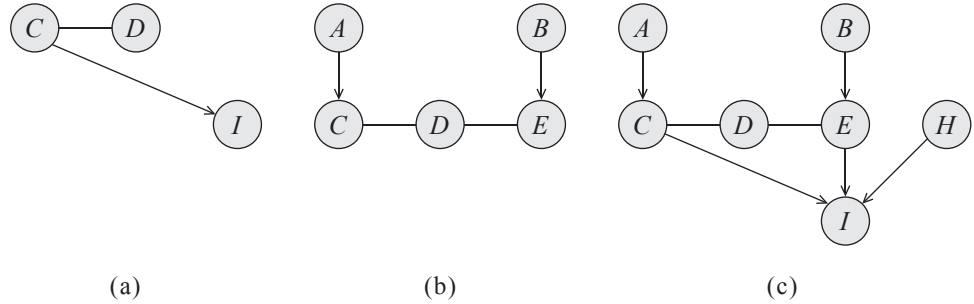
degree
indegree

Given a graph $\mathcal{K} = (\mathcal{X}, \mathcal{E})$, its undirected version is a graph $\mathcal{H} = (\mathcal{X}, \mathcal{E}')$ where $\mathcal{E}' = \{X - Y : X \rightleftharpoons Y \in \mathcal{E}\}$. ■

Whenever we have that $X_i \rightarrow X_j \in \mathcal{E}$, we say that X_j is the *child* of X_i in $\mathcal{K}$, and that X_i is the *parent* of X_j in $\mathcal{K}$. When we have $X_i - X_j \in \mathcal{E}$, we say that X_i is a *neighbor* of X_j in $\mathcal{K}$ (and vice versa). We say that X and Y are adjacent whenever $X \rightleftharpoons Y \in \mathcal{E}$. We use Pa_X to denote the parents of X , Ch_X to denote its children, and Nb_X to denote its neighbors. We define the *boundary* of X , denoted Boundary_X , to be $\text{Pa}_X \cup \text{Nb}_X$; for DAGs, this set is simply X 's parents, and for undirected graphs X 's neighbors.³ Figure 2.3 shows an example of a graph $\mathcal{K}$. There, we have that A is the only parent of C , and F, I are the children of C . The only neighbor of C is D , but its adjacent nodes are A, D, F, I . The *degree* of a node X is the number of edges in which it participates. Its *indegree* is the number of directed edges $Y \rightarrow X$. The *degree* of a graph is the maximal degree of a node in the graph.

2. Note that our definition is somewhat restricted, in that it disallows cycles of length two, where $X_i \rightarrow X_j \rightarrow X_i$, and allows self-loops where $X_i \rightarrow X_i$.

3. When the graph is not clear from context, we often add the graph as an additional argument.

Figure 2.3 An example of a partially directed graph $\mathcal{K}$ Figure 2.4 Induced graphs and their upward closure: (a) The induced subgraph $\mathcal{K}[C, D, I]$. (b) The upwardly closed subgraph $\mathcal{K}^+[C]$. (c) The upwardly closed subgraph $\mathcal{K}^+[C, D, I]$.

2.2.2 Subgraphs

In many cases, we want to consider only the part of the graph that is associated with a particular subset of the nodes.

Definition 2.12

induced
subgraph

Let $\mathcal{K} = (\mathcal{X}, \mathcal{E})$, and let $\mathbf{X} \subset \mathcal{X}$. We define the induced subgraph $\mathcal{K}[\mathbf{X}]$ to be the graph $(\mathbf{X}, \mathcal{E}')$ where $\mathcal{E}'$ are all the edges $X \rightleftharpoons Y \in \mathcal{E}$ such that $X, Y \in \mathbf{X}$. ■

For example, figure 2.4a shows the induced subgraph $\mathcal{K}[C, D, I]$.

A type of subgraph that is often of particular interest is one that contains all possible edges.

Definition 2.13

complete
subgraph

clique

A subgraph over $\mathbf{X}$ is complete if every two nodes in $\mathbf{X}$ are connected by some edge. The set $\mathbf{X}$ is often called a clique; we say that a clique $\mathbf{X}$ is maximal if for any superset of nodes $\mathbf{Y} \supset \mathbf{X}$, $\mathbf{Y}$ is not a clique. ■

Although the subset of nodes $\mathbf{X}$ can be arbitrary, we are often interested in sets of nodes that preserve certain aspects of the graph structure.

Definition 2.14

upward closure

We say that a subset of nodes $\mathbf{X} \in \mathcal{X}$ is upwardly closed in $\mathcal{K}$ if, for any $X \in \mathbf{X}$, we have that $\text{Boundary}_X \subset \mathbf{X}$. We define the upward closure of $\mathbf{X}$ to be the minimal upwardly closed subset

Y that contains X . We define the upwardly closed subgraph of X , denoted $\mathcal{K}^+[X]$, to be the induced subgraph over Y , $\mathcal{K}[Y]$. ■

For example, the set A, B, C, D, E is the upward closure of the set $\{C\}$ in $\mathcal{K}$. The upwardly closed subgraph of $\{C\}$ is shown in figure 2.4b. The upwardly closed subgraph of $\{C, D, I\}$ is shown in figure 2.4c.

2.2.3 Paths and Trails

Using the basic notion of edges, we can define different types of longer-range connections in the graph.

Definition 2.15
path

We say that $X_1, \dots, X_k$ form a path in the graph $\mathcal{K} = (\mathcal{X}, \mathcal{E})$ if, for every $i = 1, \dots, k - 1$, we have that either $X_i \rightarrow X_{i+1}$ or $X_i \dashrightarrow X_{i+1}$. A path is directed if, for at least one i , we have $X_i \rightarrow X_{i+1}$. ■

Definition 2.16
trail

We say that $X_1, \dots, X_k$ form a trail in the graph $\mathcal{K} = (\mathcal{X}, \mathcal{E})$ if, for every $i = 1, \dots, k - 1$, we have that $X_i \neq X_{i+1}$. ■

In the graph $\mathcal{K}$ of figure 2.3, A, C, D, E, I is a path, and hence also a trail. On the other hand, A, C, F, G, D is a trail, which is not a path.

Definition 2.17
connected graph

A graph is connected if for every X_i, X_j there is a trail between X_i and X_j . ■

We can now define longer-range relationships in the graph.

Definition 2.18
ancestor
descendant

We say that X is an ancestor of Y in $\mathcal{K} = (\mathcal{X}, \mathcal{E})$, and that Y is a descendant of X , if there exists a directed path $X_1, \dots, X_k$ with $X_1 = X$ and $X_k = Y$. We use Descendants_X to denote X 's descendants, Ancestors_X to denote X 's ancestors, and NonDescendants_X to denote the set of nodes in $\mathcal{X} - \text{Descendants}_X$. ■

In our example graph $\mathcal{K}$, we have that F, G, I are descendants of C . The ancestors of C are A , via the path A, C , and B , via the path B, E, D, C .

A final useful notion is that of an ordering of the nodes in a directed graph that is consistent with the directionality its edges.

Definition 2.19
topological
ordering

Let $\mathcal{G} = (\mathcal{X}, \mathcal{E})$ be a graph. An ordering of the nodes $X_1, \dots, X_n$ is a topological ordering relative to $\mathcal{K}$ if, whenever we have $X_i \rightarrow X_j \in \mathcal{E}$, then $i < j$. ■

Appendix A.3.1 presents an algorithm for finding such a topological ordering.

2.2.4 Cycles and Loops

Note that, in general, we can have a cyclic path that leads from a node to itself, making that node its own descendant.

Definition 2.20

cycle

A cycle in $\mathcal{K}$ is a directed path $X_1, \dots, X_k$ where $X_1 = X_k$. A graph is acyclic if it contains no cycles. ■

acyclic

For most of this book, we will restrict attention to graphs that do not allow such cycles, since it is quite difficult to define a coherent probabilistic model over graphs with directed cycles.

DAG

A *directed acyclic graph (DAG)* is one of the central concepts in this book, as DAGs are the basic graphical representation that underlies Bayesian networks. For some of this book, we also use acyclic graphs that are partially directed. The graph $\mathcal{K}$ of figure 2.3 is acyclic. However, if we add the undirected edge $A-E$ to $\mathcal{K}$, we have a path A, C, D, E, A from A to itself. Clearly, adding a directed edge $E \rightarrow A$ would also lead to a cycle. Note that prohibiting cycles does not imply that there is no *trail* from a node to itself. For example, $\mathcal{K}$ contains several trails: C, D, E, I, C as well as C, D, G, F, C .

PDAG

chain component

An acyclic graph containing both directed and undirected edges is called a *partially directed acyclic graph* or *PDAG*. The acyclicity requirement on a PDAG implies that the graph can be decomposed into a directed graph of *chain components*, where the nodes within each chain component are connected to each other only with undirected edges. The acyclicity of a PDAG guarantees us that we can order the components so that all edges point from lower-numbered components to higher-numbered ones.

Definition 2.21

Let $\mathcal{K}$ be a PDAG over $\mathcal{X}$. Let $\mathbf{K}_1, \dots, \mathbf{K}_\ell$ be a disjoint partition of $\mathcal{X}$ such that:

- the induced subgraph over $\mathbf{K}_i$ contains no directed edges;
- for any pair of nodes $X \in \mathbf{K}_i$ and $Y \in \mathbf{K}_j$ for $i < j$, an edge between X and Y can only be a directed edge $X \rightarrow Y$.

chain component

Each component $\mathbf{K}_i$ is called a chain component. ■

chain graph

Because of its chain structure, a PDAG is also called a *chain graph*.

Example 2.6

In the PDAG of figure 2.3, we have six chain components: $\{A\}$, $\{B\}$, $\{C, D, E\}$, $\{F, G\}$, $\{H\}$, and $\{I\}$. This ordering of the chain components is one of several possible legal orderings. ■

Note that when the PDAG is an undirected graph, the entire graph forms a single chain component. Conversely, when the PDAG is a directed graph (and therefore acyclic), each node in the graph is its own chain component.

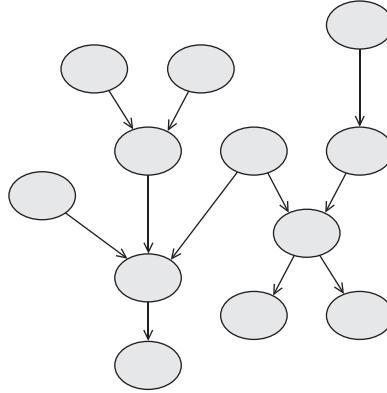


Figure 2.5 An example of a polytree

Different from a cycle is the notion of a loop:

Definition 2.22

loop

singly connected

leaf

polytree

forest

tree

A loop in $\mathcal{K}$ is a trail $X_1, \dots, X_k$ where $X_1 = X_k$. A graph is singly connected if it contains no loops. A node in a singly connected graph is called a leaf if it has exactly one adjacent node. A singly connected directed graph is also called a polytree. A singly connected undirected graph is called a forest; if it is also connected, it is called a tree. ■

We can also define a notion of a forest, or of a tree, for directed graphs.

A directed graph is a forest if each node has at most one parent. A directed forest is a tree if it is also connected. ■

Definition 2.23

Note that polytrees are very different from trees. For example, figure 2.5 shows a graph that is a polytree but is not a tree, because several nodes have more than one parent. As we will discuss later in the book, loops in the graph increase the computational cost of various tasks.

We conclude this section with a final definition relating to loops in the graph. This definition will play an important role in evaluating the cost of reasoning using graph-based representations.

Definition 2.24

chordal graph

Let $X_1 - X_2 - \dots - X_k - X_1$ be a loop in the graph; a chord in the loop is an edge connecting X_i and X_j for two nonconsecutive nodes X_i, X_j . An undirected graph $\mathcal{H}$ is said to be chordal if any loop $X_1 - X_2 - \dots - X_k - X_1$ for $k \geq 4$ has a chord. ■

Thus, for example, a loop $A - B - C - D - A$ (as in figure 1.1b) is nonchordal, but adding an edge $A - C$ would render it chordal. In other words, in a chordal graph, the longest “minimal loop” (one that has no shortcut) is a triangle. Thus, chordal graphs are often also called *triangulated*.

triangulated graph

We can extend the notion of chordal graphs to graphs that contain directed edges.

Definition 2.25

A graph $\mathcal{K}$ is said to be chordal if its underlying undirected graph is chordal. ■

PART I

Representation

3

The Bayesian Network Representation



Our goal is to represent a joint distribution P over some set of random variables $\mathcal{X} = \{X_1, \dots, X_n\}$. Even in the simplest case where these variables are binary-valued, a joint distribution requires the specification of $2^n - 1$ numbers — the probabilities of the 2^n different assignments of values $x_1, \dots, x_n$. For all but the smallest n , **the explicit representation of the joint distribution is unmanageable from every perspective. Computationally, it is very expensive to manipulate and generally too large to store in memory. Cognitively, it is impossible to acquire so many numbers from a human expert; moreover, the numbers are very small and do not correspond to events that people can reasonably contemplate. Statistically, if we want to learn the distribution from data, we would need ridiculously large amounts of data to estimate this many parameters robustly. These problems were the main barrier to the adoption of probabilistic methods for expert systems until the development of the methodologies described in this book.**

In this chapter, we first show how independence properties in the distribution can be used to represent such high-dimensional distributions much more compactly. We then show how a combinatorial data structure — a directed acyclic graph — can provide us with a general-purpose modeling language for exploiting this type of structure in our representation.

3.1 Exploiting Independence Properties

The compact representations we explore in this chapter are based on two key ideas: the representation of independence properties of the distribution, and the use of an alternative parameterization that allows us to exploit these finer-grained independencies.

3.1.1 Independent Random Variables

To motivate our discussion, consider a simple setting where we know that each X_i represents the outcome of a toss of coin i . In this case, we typically assume that the different coin tosses are marginally independent (definition 2.4), so that our distribution P will satisfy $(X_i \perp X_j)$ for any i, j . More generally (strictly more generally — see exercise 3.1), we assume that the distribution satisfies $(\mathbf{X} \perp \mathbf{Y})$ for any disjoint subsets of the variables $\mathbf{X}$ and $\mathbf{Y}$. Therefore, we have that:

$$P(X_1, \dots, X_n) = P(X_1)P(X_2) \cdots P(X_n).$$

parameters

If we use the standard parameterization of the joint distribution, this independence structure is obscured, and the representation of the distribution requires 2^n parameters. However, we can use a more natural set of parameters for specifying this distribution: If θ_i is the probability with which coin i lands heads, the joint distribution P can be specified using the n *parameters* $\theta_1, \dots, \theta_n$. These parameters implicitly specify the 2^n probabilities in the joint distribution. For example, the probability that all of the coin tosses land heads is simply $\theta_1 \cdot \theta_2 \cdot \dots \cdot \theta_n$. More generally, letting $\theta_{x_i} = \theta_i$ when $x_i = x_i^1$ and $\theta_{x_i} = 1 - \theta_i$ when $x_i = x_i^0$, we can define:

$$P(x_1, \dots, x_n) = \prod_i \theta_{x_i}. \quad (3.1)$$

independent
parameters

This representation is limited, and there are many distributions that we cannot capture by choosing values for $\theta_1, \dots, \theta_n$. This fact is obvious not only from intuition, but also from a somewhat more formal perspective. The space of all joint distributions is a $2^n - 1$ dimensional subspace of $\mathbb{R}^{2^n}$ — the set $\{(p_1, \dots, p_{2^n}) \in \mathbb{R}^{2^n} : p_1 + \dots + p_{2^n} = 1\}$. On the other hand, the space of all joint distributions specified in a factorized way as in equation (3.1) is an n -dimensional manifold in $\mathbb{R}^{2^n}$.

A key concept here is the notion of *independent parameters* — parameters whose values are not determined by others. For example, when specifying an arbitrary multinomial distribution over a k dimensional space, we have $k - 1$ independent parameters: the last probability is fully determined by the first $k - 1$. In the case where we have an arbitrary joint distribution over n binary random variables, the number of independent parameters is $2^n - 1$. On the other hand, the number of independent parameters for distributions represented as n independent binomial coin tosses is n . Therefore, the two spaces of distributions cannot be the same. (While this argument might seem trivial in this simple case, it turns out to be an important tool for comparing the expressive power of different representations.)

As this simple example shows, certain families of distributions — in this case, the distributions generated by n independent random variables — permit an alternative parameterization that is substantially more compact than the naive representation as an explicit joint distribution. Of course, in most real-world applications, the random variables are not marginally independent. However, a generalization of this approach will be the basis for our solution.

3.1.2 The Conditional Parameterization

Let us begin with a simple example that illustrates the basic intuition. Consider the problem faced by a company trying to hire a recent college graduate. The company's goal is to hire intelligent employees, but there is no way to test intelligence directly. However, the company has access to the student's SAT scores, which are informative but not fully indicative. Thus, our probability space is induced by the two random variables *Intelligence* (I) and *SAT* (S). For simplicity, we assume that each of these takes two values: $Val(I) = \{i^1, i^0\}$, which represent the values high intelligence (i^1) and low intelligence (i^0); similarly $Val(S) = \{s^1, s^0\}$, which also represent the values *high* (score) and *low* (score), respectively.

Thus, our joint distribution in this case has four entries. For example, one possible joint

distribution P would be

I	S	$P(I, S)$
i^0	s^0	0.665
i^0	s^1	0.035
i^1	s^0	0.06
i^1	s^1	0.24.

(3.2)

There is, however, an alternative, and even more natural way of representing the same joint distribution. Using the chain rule of conditional probabilities (see equation (2.5)), we have that

$$P(I, S) = P(I)P(S | I).$$

Intuitively, we are representing the process in a way that is more compatible with causality. Various factors (genetics, upbringing, ...) first determined (stochastically) the student's intelligence. His performance on the SAT is determined (stochastically) by his intelligence. We note that the models we construct are not required to follow causal intuitions, but they often do. We return to this issue later on.

From a mathematical perspective, this equation leads to the following alternative way of representing the joint distribution. Instead of specifying the various joint entries $P(I, S)$, we would specify it in the form of $P(I)$ and $P(S | I)$. Thus, for example, we can represent the joint distribution of equation (3.2) using the following two tables, one representing the *prior distribution* over I and the other the *conditional probability distribution (CPD)* of S given I :

prior distribution
CPD

i^0	i^1	I	s^0	s^1
0.7	0.3	i^0	0.95	0.05
		i^1	0.2	0.8

(3.3)

The CPD $P(S | I)$ represents the probability that the student will succeed on his SATs in the two possible cases: the case where the student's intelligence is low, and the case where it is high. The CPD asserts that a student of low intelligence is extremely unlikely to get a high SAT score ($P(s^1 | i^0) = 0.05$); on the other hand, a student of high intelligence is likely, but far from certain, to get a high SAT score ($P(s^1 | i^1) = 0.8$).

It is instructive to consider how we could parameterize this alternative representation. Here, we are using three binomial distributions, one for $P(I)$, and two for $P(S | i^0)$ and $P(S | i^1)$. Hence, we can parameterize this representation using three independent parameters, say θ_{i^1} , $\theta_{s^1|i^1}$, and $\theta_{s^1|i^0}$. Our representation of the joint distribution as a four-outcome multinomial also required three parameters. Thus, although the conditional representation is more natural than the explicit representation of the joint, it is not more compact. However, as we will soon see, the conditional parameterization provides a basis for our compact representations of more complex distributions.

Although we will only define Bayesian networks formally in section 3.2.2, it is instructive to see how this example would be represented as one. The Bayesian network, as shown in figure 3.1a, would have a node for each of the two random variables I and S , with an edge from I to S representing the direction of the dependence in this model.

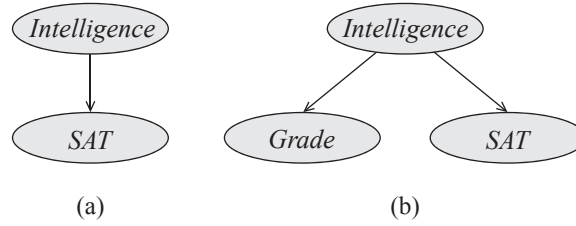


Figure 3.1 Simple Bayesian networks for the student example

3.1.3 The Naive Bayes Model

We now describe perhaps the simplest example where a conditional parameterization is combined with conditional independence assumptions to produce a very compact representation of a high-dimensional probability distribution. Importantly, unlike the previous example of fully independent random variables, none of the variables in this distribution are (marginally) independent.

3.1.3.1 The Student Example

Elaborating our example, we now assume that the company also has access to the student's grade G in some course. In this case, our probability space is the joint distribution over the three relevant random variables I , S , and G . Assuming that I and S are as before, and that G takes on three values g^1, g^2, g^3 , representing the grades A , B , and C , respectively, then the joint distribution has twelve entries.

Before we even consider the specific numerical aspects of our distribution P in this example, we can see that independence does not help us: for any reasonable P , there are no independencies that hold. The student's intelligence is clearly correlated both with his SAT score and with his grade. The SAT score and grade are also not independent: if we condition on the fact that the student received a high score on his SAT, the chances that he gets a high grade in his class are also likely to increase. Thus, we may assume that, for our particular distribution P , $P(g^1 | s^1) > P(g^1 | s^0)$.

However, it is quite plausible that our distribution P in this case satisfies a conditional independence property. If we know that the student has high intelligence, a high grade on the SAT no longer gives us information about the student's performance in the class. More formally:

$$P(g | i^1, s^1) = P(g | i^1).$$

More generally, we may well assume that

$$P \models (S \perp G | I). \quad (3.4)$$

Note that this independence statement holds only if we assume that the student's intelligence is the only reason why his grade and SAT score might be correlated. In other words, it assumes that there are no correlations due to other factors, such as the student's ability to take timed exams. These assumptions are also not "true" in any formal sense of the word, and they are often only approximations of our true beliefs. (See box 3.C for some further discussion.)

As in the case of marginal independence, conditional independence allows us to provide a compact specification of the joint distribution. Again, the compact representation is based on a very natural alternative parameterization. By simple probabilistic reasoning (as in equation (2.5)), we have that

$$P(I, S, G) = P(S, G \mid I)P(I).$$

But now, the conditional independence assumption of equation (3.4) implies that

$$P(S, G \mid I) = P(S \mid I)P(G \mid I).$$

Hence, we have that

$$P(I, S, G) = P(S \mid I)P(G \mid I)P(I). \quad (3.5)$$

Thus, we have factorized the joint distribution $P(I, S, G)$ as a product of three conditional probability distributions (CPDs). This factorization immediately leads us to the desired alternative parameterization. In order to specify fully a joint distribution satisfying equation (3.4), we need the following three CPDs: $P(I)$, $P(S \mid I)$, and $P(G \mid I)$. The first two might be the same as in equation (3.3). The latter might be

I	g^1	g^2	g^3
i^0	0.2	0.34	0.46
i^1	0.74	0.17	0.09

Together, these three CPDs fully specify the joint distribution (assuming the conditional independence of equation (3.4)). For example,

$$\begin{aligned} P(i^1, s^1, g^2) &= P(i^1)P(s^1 \mid i^1)P(g^2 \mid i^1) \\ &= 0.3 \cdot 0.8 \cdot 0.17 = 0.0408. \end{aligned}$$

Once again, we note that this probabilistic model would be represented using the Bayesian network shown in figure 3.1b.

In this case, the alternative parameterization is more compact than the joint. We now have three binomial distributions — $P(I)$, $P(S \mid i^1)$ and $P(S \mid i^0)$, and two three-valued multinomial distributions — $P(G \mid i^1)$ and $P(G \mid i^0)$. Each of the binomials requires one independent parameter, and each three-valued multinomial requires two independent parameters, for a total of seven. By contrast, our joint distribution has twelve entries, so that eleven independent parameters are required to specify an arbitrary joint distribution over these three variables.

It is important to note another advantage of this way of representing the joint: modularity. When we added the new variable G , the joint distribution changed entirely. Had we used the explicit representation of the joint, we would have had to write down twelve new numbers. In the factored representation, we could reuse our local probability models for the variables I and S , and specify only the probability model for G — the CPD $P(G \mid I)$. This property will turn out to be invaluable in modeling real-world systems.

3.1.3.2 The General Model

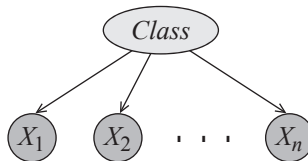


Figure 3.2 The Bayesian network graph for a naive Bayes model

model (also known as the *Idiot Bayes model*). The naive Bayes model assumes that instances fall into one of a number of mutually exclusive and exhaustive *classes*. Thus, we have a class variable C that takes on values in some set $\{c^1, \dots, c^k\}$. In our example, the class variable is the student's intelligence I , and there are two classes of instances — students with high intelligence and students with low intelligence.

features The model also includes some number of *features* $X_1, \dots, X_n$ whose values are typically observed. The *naive Bayes assumption* is that the features are conditionally independent given the instance's class. In other words, within each class of instances, the different properties can be determined independently. More formally, we have that

$$(X_i \perp \mathbf{X}_{-i} \mid C) \quad \text{for all } i, \quad (3.6)$$

where $\mathbf{X}_{-i} = \{X_1, \dots, X_n\} - \{X_i\}$. This model can be represented using the Bayesian network of figure 3.2. In this example, and later on in the book, we use a darker oval to represent variables that are always observed when the network is used.

factorization Based on these independence assumptions, we can show that the model *factorizes* as:

$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i \mid C). \quad (3.7)$$

(See exercise 3.2.) Thus, in this model, we can represent the joint distribution using a small set of factors: a prior distribution $P(C)$, specifying how likely an instance is to belong to different classes a priori, and a set of CPDs $P(X_j \mid C)$, one for each of the n finding variables. These factors can be encoded using a very small number of parameters. For example, if all of the variables are binary, the number of independent parameters required to specify the distribution is $2n + 1$ (see exercise 3.6). Thus, the number of parameters is linear in the number of variables, as opposed to exponential for the explicit representation of the joint.

Box 3.A — Concept: The Naive Bayes Model. The naive Bayes model, despite the strong assumptions that it makes, is often used in practice, because of its simplicity and the small number of parameters required. The model is generally used for classification — deciding, based on the values of the evidence variables for a given instance, the class to which the instance is most likely to belong. We might also want to compute our confidence in this decision, that is, the extent to which our model favors one class c^1 over another c^2 . Both queries can be addressed by the following ratio:

$$\frac{P(C = c^1 \mid x_1, \dots, x_n)}{P(C = c^2 \mid x_1, \dots, x_n)} = \frac{P(C = c^1)}{P(C = c^2)} \prod_{i=1}^n \frac{P(x_i \mid C = c^1)}{P(x_i \mid C = c^2)}; \quad (3.8)$$

medical diagnosis

see exercise 3.2). This formula is very natural, since it computes the posterior probability ratio of c^1 versus c^2 as a product of their prior probability ratio (the first term), multiplied by a set of terms $\frac{P(x_i|C=c^1)}{P(x_i|C=c^2)}$ that measure the relative support of the finding x_i for the two classes.

This model was used in the early days of medical diagnosis, where the different values of the class variable represented different diseases that the patient could have. The evidence variables represented different symptoms, test results, and the like. Note that the model makes several strong assumptions that are not generally true, specifically that the patient can have at most one disease, and that, given the patient's disease, the presence or absence of different symptoms, and the values of different tests, are all independent. This model was used for medical diagnosis because the small number of interpretable parameters made it easy to elicit from experts. For example, it is quite natural to ask of an expert physician what the probability is that a patient with pneumonia has high fever. Indeed, several early medical diagnosis systems were based on this technology, and some were shown to provide better diagnoses than those made by expert physicians.

However, later experience showed that the strong assumptions underlying this model decrease its diagnostic accuracy. In particular, the model tends to overestimate the impact of certain evidence by “overcounting” it. For example, both hypertension (high blood pressure) and obesity are strong indicators of heart disease. However, because these two symptoms are themselves highly correlated, equation (3.8), which contains a multiplicative term for each of them, double-counts the evidence they provide about the disease. Indeed, some studies showed that the diagnostic performance of a naive Bayes model degraded as the number of features increased; this degradation was often traced to violations of the strong conditional independence assumption. This phenomenon led to the use of more complex Bayesian networks, with more realistic independence assumptions, for this application (see box 3.D).

Nevertheless, the naive Bayes model is still useful in a variety of applications, particularly in the context of models learned from data in domains with a large number of features and a relatively small number of instances, such as classifying documents into topics using the words in the documents as features; see box 17.E).

3.2 Bayesian Networks

Bayesian networks build on the same intuitions as the naive Bayes model by exploiting conditional independence properties of the distribution in order to allow a compact and natural representation. However, they are not restricted to representing distributions satisfying the strong independence assumptions implicit in the naive Bayes model. They allow us the flexibility to tailor our representation of the distribution to the independence properties that appear reasonable in the current setting.

The core of the Bayesian network representation is a directed acyclic graph (DAG) $\mathcal{G}$, whose nodes are the random variables in our domain and whose edges correspond, intuitively, to direct influence of one node on another. **This graph $\mathcal{G}$ can be viewed in two very different ways:**



- **as a data structure that provides the skeleton for representing a joint distribution compactly in a factorized way;**

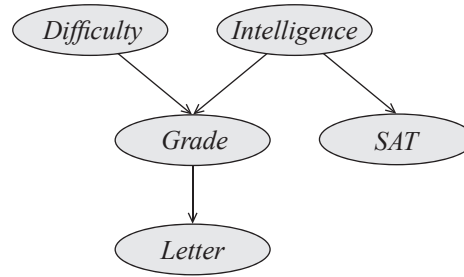


Figure 3.3 The Bayesian Network graph for the Student example

- as a compact representation for a set of conditional independence assumptions about a distribution.

As we will see, these two views are, in a strong sense, equivalent.

3.2.1 The Student Example Revisited

We begin our discussion with a simple toy example, which will accompany us, in various versions, throughout much of this book.

3.2.1.1 The Model

Consider our student from before, but now consider a slightly more complex scenario. The student's grade, in this case, depends not only on his intelligence but also on the difficulty of the course, represented by a random variable D whose domain is $Val(D) = \{easy, hard\}$. Our student asks his professor for a recommendation letter. The professor is absentminded and never remembers the names of her students. She can only look at his grade, and she writes her letter for him based on that information alone. The quality of her letter is a random variable L , whose domain is $Val(L) = \{strong, weak\}$. The actual quality of the letter depends stochastically on the grade. (It can vary depending on how stressed the professor is and the quality of the coffee she had that morning.)

We therefore have five random variables in this domain: the student's intelligence (I), the course difficulty (D), the grade (G), the student's SAT score (S), and the quality of the recommendation letter (L). All of the variables except G are binary-valued, and G is ternary-valued. Hence, the joint distribution has 48 entries.

As we saw in our simple illustrations of figure 3.1, a Bayesian network is represented using a directed graph whose nodes represent the random variables and whose edges represent direct influence of one variable on another. We can view the graph as encoding a generative sampling process executed by nature, where the value for each variable is selected by nature using a distribution that depends only on its parents. In other words, each variable is a stochastic function of its parents.

Based on this intuition, perhaps the most natural network structure for the distribution in this example is the one presented in figure 3.3. The edges encode our intuition about

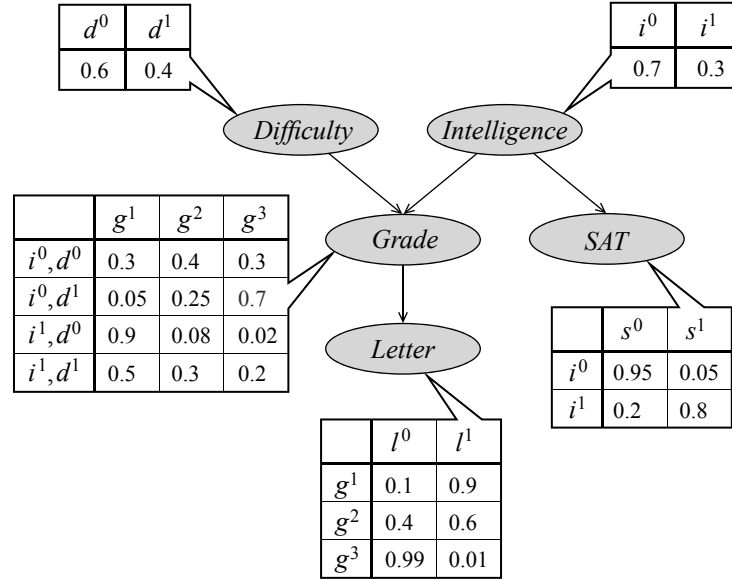


Figure 3.4 Student Bayesian network $\mathcal{B}^{student}$ with CPDs

the way the world works. The course difficulty and the student's intelligence are determined independently, and before any of the variables in the model. The student's grade depends on both of these factors. The student's SAT score depends only on his intelligence. The quality of the professor's recommendation letter depends (by assumption) only on the student's grade in the class. Intuitively, each variable in the model depends directly only on its parents in the network. We formalize this intuition later.

The second component of the Bayesian network representation is a set of *local probability models* that represent the nature of the dependence of each variable on its parents. One such model, $P(I)$, represents the distribution in the population of intelligent versus less intelligent student. Another, $P(D)$, represents the distribution of difficult and easy classes. The distribution over the student's grade is a conditional distribution $P(G \mid I, D)$. It specifies the distribution over the student's grade, inasmuch as it depends on the student's intelligence and the course difficulty. Specifically, we would have a different distribution for each assignment of values i, d . For example, we might believe that a smart student in an easy class is 90 percent likely to get an A, 8 percent likely to get a B, and 2 percent likely to get a C. Conversely, a smart student in a hard class may only be 50 percent likely to get an A. In general, each variable X in the model is associated with a *conditional probability distribution (CPD)* that specifies a distribution over the values of X given each possible joint assignment of values to its parents in the model. For a node with no parents, the CPD is conditioned on the empty set of variables. Hence, the CPD turns into a marginal distribution, such as $P(D)$ or $P(I)$. One possible choice of CPDs for this domain is shown in figure 3.4. The network structure together with its CPDs is a *Bayesian network* $\mathcal{B}$; we use $\mathcal{B}^{student}$ to refer to the Bayesian network for our student example.

local probability
model

CPD

How do we use this data structure to specify the joint distribution? Consider some particular state in this space, for example, i^1, d^0, g^2, s^1, l^0 . Intuitively, the probability of this event can be computed from the probabilities of the basic events that comprise it: the probability that the student is intelligent; the probability that the course is easy; the probability that a smart student gets a B in an easy class; the probability that a smart student gets a high score on his SAT; and the probability that a student who got a B in the class gets a weak letter. The total probability of this state is:

$$\begin{aligned} P(i^1, d^0, g^2, s^1, l^0) &= P(i^1)P(d^0)P(g^2 | i^1, d^0)P(s^1 | i^1)P(l^0 | g^2) \\ &= 0.3 \cdot 0.6 \cdot 0.08 \cdot 0.8 \cdot 0.4 = 0.004608. \end{aligned}$$

Clearly, we can use the same process for any state in the joint probability space. In general, we will have that

$$P(I, D, G, S, L) = P(I)P(D)P(G | I, D)P(S | I)P(L | G). \quad (3.9)$$

chain rule for
Bayesian
networks

This equation is our first example of the *chain rule for Bayesian networks* which we will define in a general setting in section 3.2.3.2.

3.2.1.2 Reasoning Patterns

A joint distribution $P_{\mathcal{B}}$ specifies (albeit implicitly) the probability $P_{\mathcal{B}}(\mathbf{Y} = \mathbf{y} | \mathbf{E} = \mathbf{e})$ of any event $\mathbf{y}$ given any observations $\mathbf{e}$, as discussed in section 2.1.3.3: We condition the joint distribution on the event $\mathbf{E} = \mathbf{e}$ by eliminating the entries in the joint inconsistent with our observation $\mathbf{e}$, and renormalizing the resulting entries to sum to 1; we compute the probability of the event $\mathbf{y}$ by summing the probabilities of all of the entries in the resulting posterior distribution that are consistent with $\mathbf{y}$. To illustrate this process, let us consider our $\mathcal{B}^{student}$ network and see how the probabilities of various events change as evidence is obtained.

Consider a particular student, George, about whom we would like to reason using our model. We might ask how likely George is to get a strong recommendation (l^1) from his professor in Econ101. Knowing nothing else about George or Econ101, this probability is about 50.2 percent. More precisely, let $P_{\mathcal{B}^{student}}$ be the joint distribution defined by the preceding BN; then we have that $P_{\mathcal{B}^{student}}(l^1) \approx 0.502$. We now find out that George is not so intelligent (i^0); the probability that he gets a strong letter from the professor of Econ101 goes down to around 38.9 percent; that is, $P_{\mathcal{B}^{student}}(l^1 | i^0) \approx 0.389$. We now further discover that Econ101 is an easy class (d^0). The probability that George gets a strong letter from the professor is now $P_{\mathcal{B}^{student}}(l^1 | i^0, d^0) \approx 0.513$. Queries such as these, where we predict the “downstream” effects of various factors (such as George’s intelligence), are instances of *causal reasoning* or *prediction*.

causal reasoning

Now, consider a recruiter for Acme Consulting, trying to decide whether to hire George based on our previous model. A priori, the recruiter believes that George is 30 percent likely to be intelligent. He obtains George’s grade record for a particular class Econ101 and sees that George received a C in the class (g^3). His probability that George has high intelligence goes down significantly, to about 7.9 percent; that is, $P_{\mathcal{B}^{student}}(i^1 | g^3) \approx 0.079$. We note that the probability that the class is a difficult one also goes up, from 40 percent to 62.9 percent.

Now, assume that the recruiter fortunately (for George) lost George’s transcript, and has only the recommendation letter from George’s professor in Econ101, which (not surprisingly) is

evidential
reasoning

weak. The probability that George has high intelligence still goes down, but only to 14 percent: $P_{\mathcal{B}^{student}}(i^1 | l^0) \approx 0.14$. Note that if the recruiter has both the grade and the letter, we have the same probability as if he had only the grade: $P_{\mathcal{B}^{student}}(i^1 | g^3, l^0) \approx 0.079$; we will revisit this issue. Queries such as this, where we reason from effects to causes, are instances of *evidential reasoning* or *explanation*.

Finally, George submits his SAT scores to the recruiter, and astonishingly, his SAT score is high. The probability that George has high intelligence goes up dramatically, from 7.9 percent to 57.8 percent: $P_{\mathcal{B}^{student}}(i^1 | g^3, s^1) \approx 0.578$. Intuitively, the reason that the high SAT score outweighs the poor grade is that students with low intelligence are extremely unlikely to get good scores on their SAT, whereas students with high intelligence can still get C's. However, smart students are much more likely to get C's in hard classes. Indeed, we see that the probability that Econ101 is a difficult class goes up from the 62.9 percent we saw before to around 76 percent.

This last pattern of reasoning is a particularly interesting one. The information about the SAT gave us information about the student's intelligence, which, in conjunction with the student's grade in the course, told us something about the difficulty of the course. In effect, we have one causal factor for the *Grade* variable — *Intelligence* — giving us information about another — *Difficulty*.

explaining away



intercausal
reasoning

Let us examine this pattern in its pure form. As we said, $P_{\mathcal{B}^{student}}(i^1 | g^3) \approx 0.079$. On the other hand, if we now discover that Econ101 is a hard class, we have that $P_{\mathcal{B}^{student}}(i^1 | g^3, d^1) \approx 0.11$. In effect, we have provided at least a partial explanation for George's grade in Econ101. To take an even more striking example, if George gets a B in Econ 101, we have that $P_{\mathcal{B}^{student}}(i^1 | g^2) \approx 0.175$. On the other hand, if Econ101 is a hard class, we get $P_{\mathcal{B}^{student}}(i^1 | g^2, d^1) \approx 0.34$. In effect we have *explained away* the poor grade via the difficulty of the class. **Explaining away is an instance of a general reasoning pattern called *intercausal reasoning*, where different causes of the same effect can interact. This type of reasoning is a very common pattern in human reasoning.** For example, when we have fever and a sore throat, and are concerned about mononucleosis, we are greatly relieved to be told we have the flu. Clearly, having the flu does not prohibit us from having mononucleosis. Yet, having the flu provides an alternative explanation of our symptoms, thereby reducing substantially the probability of mononucleosis.

This intuition of providing an alternative explanation for the evidence can be made very precise. As shown in exercise 3.3, if the flu deterministically causes the symptoms, the probability of mononucleosis goes down to its prior probability (the one prior to the observations of any symptoms). On the other hand, if the flu might occur without causing these symptoms, the probability of mononucleosis goes down, but it still remains somewhat higher than its base level. Explaining away, however, is not the only form of intercausal reasoning. The influence can go in any direction. Consider, for example, a situation where someone is found dead and may have been murdered. The probabilities that a suspect has motive and opportunity both go up. If we now discover that the suspect has motive, the probability that he has opportunity goes up. (See exercise 3.4.)

It is important to emphasize that, although our explanations used intuitive concepts such as cause and evidence, there is nothing mysterious about the probability computations we performed. They can be replicated simply by generating the joint distribution, as defined in equation (3.9), and computing the probabilities of the various events directly from that.

3.2.2 Basic Independencies in Bayesian Networks

As we discussed, a Bayesian network graph $\mathcal{G}$ can be viewed in two ways. In the previous section, we showed, by example, how it can be used as a skeleton data structure to which we can attach local probability models that together define a joint distribution. In this section, we provide a formal semantics for a Bayesian network, starting from the perspective that the graph encodes a set of conditional independence assumptions. We begin by understanding, intuitively, the basic conditional independence assumptions that we want a directed graph to encode. We then formalize these desired assumptions in a definition.

3.2.2.1 Independencies in the Student Example

In the Student example, we used the intuition that edges represent direct dependence. For example, we made intuitive statements such as “the professor’s recommendation letter depends only on the student’s grade in the class”; this statement was encoded in the graph by the fact that there are no direct edges into the L node except from G . This intuition, that “a node depends directly only on its parents,” lies at the heart of the semantics of Bayesian networks.

We give formal semantics to this assertion using conditional independence statements. For example, the previous assertion can be stated formally as the assumption that L is conditionally independent of all other nodes in the network given its parent G :

$$(L \perp I, D, S \mid G). \quad (3.10)$$

In other words, once we know the student’s grade, our beliefs about the quality of his recommendation letter are not influenced by information about any other variable. Similarly, to formalize our intuition that the student’s SAT score depends only on his intelligence, we can say that S is conditionally independent of all other nodes in the network given its parent I :

$$(S \perp D, G, L \mid I). \quad (3.11)$$

Now, let us consider the G node. Following the pattern blindly, we may be tempted to assert that G is conditionally independent of all other variables in the network given its parents. However, this assumption is false both at an intuitive level and for the specific example distribution we used earlier. Assume, for example, that we condition on i^1, d^1 ; that is, we have a smart student in a difficult class. In this setting, is G independent of L ? Clearly, the answer is no: if we observe l^1 (the student got a strong letter), then our probability in g^1 (the student received an A in the course) should go up; that is, we would expect

$$P(g^1 \mid i^1, d^1, l^1) > P(g^1 \mid i^1, d^1).$$

Indeed, if we examine our distribution, the latter probability is 0.5 (as specified in the CPD), whereas the former is a much higher 0.712.

Thus, we see that we do not expect a node to be conditionally independent of all other nodes given its parents. In particular, even given its parents, it can still depend on its descendants. Can it depend on other nodes? For example, do we expect G to depend on S given I and D ? Intuitively, the answer is no. Once we know, say, that the student has high intelligence, his SAT score gives us no additional information that is relevant toward predicting his grade. Thus, we

would want the property that:

$$(G \perp S \mid I, D). \quad (3.12)$$

It remains only to consider the variables I and D , which have no parents in the graph. Thus, in our search for independencies given a node's parents, we are now looking for marginal independencies. As the preceding discussion shows, in our distribution $P_{\mathcal{B}^{student}}$, I is not independent of its descendants G , L , or S . Indeed, the only nondescendant of I is D . Indeed, we assumed implicitly that *Intelligence* and *Difficulty* are independent. Thus, we expect that:

$$(I \perp D). \quad (3.13)$$

This analysis might seem somewhat surprising in light of our earlier examples, where learning something about the course difficulty drastically changed our beliefs about the student's intelligence. In that situation, however, we were reasoning in the presence of information about the student's grade. In other words, we were demonstrating the dependence of I and D *given* G . This phenomenon is a very important one, and we will return to it.

For the variable D , both I and S are nondescendants. Recall that, if $(I \perp D)$ then $(D \perp I)$. The variable S increases our beliefs in the student's intelligence, but knowing that the student is smart (or not) does not influence our beliefs in the difficulty of the course. Thus, we have that

$$(D \perp I, S). \quad (3.14)$$

We can see a pattern emerging. Our intuition tells us that the parents of a variable “shield” it from probabilistic influence that is causal in nature. In other words, once I know the value of the parents, no information relating directly or indirectly to its parents or other ancestors can influence my beliefs about it. However, information about its descendants *can* change my beliefs about it, via an evidential reasoning process.

3.2.2.2 Bayesian Network Semantics

We are now ready to provide the formal definition of the semantics of a Bayesian network structure. We would like the formal definition to match the intuitions developed in our example.

Definition 3.1

Bayesian network
structure

local
independencies

A Bayesian network structure $\mathcal{G}$ is a directed acyclic graph whose nodes represent random variables $X_1, \dots, X_n$. Let $\text{Pa}_{X_i}^{\mathcal{G}}$ denote the parents of X_i in $\mathcal{G}$, and $\text{NonDescendants}_{X_i}$ denote the variables in the graph that are not descendants of X_i . Then $\mathcal{G}$ encodes the following set of conditional independence assumptions, called the local independencies, and denoted by $\mathcal{I}_{\ell}(\mathcal{G})$:

$$\text{For each variable } X_i: (X_i \perp \text{NonDescendants}_{X_i} \mid \text{Pa}_{X_i}^{\mathcal{G}}). \quad \blacksquare$$

In other words, the local independencies state that each node X_i is conditionally independent of its nondescendants given its parents.

Returning to the Student network $G^{student}$, the local Markov independencies are precisely the ones dictated by our intuition, and specified in equation (3.10) – equation (3.14).

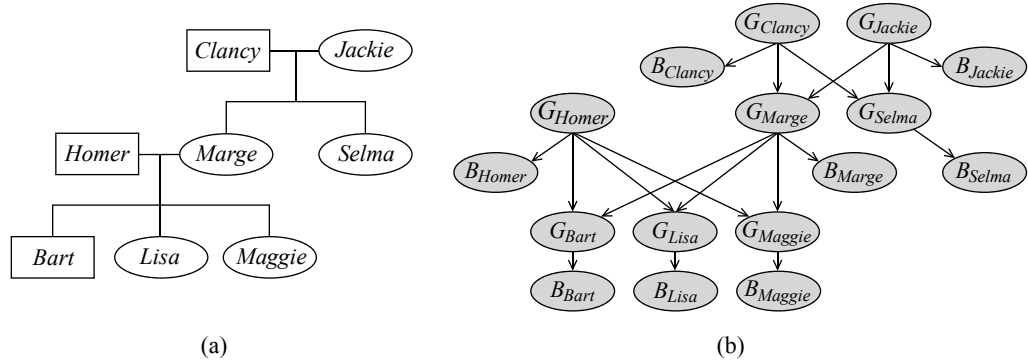


Figure 3.B.1 — Modeling Genetic Inheritance (a) A small family tree. (b) A simple BN for genetic inheritance in this domain. The G variables represent a person's genotype, and the B variables the result of a blood-type test.

Box 3.B — Case Study: The Genetics Example. One of the very earliest uses of a Bayesian network model (long before the general framework was defined) is in the area of genetic pedigrees. In this setting, the local independencies are particularly intuitive. In this application, we want to model the transmission of a certain property, say blood type, from parent to child. The blood type of a person is an observable quantity that depends on her genetic makeup. Such properties are called phenotypes. The genetic makeup of a person is called genotype.

To model this scenario properly, we need to introduce some background on genetics. The human genetic material consists of 22 pairs of autosomal chromosomes and a pair of the sex chromosomes (X and Y). Each chromosome contains a set of genetic material, consisting (among other things) of genes that determine a person's properties. A region of the chromosome that is of interest is called a locus; a locus can have several variants, called alleles.

For concreteness, we focus on autosomal chromosome pairs. In each autosomal pair, one chromosome is the paternal chromosome, inherited from the father, and the other is the maternal chromosome, inherited from the mother. For genes in an autosomal pair, a person has two copies of the gene, one on each copy of the chromosome. Thus, one of the gene's alleles is inherited from the person's mother, and the other from the person's father. For example, the region containing the gene that encodes a person's blood type is a locus. This gene comes in three variants, or alleles: A, B, and O. Thus, a person's genotype is denoted by an ordered pair, such as $\langle A, B \rangle$; with three choices for each entry in the pair, there are 9 possible genotypes. The blood type phenotype is a function of both copies of the gene. For example, if the person has an A allele and an O allele, her observed blood type is "A." If she has two O alleles, her observed blood type is "O."

To represent this domain, we would have, for each person, two variables: one representing the person's genotype, and the other her phenotype. We use the name $G(p)$ to represent person p 's genotype, and $B(p)$ to represent her blood type.

In this example, the independence assumptions arise immediately from the biology. Since the

blood type is a function of the genotype, once we know the genotype of a person, additional evidence about other members of the family will not provide new information about the blood type. Similarly, the process of genetic inheritance implies independence assumption. Once we know the genotype of both parents, we know what each of them can pass on to the offspring. Thus, learning new information about ancestors (or nondescendants) does not provide new information about the genotype of the offspring. These are precisely the local independencies in the resulting network structure, shown for a simple family tree in figure 3.B.1. The intuition here is clear; for example, Bart's blood type is correlated with that of his aunt Selma, but once we know Homer's and Marge's genotype, the two become independent.

To define the probabilistic model fully, we need to specify the CPDs. There are three types of CPDs in this model:

- The penetrance model $P(B(c) \mid G(c))$, which describes the probability of different variants of a particular phenotype (say different blood types) given the person's genotype. In the case of the blood type, this CPD is a deterministic function, but in other cases, the dependence can be more complex.
- The transmission model $P(G(c) \mid G(p), G(m))$, where c is a person and p, m her father and mother, respectively. Each parent is equally likely to transmit either of his or her two alleles to the child.
- Genotype priors $P(G(c))$, used when person c has no parents in the pedigree. These are the general genotype frequencies within the population.

Our discussion of blood type is simplified for several reasons. First, some phenotypes, such as late-onset diseases, are not a deterministic function of the genotype. Rather, an individual with a particular genotype might be more likely to have the disease than an individual with other genotypes. Second, the genetic makeup of an individual is defined by many genes. Some phenotypes might depend on multiple genes. In other settings, we might be interested in multiple phenotypes, which (naturally) implies a dependence on several genes. Finally, as we now discuss, the inheritance patterns of different genes are not independent of each other.

Recall that each of the person's autosomal chromosomes is inherited from one of her parents. However, each of the parents also has two copies of each autosomal chromosome. These two copies, within each parent, recombine to produce the chromosome that is transmitted to the child. Thus, the maternal chromosome inherited by Bart is a combination of the chromosomes inherited by his mother Marge from her mother Jackie and her father Clancy. The recombination process is stochastic, but only a handful recombination events take place within a chromosome in a single generation. Thus, if Bart inherited the allele for some locus from the chromosome his mother inherited from her mother Jackie, he is also much more likely to inherit Jackie's copy for a nearby locus. Thus, to construct an appropriate model for multilocus inheritance, we must take into consideration the probability of a recombination taking place between pairs of adjacent loci.

We can facilitate this modeling by introducing selector variables that capture the inheritance pattern along the chromosome. In particular, for each locus ℓ and each child c , we have a variable $S(\ell, c, m)$ that takes the value 1 if the locus ℓ in c 's maternal chromosome was inherited from c 's maternal grandmother, and 2 if this locus was inherited from c 's maternal grandfather. We have a similar selector variable $S(\ell, c, p)$ for c 's paternal chromosome. We can now model correlations induced by low recombination frequency by correlating the variables $S(\ell, c, m)$ and $S(\ell', c, m)$ for adjacent loci ℓ, ℓ' .

This type of model has been used extensively for many applications. In genetic counseling and prediction, one takes a phenotype with known loci and a set of observed phenotype and genotype data for some individuals in the pedigree to infer the genotype and phenotype for another person in the pedigree (say, a planned child). The genetic data can consist of direct measurements of the relevant disease loci (for some individuals) or measurements of nearby loci, which are correlated with the disease loci.

In linkage analysis, the task is a harder one: identifying the location of disease genes from pedigree data using some number of pedigrees where a large fraction of the individuals exhibit a disease phenotype. Here, the available data includes phenotype information for many individuals in the pedigree, as well as genotype information for loci whose location in the chromosome is known. Using the inheritance model, the researchers can evaluate the likelihood of these observations under different hypotheses about the location of the disease gene relative to the known loci. By repeated calculation of the probabilities in the network for different hypotheses, researchers can pinpoint the area that is “linked” to the disease. This much smaller region can then be used as the starting point for more detailed examination of genes in that area. This process is crucial, for it can allow the researchers to focus on a small area (for example, 1/10,000 of the genome).

As we will see in later chapters, the ability to describe the genetic inheritance process using a sparse Bayesian network provides us the capability to use sophisticated inference algorithms that allow us to reason about large pedigrees and multiple loci. It also allows us to use algorithms for model learning to obtain a deeper understanding of the genetic inheritance process, such as recombination rates in different regions or penetrance probabilities for different diseases.

3.2.3 Graphs and Distributions

The formal semantics of a Bayesian network graph is as a set of independence assertions. On the other hand, our Student BN was a graph annotated with CPDs, which defined a joint distribution via the chain rule for Bayesian networks. In this section, we show that these two definitions are, in fact, equivalent. A distribution P satisfies the local independencies associated with a graph $\mathcal{G}$ if and only if P is representable as a set of CPDs associated with the graph $\mathcal{G}$. We begin by formalizing the basic concepts.

3.2.3.1 I-Maps

We first define the set of independencies associated with a distribution P .

Definition 3.2

independencies
in P

Let P be a distribution over $\mathcal{X}$. We define $\mathcal{I}(P)$ to be the set of independence assertions of the form $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$ that hold in P . ■

We can now rewrite the statement that “ P satisfies the local independencies associated with $\mathcal{G}$ ” simply as $\mathcal{I}_\ell(\mathcal{G}) \subseteq \mathcal{I}(P)$. In this case, we say that $\mathcal{G}$ is an *I-map* (independency map) for P . However, it is useful to define this concept more broadly, since different variants of it will be used throughout the book.

Definition 3.3

I-map

Let $\mathcal{K}$ be any graph object associated with a set of independencies $\mathcal{I}(\mathcal{K})$. We say that $\mathcal{K}$ is an I-map for a set of independencies $\mathcal{I}$ if $\mathcal{I}(\mathcal{K}) \subseteq \mathcal{I}$. ■

We now say that $\mathcal{G}$ is an I-map for P if $\mathcal{G}$ is an I-map for $\mathcal{I}(P)$.

As we can see from the direction of the inclusion, for $\mathcal{G}$ to be an I-map of P , it is necessary that $\mathcal{G}$ does not mislead us regarding independencies in P : any independence that $\mathcal{G}$ asserts must also hold in P . Conversely, P may have additional independencies that are not reflected in $\mathcal{G}$.

Let us illustrate the concept of an I-map on a very simple example.

Example 3.1

Consider a joint probability space over two independent random variables X and Y . There are three possible graphs over these two nodes: $\mathcal{G}_\emptyset$, which is a disconnected pair $X \quad Y$; $\mathcal{G}_{X \rightarrow Y}$, which has the edge $X \rightarrow Y$; and $\mathcal{G}_{Y \rightarrow X}$, which contains $Y \rightarrow X$. The graph $\mathcal{G}_\emptyset$ encodes the assumption that $(X \perp Y)$. The latter two encode no independence assumptions.

Consider the following two distributions:

X	Y	$P(X, Y)$	X	Y	$P(X, Y)$
x^0	y^0	0.08	x^0	y^0	0.4
x^0	y^1	0.32	x^0	y^1	0.3
x^1	y^0	0.12	x^1	y^0	0.2
x^1	y^1	0.48	x^1	y^1	0.1

In the example on the left, X and Y are independent in P ; for example, $P(x^1) = 0.48 + 0.12 = 0.6$, $P(y^1) = 0.8$, and $P(x^1, y^1) = 0.48 = 0.6 \cdot 0.8$. Thus, $(X \perp Y) \in \mathcal{I}(P)$, and we have that $\mathcal{G}_\emptyset$ is an I-map of P . In fact, all three graphs are I-maps of P : $\mathcal{I}_\ell(\mathcal{G}_{X \rightarrow Y})$ is empty, so that trivially P satisfies all the independencies in it (similarly for $\mathcal{G}_{Y \rightarrow X}$). In the example on the right, $(X \perp Y) \notin \mathcal{I}(P)$, so that $\mathcal{G}_\emptyset$ is not an I-map of P . Both other graphs are I-maps of P . ■

3.2.3.2 I-Map to Factorization

A BN structure $\mathcal{G}$ encodes a set of conditional independence assumptions; every distribution for which $\mathcal{G}$ is an I-map must satisfy these assumptions. This property is the key to allowing the compact factorized representation that we saw in the Student example in section 3.2.1. The basic principle is the same as the one we used in the naive Bayes decomposition in section 3.1.3.

Consider any distribution P for which our Student BN $\mathcal{G}_{student}$ is an I-map. We will decompose the joint distribution and show that it factorizes into local probabilistic models, as in section 3.2.1. Consider the joint distribution $P(I, D, G, L, S)$; from the chain rule for probabilities (equation (2.5)), we can decompose this joint distribution in the following way:

$$P(I, D, G, L, S) = P(I)P(D | I)P(G | I, D)P(L | I, D, G)P(S | I, D, G, L). \quad (3.15)$$

This transformation relies on no assumptions; it holds for any joint distribution P . However, it is also not very helpful, since the conditional probabilities in the factorization on the right-hand side are neither natural nor compact. For example, the last factor requires the specification of 24 conditional probabilities: $P(s^1 | i, d, g, l)$ for every assignment of values i, d, g, l .

This form, however, allows us to apply the conditional independence assumptions induced from the BN. Let us assume that $\mathcal{G}_{student}$ is an I-map for our distribution P . In particular, from equation (3.13), we have that $(D \perp I) \in \mathcal{I}(P)$. From that, we can conclude that $P(D | I) = P(D)$, allowing us to simplify the second factor on the right-hand side. Similarly, we know from

equation (3.10) that $(L \perp I, D \mid G) \in \mathcal{I}(P)$. Hence, $P(L \mid I, D, G) = P(L \mid G)$, allowing us to simplify the third term. Using equation (3.11) in a similar way, we obtain that

$$P(I, D, G, L, S) = P(I)P(D)P(G \mid I, D)P(L \mid G)P(S \mid I). \quad (3.16)$$

This factorization is precisely the one we used in section 3.2.1.

This result tells us that any entry in the joint distribution can be computed as a product of factors, one for each variable. Each factor represents a conditional probability of the variable given its parents in the network. This factorization applies to *any* distribution P for which $G_{student}$ is an I-map.

We now state and prove this fundamental result more formally.

Definition 3.4
factorization

Let $\mathcal{G}$ be a BN graph over the variables $X_1, \dots, X_n$. We say that a distribution P over the same space factorizes according to $\mathcal{G}$ if P can be expressed as a product

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Pa}_{X_i}^{\mathcal{G}}). \quad (3.17)$$

chain rule for
Bayesian
networks

This equation is called the chain rule for Bayesian networks. The individual factors $P(X_i \mid \text{Pa}_{X_i}^{\mathcal{G}})$ are called conditional probability distributions (CPDs) or local probabilistic models. ■

CPD

Definition 3.5
Bayesian network

A Bayesian network is a pair $\mathcal{B} = (\mathcal{G}, P)$ where P factorizes over $\mathcal{G}$, and where P is specified as a set of CPDs associated with $\mathcal{G}$'s nodes. The distribution P is often annotated $P_{\mathcal{B}}$. ■

We can now prove that the phenomenon we observed for $G_{student}$ holds more generally.

Theorem 3.1

Let $\mathcal{G}$ be a BN structure over a set of random variables $\mathcal{X}$, and let P be a joint distribution over the same space. If $\mathcal{G}$ is an I-map for P , then P factorizes according to $\mathcal{G}$.

topological
ordering

PROOF Assume, without loss of generality, that $X_1, \dots, X_n$ is a *topological ordering* of the variables in $\mathcal{X}$ relative to $\mathcal{G}$ (see definition 2.19). As in our example, we first use the chain rule for probabilities:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid X_1, \dots, X_{i-1}).$$

Now, consider one of the factors $P(X_i \mid X_1, \dots, X_{i-1})$. As $\mathcal{G}$ is an I-map for P , we have that $(X_i \perp \text{NonDescendants}_{X_i} \mid \text{Pa}_{X_i}^{\mathcal{G}}) \in \mathcal{I}(P)$. By assumption, all of X_i 's parents are in the set $X_1, \dots, X_{i-1}$. Furthermore, none of X_i 's descendants can possibly be in the set. Hence,

$$\{X_1, \dots, X_{i-1}\} = \text{Pa}_{X_i} \cup \mathbf{Z}$$

where $\mathbf{Z} \subseteq \text{NonDescendants}_{X_i}$. From the local independencies for X_i and from the decomposition property (equation (2.8)) it follows that $(X_i \perp \mathbf{Z} \mid \text{Pa}_{X_i})$. Hence, we have that

$$P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \text{Pa}_{X_i}).$$

Applying this transformation to all of the factors in the chain rule decomposition, the result follows. ■

Thus, the conditional independence assumptions implied by a BN structure $\mathcal{G}$ allow us to factorize a distribution P for which $\mathcal{G}$ is an I-map into small CPDs. Note that the proof is constructive, providing a precise algorithm for constructing the factorization given the distribution P and the graph $\mathcal{G}$.

The resulting factorized representation can be substantially more compact, particularly for sparse structures.

Example 3.2

In our Student example, the number of independent parameters is fifteen: we have two binomial distributions $P(I)$ and $P(D)$, with one independent parameter each; we have four multinomial distributions over G — one for each assignment of values to I and D — each with two independent parameters; we have three binomial distributions over L , each with one independent parameter; and similarly two binomial distributions over S , each with an independent parameter. The specification of the full joint distribution would require $48 - 1 = 47$ independent parameters. ■

More generally, in a distribution over n binary random variables, the specification of the joint distribution requires $2^n - 1$ independent parameters. If the distribution factorizes according to a graph $\mathcal{G}$ where each node has at most k parents, the total number of independent parameters required is less than $n \cdot 2^k$ (see exercise 3.6). In many applications, we can assume a certain locality of influence between variables: although each variable is generally correlated with many of the others, it often depends *directly* on only a small number of other variables. Thus, in many cases, k will be very small, even though n is large. As a consequence, the number of parameters in the Bayesian network representation is typically exponentially smaller than the number of parameters of a joint distribution. This property is one of the main benefits of the Bayesian network representation.

3.2.3.3 Factorization to I-Map

Theorem 3.1 shows one direction of the fundamental connection between the conditional independencies encoded by the BN structure and the factorization of the distribution into local probability models: that the conditional independencies imply factorization. The converse also holds: factorization according to $\mathcal{G}$ implies the associated conditional independencies.

Theorem 3.2

Let $\mathcal{G}$ be a BN structure over a set of random variables $\mathcal{X}$ and let P be a joint distribution over the same space. If P factorizes according to $\mathcal{G}$, then $\mathcal{G}$ is an I-map for P .

We illustrate this theorem by example, leaving the proof as an exercise (exercise 3.9). Let P be some distribution that factorizes according to G_{student} . We need to show that $\mathcal{I}_{\mathcal{G}}(G_{\text{student}})$ holds in P . Consider the independence assumption for the random variable S — $(S \perp D, G, L \mid I)$. To prove that it holds for P , we need to show that

$$P(S \mid I, D, G, L) = P(S \mid I).$$

By definition,

$$P(S \mid I, D, G, L) = \frac{P(S, I, D, G, L)}{P(I, D, G, L)}.$$

By the chain rule for BNs equation (3.16), the numerator is equal to $P(I)P(D)P(G | I, D)P(L | G)P(S | I)$. By the process of marginalizing over a joint distribution, we have that the denominator is:

$$\begin{aligned}
 P(I, D, G, L) &= \sum_S P(I, D, G, L, S) \\
 &= \sum_S P(I)P(D)P(G | I, D)P(L | G)P(S | I) \\
 &= P(I)P(D)P(G | I, D)P(L | G) \sum_S P(S | I) \\
 &= P(I)P(D)P(G | I, D)P(L | G),
 \end{aligned}$$

where the last step is a consequence of the fact that $P(S | I)$ is a distribution over values of S , and therefore it sums to 1. We therefore have that

$$\begin{aligned}
 P(S | I, D, G, L) &= \frac{P(S, I, D, G, L)}{P(I, D, G, L)} \\
 &= \frac{P(I)P(D)P(G | I, D)P(L | G)P(S | I)}{P(I)P(D)P(G | I, D)P(L | G)} \\
 &= P(S | I).
 \end{aligned}$$

Box 3.C — Skill: Knowledge Engineering. Our discussion of Bayesian network construction focuses on the process of going from a given distribution to a Bayesian network. Real life is not like that. We have a vague model of the world, and we need to crystallize it into a network structure and parameters. This task breaks down into several components, each of which can be quite subtle. Unfortunately, modeling mistakes can have significant consequences for the quality of the answers obtained from the network, or to the cost of using the network in practice.

Picking variables When we model a domain, there are many possible ways to describe the relevant entities and their attributes. Choosing which random variables to use in the model is often one of the hardest tasks, and this decision has implications throughout the model. A common problem is using ill-defined variables. For example, deciding to include the variable *Fever* to describe a patient in a medical domain seems fairly innocuous. However, does this random variable relate to the internal temperature of the patient? To the thermometer reading (if one is taken by the medical staff)? Does it refer to the temperature of the patient at a specific moment (for example, the time of admission to the hospital) or to occurrence of a fever over a prolonged period? Clearly, each of these might be a reasonable attribute to model, but the interaction of *Fever* with other variables depends on the specific interpretation we use.

clarity test

As this example shows, we must be precise in defining the variables in the model. The clarity test is a good way of evaluating whether they are sufficiently well defined. Assume that we are a million years after the events described in the domain; can an omniscient being, one who saw everything, determine the value of the variable? For example, consider a *Weather* variable with a value *sunny*. To be absolutely precise, we must define where we check the weather, at what time,

and what fraction of the sky must be clear in order for it to be sunny. For a variable such as Heart-attack, we must specify how large the heart attack has to be, during what period of time it has to happen, and so on. By contrast, a variable such as Risk-of-heart-attack is meaningless, as even an omniscient being cannot evaluate whether a person had high risk or low risk, only whether the heart attack occurred or not. Introducing variables such as this confounds actual events and their probability. Note, however, that we can use a notion of “risk group,” as long as it is defined in terms of clearly specified attributes such as age or lifestyle.

If we are not careful in our choice of variables, we will have a hard time making sure that evidence observed and conclusions made are coherent.

hidden variable

Generally speaking, we want our model to contain variables that we can potentially observe or that we may want to query. However, sometimes we want to put in a hidden variable that is neither observed nor directly of interest. Why would we want to do that? Let us consider an example relating to a cholesterol test. Assume that, for the answers to be accurate, the subject has to have eaten nothing after 10:00 PM the previous evening. If the person eats (having no willpower), the results are consistently off. We do not really care about a Willpower variable, nor can we observe it. However, without it, all of the different cholesterol tests become correlated. To avoid graphs where all the tests are correlated, it is better to put in this additional hidden variable, rendering them conditionally independent given the true cholesterol level and the person's willpower.

On the other hand, it is not necessary to add every variable that might be relevant. In our Student example, the student's SAT score may be affected by whether he goes out for drinks on the night before the exam. Is this variable important to represent? The probabilities already account for the fact that he may achieve a poor score despite being intelligent. It might not be worthwhile to include this variable if it cannot be observed.

It is also important to specify a reasonable domain of values for our variables. In particular, if our partition is not fine enough, conditional independence assumptions may be false. For example, we might want to construct a model where we have a person's cholesterol level, and two cholesterol tests that are conditionally independent given the person's true cholesterol level. We might choose to define the value normal to correspond to levels up to 200, and high to levels above 200. But it may be the case that both tests are more likely to fail if the person's cholesterol is marginal (200–240). In this case, the assumption of conditional independence given the value (high/normal) of the cholesterol test is false. It is only true if we add a marginal value.

Picking structure As we saw, there are many structures that are consistent with the same set of independencies. One successful approach is to choose a structure that reflects the causal order and dependencies, so that causes are parents of the effect. Such structures tend to work well. Either because of some real locality of influence in the world, or because of the way people perceive the world, causal graphs tend to be sparser. It is important to stress that the causality is in the world, not in our inference process. For example, in an automobile insurance network, it is tempting to put Previous-accident as a parent of Good-driver, because that is how the insurance company thinks about the problem. This is not the causal order in the world, because being a bad driver causes previous (and future) accidents. In principle, there is nothing to prevent us from directing the edges in this way. However, a noncausal ordering often requires that we introduce many additional edges to account for induced dependencies (see section 3.4.1).

One common approach to constructing a structure is a backward construction process. We begin with a variable of interest, say Lung-Cancer. We then try to elicit a prior probability for that

variable. If our expert responds that this probability is not determinable, because it depends on other factors, that is a good indication that these other factors should be added as parents for that variable (and as variables into the network). For example, we might conclude using this process that Lung-Cancer really should have Smoking as a parent, and (perhaps not as obvious) that Smoking should have Gender and Age as a parent. This approach, called extending the conversation, avoids probability estimates that result from an average over a heterogeneous population, and therefore leads to more precise probability estimates.



When determining the structure, however, we must also keep in mind that approximations are inevitable. For many pairs of variables, we can construct a scenario where one depends on the other. For example, perhaps Difficulty depends on Intelligence, because the professor is more likely to make a class difficult if intelligent students are registered. In general, **there are many weak influences that we might choose to model, but if we put in all of them, the network can become very complex.** Such networks are problematic from a representational perspective: they are hard to understand and hard to debug, and eliciting (or learning) parameters can get very difficult. Moreover, as reasoning in Bayesian networks depends strongly on their connectivity (see section 9.4), adding such edges can make the network too expensive to use.

This final consideration may lead us, in fact, to make approximations that we know to be wrong. For example, in networks for fault or medical diagnosis, the correct approach is usually to model each possible fault as a separate random variable, allowing for multiple failures. However, such networks might be too complex to perform effective inference in certain settings, and so we may sometimes resort to a single fault approximation, where we have a single random variable encoding the primary fault or disease.

Picking probabilities One of the most challenging tasks in constructing a network manually is eliciting probabilities from people. This task is somewhat easier in the context of causal models, since the parameters tend to be natural and more interpretable. Nevertheless, people generally dislike committing to an exact estimate of probability.

One approach is to elicit estimates qualitatively, using abstract terms such as “common,” “rare,” and “surprising,” and then assign these to numbers using a predefined scale. This approach is fairly crude, and often can lead to misinterpretation. There are several approaches developed for assisting in eliciting probabilities from people. For example, one can visualize the probability of the event as an area (slice of a pie), or ask people how they would compare the probability in question to certain predefined lotteries. Nevertheless, probability elicitation is a long, difficult process, and one whose outcomes are not always reliable: the elicitation method can often influence the results, and asking the same question using different phrasing can often lead to significant differences in the answer. For example, studies show that people’s estimates for an event such as “Death by disease” are significantly lower than their estimates for this event when it is broken down into different possibilities such as “Death from cancer,” “Death from heart disease,” and so on.

How important is it that we get our probability estimates exactly right? In some cases, small errors have very little effect. For example, changing a conditional probability of 0.7 to 0.75 generally does not have a significant effect. Other errors, however, can have a significant effect:

- **Zero probabilities:** A common mistake is to assign a probability of zero to an event that is extremely unlikely, but not impossible. The problem is that **one can never condition away a zero probability, no matter how much evidence we get. When an event is unlikely**



but not impossible, giving it probability zero is guaranteed to lead to irrecoverable errors. For example, in one of the early versions of the the Pathfinder system (box 3.D), 10 percent of the misdiagnoses were due to zero probability estimates given by the expert to events that were unlikely but not impossible. As a general rule, very few things (except definitions) have probability zero, and we must be careful in assigning zeros.

- **Orders of magnitude:** Small differences in very low probability events can make a large difference to the network conclusions. Thus, a (conditional) probability of 10^{-4} is very different from 10^{-5} .
- **Relative values:** The qualitative behavior of the conclusions reached by the network — the value that has the highest probability — is fairly sensitive to the relative sizes of $P(x \mid \mathbf{y})$ for different values $\mathbf{y}$ of Pa_X . For example, it is important that the network encode correctly that the probability of having a high fever is greater when the patient has pneumonia than when he has the flu.

sensitivity
analysis

A very useful tool for estimating network parameters is sensitivity analysis, which allows us to determine the extent to which a given probability parameter affects the outcome. This process allows us to evaluate whether it is important to get a particular CPD entry right. It also helps us figure out which CPD entries are responsible for an answer to some query that does not match our intuitions.

medical diagnosis
expert system

Box 3.D — Case Study: Medical Diagnosis Systems. One of the earliest applications of Bayesian networks was to the task of medical diagnosis. In the 1980s, a very active area of research was the construction of expert systems — computer-based systems that replace or assist an expert in performing a complex task. One such task that was tackled in several ways was medical diagnosis. This task, more than many others, required a treatment of uncertainty, due to the complex, nondeterministic relationships between findings and diseases. Thus, it formed the basis for experimentation with various formalisms for uncertain reasoning.

Pathfinder

The Pathfinder expert system was designed by Heckerman and colleagues (Heckerman and Nathuwani 1992a; Heckerman et al. 1992; Heckerman and Nathuwani 1992b) to help a pathologist diagnose diseases in lymph nodes. Ultimately, the model contained more than sixty different diseases and around a hundred different features. It evolved through several versions, including some based on nonprobabilistic formalisms, and several that used variants of Bayesian networks. Its diagnostic ability was evaluated over real pathological cases and compared to the diagnoses of pathological experts.

One of the first models used was a simple naive Bayes model, which was compared to the models based on alternative uncertainty formalisms, and judged to be superior in its diagnostic ability. It therefore formed the basis for subsequent development of the system.

The same evaluation pointed out important problems in the way in which parameters were elicited from the expert. First, it was shown that 10 percent of the cases were diagnosed incorrectly, because the correct disease was ruled out by a finding that was unlikely, but not impossible, to manifest in that disease. Second, in the original construction, the expert estimated the probabilities $P(\text{Finding} \mid \text{Disease})$ by fixing a single disease and evaluating the probabilities of all its findings.

It was found that the expert was more comfortable considering a single finding and evaluating its probability across all diseases. This approach allows the expert to compare the relative values of the same finding across multiple diseases, as described in box 3.C.

With these two lessons in mind, another version of Pathfinder — Pathfinder III — was constructed, still using the naive Bayes model. Finally, Pathfinder IV used a full Bayesian network, with a single disease hypothesis but with dependencies between the features. Pathfinder IV was constructed using a similarity network (see box 5.B), significantly reducing the number of parameters that must be elicited. Pathfinder IV, viewed as a Bayesian network, had a total of around 75,000 parameters, but the use of similarity networks allowed the model to be constructed with fewer than 14,000 distinct parameters. Overall, the structure of Pathfinder IV took about 35 hours to define, and the parameters 40 hours.

A comprehensive evaluation of the performance of the two models revealed some important insights. First, the Bayesian network performed as well or better on most cases than the naive Bayes model. In most of the cases where the Bayesian network performed better, the use of richer dependency models was a contributing factor. As expected, these models were useful because they address the strong conditional independence assumptions of the naive Bayes model, as described in box 3.A. Somewhat more surprising, they also helped in allowing the expert to condition the probabilities on relevant factors other than the disease, using the process of extending the conversation described in box 3.C, leading to more accurate elicited probabilities. Finally, the use of similarity networks led to more accurate models, for the smaller number of elicited parameters reduced irrelevant fluctuations in parameter values (due to expert inconsistency) that can lead to spurious dependencies.

Overall, the Bayesian network model agreed with the predictions of an expert pathologist in 50/53 cases, as compared with 47/53 cases for the naive Bayes model, with significant therapeutic implications. A later evaluation showed that the diagnostic accuracy of Pathfinder IV was at least as good as that of the expert used to design the system. When used with less expert pathologists, the system significantly improved the diagnostic accuracy of the physicians alone. Moreover, the system showed greater ability to identify important findings and to integrate these findings into a correct diagnosis.

Unfortunately, multiple reasons prevent the widespread adoption of Bayesian networks as an aid for medical diagnosis, including legal liability issues for misdiagnoses and incompatibility with the physicians' workflow. However, several such systems have been fielded, with significant success. Moreover, similar technology is being used successfully in a variety of other diagnosis applications (see box 23.C).

3.3 Independencies in Graphs

Dependencies and independencies are key properties of a distribution and are crucial for understanding its behavior. As we will see, independence properties are also important for answering queries: they can be exploited to reduce substantially the computation cost of inference. Therefore, it is important that our representations make these properties clearly visible both to a user and to algorithms that manipulate the BN data structure.

As we discussed, a graph structure $\mathcal{G}$ encodes a certain set of conditional independence assumptions $\mathcal{I}_\ell(\mathcal{G})$. Knowing only that a distribution P factorizes over $\mathcal{G}$, we can conclude that it satisfies $\mathcal{I}_\ell(\mathcal{G})$. An immediate question is whether there are other independencies that we can “read off” directly from $\mathcal{G}$. That is, are there other independencies that hold for *every* distribution P that factorizes over $\mathcal{G}$?

3.3.1 D-separation

Our aim in this section is to understand when we can *guarantee* that an independence $(X \perp Y \mid Z)$ holds in a distribution associated with a BN structure $\mathcal{G}$. To understand when a property is guaranteed to hold, it helps to consider its converse: “Can we imagine a case where it does not?” Thus, we focus our discussion on analyzing when it is *possible* that X can influence Y given Z . If we construct an example where this influence occurs, then the converse property $(X \perp Y \mid Z)$ cannot hold for all of the distributions that factorize over $\mathcal{G}$, and hence the independence property $(X \perp Y \mid Z)$ cannot follow from $\mathcal{I}_\ell(\mathcal{G})$.

We therefore begin with an intuitive case analysis: Here, we try to understand when an observation regarding a variable X can possibly change our beliefs about Y , in the presence of evidence about the variables Z . Although this analysis will be purely intuitive, we will show later that our conclusions are actually provably correct.

Direct connection We begin with the simple case, when X and Y are directly connected via an edge, say $X \rightarrow Y$. For any network structure $\mathcal{G}$ that contains the edge $X \rightarrow Y$, it is possible to construct a distribution where X and Y are correlated regardless of any evidence about any of the other variables in the network. In other words, if X and Y are directly connected, we can always get examples where they influence each other, regardless of Z .

In particular, assume that $Val(X) = Val(Y)$; we can simply set $X = Y$. That, by itself, however, is not enough; if (given the evidence Z) X deterministically takes some particular value, say 0, then X and Y both deterministically take that value, and are uncorrelated. We therefore set the network so that X is (for example) uniformly distributed, regardless of the values of any of its parents. This construction suffices to induce a correlation between X and Y , regardless of the evidence.

Indirect connection Now consider the more complicated case when X and Y are not directly connected, but there is a trail between them in the graph. We begin by considering the simplest such case: a three-node network, where X and Y are not directly connected, but where there is a trail between them via Z . It turns out that this simple case is the key to understanding the whole notion of indirect interaction in Bayesian networks.

There are four cases where X and Y are connected via Z , as shown in figure 3.5. The first two correspond to causal chains (in either direction), the third to a common cause, and the fourth to a common effect. We analyze each in turn.

Indirect causal effect (figure 3.5a). To gain intuition, let us return to the Student example, where we had a causal trail $I \rightarrow G \rightarrow L$. Let us begin with the case where G is not observed. Intuitively, if we observe that the student is intelligent, we are more inclined to believe that he gets an A, and therefore that his recommendation letter is strong. In other words, the probability of these latter events is higher conditioned on the observation that the student is intelligent.

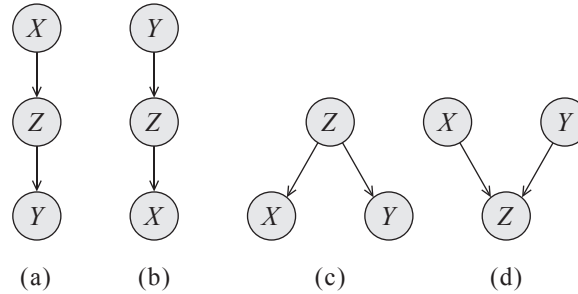


Figure 3.5 The four possible two-edge trails from X to Y via Z : (a) An indirect causal effect; (b) An indirect evidential effect; (c) A common cause; (d) A common effect.

In fact, we saw precisely this behavior in the distribution of figure 3.4. Thus, in this case, we believe that X can influence Y via Z .

Now assume that Z is observed, that is, $Z \in \mathbf{Z}$. As we saw in our analysis of the Student example, if we observe the student's grade, then (as we assumed) his intelligence no longer influences his letter. In fact, the local independencies for this network tell us that $(L \perp I \mid G)$. Thus, we conclude that X cannot influence Y via Z if Z is observed.

Indirect evidential effect (figure 3.5b). Returning to the Student example, we have a chain $I \rightarrow G \rightarrow L$. We have already seen that observing a strong recommendation letter for the student changes our beliefs in his intelligence. Conversely, once the grade is observed, the letter gives no additional information about the student's intelligence. Thus, our analysis in the case $Y \rightarrow Z \rightarrow X$ here is identical to the causal case: X can influence Y via Z , but only if Z is not observed. The similarity is not surprising, as dependence is a symmetrical notion. Specifically, if $(X \perp Y)$ does not hold, then $(Y \perp X)$ does not hold either.

Common cause (figure 3.5c). This case is one that we have analyzed extensively, both within the simple naive Bayes model of section 3.1.3 and within our Student example. Our example has the student's intelligence I as a parent of his grade G and his SAT score S . As we discussed, S and G are correlated in this model, in that observing (say) a high SAT score gives us information about a student's intelligence and hence helps us predict his grade. However, once we observe I , this correlation disappears, and S gives us no additional information about G . Once again, for this network, this conclusion follows from the local independence assumption for the node G (or for S). Thus, our conclusion here is identical to the previous two cases: X can influence Y via Z if and only if Z is not observed.

Common effect (figure 3.5d). In all of the three previous cases, we have seen a common pattern: X can influence Y via Z if and only if Z is not observed. Therefore, we might expect that this pattern is universal, and will continue through this last case. Somewhat surprisingly, this is not the case. Let us return to the Student example and consider I and D , which are parents of G . When G is not observed, we have that I and D are independent. In fact, this conclusion follows (once again) from the local independencies from the network. Thus, in this case, influence cannot “flow” along the trail $X \rightarrow Z \leftarrow Y$ if the intermediate node Z is not observed.

On the other hand, consider the behavior when Z is observed. In our discussion of the

Student example, we analyzed precisely this case, which we called intercausal reasoning; we showed, for example, that the probability that the student has high intelligence goes down dramatically when we observe that his grade is a C ($G = g^3$), but then goes up when we observe that the class is a difficult one $D = d^1$. Thus, in presence of the evidence $G = g^3$, we have that I and D are correlated.

Let us consider a variant of this last case. Assume that we do not observe the student's grade, but we do observe that he received a weak recommendation letter ($L = l^0$). Intuitively, the same phenomenon happens. The weak letter is an indicator that he received a low grade, and therefore it suffices to correlate I and D .

When influence can flow from X to Y via Z , we say that the trail $X \rightleftharpoons Z \rightleftharpoons Y$ is *active*. The results of our analysis for active two-edge trails are summarized thus:

- **Causal trail** $X \rightarrow Z \rightarrow Y$: active if and only if Z is not observed.
- **Evidential trail** $X \leftarrow Z \leftarrow Y$: active if and only if Z is not observed.
- **Common cause** $X \leftarrow Z \rightarrow Y$: active if and only if Z is not observed.
- **Common effect** $X \rightarrow Z \leftarrow Y$: active if and only if either Z or one of Z 's descendants is observed.

v-structure

A structure where $X \rightarrow Z \leftarrow Y$ (as in figure 3.5d) is also called a *v-structure*.

It is useful to view probabilistic influence as a flow in the graph. Our analysis here tells us when influence from X can “flow” through Z to affect our beliefs about Y .

General Case Now consider the case of a longer trail $X_1 \rightleftharpoons \dots \rightleftharpoons X_n$. Intuitively, for influence to “flow” from X_1 to X_n , it needs to flow through every single node on the trail. In other words, X_1 can influence X_n if every two-edge trail $X_{i-1} \rightleftharpoons X_i \rightleftharpoons X_{i+1}$ along the trail allows influence to flow.

We can summarize this intuition in the following definition:

Definition 3.6

observed variable

active trail

Let $\mathcal{G}$ be a BN structure, and $X_1 \rightleftharpoons \dots \rightleftharpoons X_n$ a trail in $\mathcal{G}$. Let $\mathbf{Z}$ be a subset of observed variables. The trail $X_1 \rightleftharpoons \dots \rightleftharpoons X_n$ is active given $\mathbf{Z}$ if

- Whenever we have a v-structure $X_{i-1} \rightarrow X_i \leftarrow X_{i+1}$, then X_i or one of its descendants are in $\mathbf{Z}$;
- no other node along the trail is in $\mathbf{Z}$. ■

Note that if X_1 or X_n are in $\mathbf{Z}$ the trail is not active.

In our Student BN, we have that $D \rightarrow G \leftarrow I \rightarrow S$ is not an active trail for $\mathbf{Z} = \emptyset$, because the v-structure $D \rightarrow G \leftarrow I$ is not activated. That same trail is active when $\mathbf{Z} = \{L\}$, because observing the descendant of G activates the v-structure. On the other hand, when $\mathbf{Z} = \{L, I\}$, the trail is not active, because observing I blocks the trail $G \leftarrow I \rightarrow S$.

What about graphs where there is more than one trail between two nodes? Our flow intuition continues to carry through: one node can influence another if there is any trail along which influence can flow. Putting these intuitions together, we obtain the notion of *d-separation*, which provides us with a notion of separation between nodes in a directed graph (hence the term d-separation, for directed separation):

d-separation

Definition 3.7

Let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ be three sets of nodes in $\mathcal{G}$. We say that $\mathbf{X}$ and $\mathbf{Y}$ are d-separated given $\mathbf{Z}$, denoted $\text{d-sep}_{\mathcal{G}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})$, if there is no active trail between any node $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ given $\mathbf{Z}$.

We use $\mathcal{I}(\mathcal{G})$ to denote the set of independencies that correspond to d-separation:

$$\mathcal{I}(\mathcal{G}) = \{(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) : \text{d-sep}_{\mathcal{G}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})\}. \quad \blacksquare$$

global Markov
independencies

This set is also called the set of *global Markov independencies*. The similarity between the notation $\mathcal{I}(\mathcal{G})$ and our notation $\mathcal{I}(P)$ is not coincidental: As we discuss later, the independencies in $\mathcal{I}(\mathcal{G})$ are precisely those that are guaranteed to hold for every distribution over $\mathcal{G}$.

3.3.2 Soundness and Completeness

So far, our definition of d-separation has been based on our intuitions regarding flow of influence, and on our one example. As yet, we have no guarantee that this analysis is “correct.” Perhaps there is a distribution over the BN where X can influence Y despite the fact that all trails between them are blocked.

soundness of
d-separation

Hence, the first property we want to ensure for d-separation as a method for determining independence is *soundness*: if we find that two nodes X and Y are d-separated given some $\mathbf{Z}$, then we are guaranteed that they are, in fact, conditionally independent given $\mathbf{Z}$.

Theorem 3.3

If a distribution P factorizes according to $\mathcal{G}$, then $\mathcal{I}(\mathcal{G}) \subseteq \mathcal{I}(P)$.

In other words, any independence reported by d-separation is satisfied by the underlying distribution. The proof of this theorem requires some additional machinery that we introduce in chapter 4, so we defer the proof to that chapter (see section 4.5.1.1).

completeness of
d-separation

A second desirable property is the complementary one — *completeness*: d-separation detects *all* possible independencies. More precisely, if we have that two variables X and Y are independent given $\mathbf{Z}$, then they are d-separated. A careful examination of the completeness property reveals that it is ill defined, inasmuch as it does not specify the distribution in which X and Y are independent.

To formalize this property, we first define the following notion:

Definition 3.8

faithful

A distribution P is faithful to $\mathcal{G}$ if, whenever $(X \perp Y \mid \mathbf{Z}) \in \mathcal{I}(P)$, then $\text{d-sep}_{\mathcal{G}}(X; Y \mid \mathbf{Z})$. In other words, any independence in P is reflected in the d-separation properties of the graph. $\blacksquare$

We can now provide one candidate formalization of the completeness property is as follows:

- For any distribution P that factorizes over $\mathcal{G}$, we have that P is faithful to $\mathcal{G}$; that is, if X and Y are *not d-separated* given $\mathbf{Z}$ in $\mathcal{G}$, then X and Y are *dependent* in *all* distributions P that factorize over $\mathcal{G}$.

This property is the obvious converse to our notion of soundness: If true, the two together would imply that, for any P that factorizes over $\mathcal{G}$, we have that $\mathcal{I}(P) = \mathcal{I}(\mathcal{G})$. Unfortunately, this highly desirable property is easily shown to be false: Even if a distribution factorizes over $\mathcal{G}$, it can still contain additional independencies that are not reflected in the structure.

Example 3.3

Consider a distribution P over two variables A and B , where A and B are independent. One possible I-map for P is the network $A \rightarrow B$. For example, we can set the CPD for B to be

	b^0	b^1
a^0	0.4	0.6
a^1	0.4	0.6

This example clearly violates the first candidate definition of completeness, because the graph $\mathcal{G}$ is an I-map for the distribution P , yet there are independencies that hold for this distribution but do not follow from d -separation. In fact, these are not independencies that we can hope to discover by examining the network structure. ■

Thus, the completeness property does not hold for this candidate definition of completeness. We therefore adopt a weaker yet still useful definition:

- If $(X \perp Y \mid \mathbf{Z})$ in *all* distributions P that factorize over $\mathcal{G}$, then $d\text{-sep}_{\mathcal{G}}(X; Y \mid \mathbf{Z})$. And the contrapositive: If X and Y are *not* d -separated given $\mathbf{Z}$ in $\mathcal{G}$, then X and Y are *dependent* in *some* distribution P that factorizes over $\mathcal{G}$.

Using this definition, we can show:

Theorem 3.4

Let $\mathcal{G}$ be a BN structure. If X and Y are not d -separated given $\mathbf{Z}$ in $\mathcal{G}$, then X and Y are dependent given $\mathbf{Z}$ in some distribution P that factorizes over $\mathcal{G}$.

PROOF The proof constructs a distribution P that makes X and Y correlated. The construction is roughly as follows. As X and Y are not d -separated, there exists an active trail $U_1, \dots, U_k$ between them. We define CPDs for the variables on the trail so as to make each pair U_i, U_{i+1} correlated; in the case of a v-structure $U_i \rightarrow U_{i+1} \leftarrow U_{i+2}$, we define the CPD of U_{i+1} so as to ensure correlation, and also define the CPDs of the path to some downstream evidence node, in a way that guarantees that the downstream evidence activates the correlation between U_i and U_{i+2} . All other CPDs in the graph are chosen to be uniform, and thus the construction guarantees that influence only flows along this single path, preventing cases where the influence of two (or more) paths cancel out. The details of the construction are quite technical and laborious, and we omit them. ■

We can view the completeness result as telling us that our definition of $\mathcal{I}(\mathcal{G})$ is the maximal one. For any independence assertion that is not a consequence of d -separation in $\mathcal{G}$, we can always find a counterexample distribution P that factorizes over $\mathcal{G}$. In fact, this result can be strengthened significantly:

Theorem 3.5

For almost all distributions P that factorize over $\mathcal{G}$, that is, for all distributions except for a set of measure zero in the space of CPD parameterizations, we have that $\mathcal{I}(P) = \mathcal{I}(\mathcal{G})$.¹

1. A set has measure zero if it is infinitesimally small relative to the overall space. For example, the set of all rationals has measure zero within the interval $[0, 1]$. A straight line has measure zero in the plane. This intuition is defined formally in the field of *measure theory*.

This result strengthens theorem 3.4 in two distinct ways: First, whereas theorem 3.4 shows that any dependency in the graph can be found in some distribution, this new result shows that there exists a single distribution that is faithful to the graph, that is, where all of the dependencies in the graph hold simultaneously. Second, not only does this property hold for a single distribution, but it also holds for almost all distributions that factorize over $\mathcal{G}$.

PROOF At a high level, the proof is based on the following argument: Each conditional independence assertion is a set of polynomial equalities over the space of CPD parameters (see exercise 3.13). A basic property of polynomials is that a polynomial is either identically zero or it is nonzero almost everywhere (its set of roots has measure zero). Theorem 3.4 implies that polynomials corresponding to assertions outside $\mathcal{I}(\mathcal{G})$ cannot be identically zero, because they have at least one counterexample. Thus, the set of distributions P , which exhibit any one of these “spurious” independence assertions, has measure zero. The set of distributions that do not satisfy $\mathcal{I}(P) = \mathcal{I}(\mathcal{G})$ is the union of these separate sets, one for each spurious independence assertion. The union of a finite number of sets of measure zero is a set of measure zero, proving the result. ■



These results state that for almost all parameterizations P of the graph $\mathcal{G}$ (that is, for almost all possible choices of CPDs for the variables), the d-separation test precisely characterizes the independencies that hold for P . In other words, even if we have a distribution P that satisfies more independencies than $\mathcal{I}(\mathcal{G})$, a slight perturbation of the CPDs of P will almost always eliminate these “extra” independencies. This guarantee seems to state that such independencies are always accidental, and we will never encounter them in practice. However, as we illustrate in example 3.7, there are cases where our CPDs have certain local structure that is not accidental, and that implies these additional independencies that are not detected by d-separation.

3.3.3 An Algorithm for d-Separation

The notion of d-separation allows us to infer independence properties of a distribution P that factorizes over $\mathcal{G}$ simply by examining the connectivity of $\mathcal{G}$. However, in order to be useful, we need to be able to determine d-separation effectively. Our definition gives us a constructive solution, but a very inefficient one: We can enumerate all trails between X and Y , and check each one to see whether it is active. The running time of this algorithm depends on the number of trails in the graph, which can be exponential in the size of the graph.

Fortunately, there is a much more efficient algorithm that requires only linear time in the size of the graph. The algorithm has two phases. We begin by traversing the graph bottom up, from the leaves to the roots, marking all nodes that are in $\mathbf{Z}$ or that have descendants in $\mathbf{Z}$. Intuitively, these nodes will serve to enable v-structures. In the second phase, we traverse breadth-first from X to Y , stopping the traversal along a trail when we get to a blocked node. A node is blocked if: (a) it is the “middle” node in a v-structure and unmarked in phase I, or (b) is not such a node and is in $\mathbf{Z}$. If our breadth-first search gets us from X to Y , then there is an active trail between them.

The precise algorithm is shown in algorithm 3.1. The first phase is straightforward. The second phase is more subtle. For efficiency, and to avoid infinite loops, the algorithm must keep track of all nodes that have been visited, so as to avoid visiting them again. However, in graphs

Algorithm 3.1 Algorithm for finding nodes reachable from X given Z via active trails

```

Procedure Reachable (
     $\mathcal{G}$ , // Bayesian network graph
     $X$ , // Source variable
     $Z$  // Observations
)
1 // Phase I: Insert all ancestors of  $Z$  into  $A$ 
2  $L \leftarrow Z$  // Nodes to be visited
3  $A \leftarrow \emptyset$  // Ancestors of  $Z$ 
4 while  $L \neq \emptyset$ 
5     Select some  $Y$  from  $L$ 
6      $L \leftarrow L - \{Y\}$ 
7     if  $Y \notin A$  then
8          $L \leftarrow L \cup \text{Pa}_Y$  //  $Y$ 's parents need to be visited
9          $A \leftarrow A \cup \{Y\}$  //  $Y$  is ancestor of evidence
10
11 // Phase II: traverse active trails starting from  $X$ 
12  $L \leftarrow \{(X, \uparrow)\}$  // (Node,direction) to be visited
13  $V \leftarrow \emptyset$  // (Node,direction) marked as visited
14  $R \leftarrow \emptyset$  // Nodes reachable via active trail
15 while  $L \neq \emptyset$ 
16     Select some  $(Y, d)$  from  $L$ 
17      $L \leftarrow L - \{(Y, d)\}$ 
18     if  $(Y, d) \notin V$  then
19         if  $Y \notin Z$  then
20              $R \leftarrow R \cup \{Y\}$  //  $Y$  is reachable
21              $V \leftarrow V \cup \{(Y, d)\}$  // Mark  $(Y, d)$  as visited
22             if  $d = \uparrow$  and  $Y \notin Z$  then // Trail up through  $Y$  active if  $Y$  not in  $Z$ 
23                 for each  $Z \in \text{Pa}_Y$ 
24                      $L \leftarrow L \cup \{(Z, \uparrow)\}$  //  $Y$ 's parents to be visited from bottom
25                 for each  $Z \in \text{Ch}_Y$ 
26                      $L \leftarrow L \cup \{(Z, \downarrow)\}$  //  $Y$ 's children to be visited from top
27             else if  $d = \downarrow$  then // Trails down through  $Y$ 
28                 if  $Y \notin Z$  then
29                     // Downward trails to  $Y$ 's children are active
30                     for each  $Z \in \text{Ch}_Y$ 
31                          $L \leftarrow L \cup \{(Z, \downarrow)\}$  //  $Y$ 's children to be visited from top
32                 if  $Y \in A$  then // v-structure trails are active
33                     for each  $Z \in \text{Pa}_Y$ 
34                          $L \leftarrow L \cup \{(Z, \uparrow)\}$  //  $Y$ 's parents to be visited from bottom
35 return  $R$ 

```

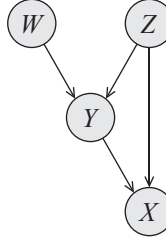


Figure 3.6 A simple example for the d-separation algorithm

with loops (multiple trails between a pair of nodes), an intermediate node Y might be involved in several trails, which may require different treatment within the algorithm:

Example 3.4

Consider the Bayesian network of figure 3.6, where our task is to find all nodes reachable from X . Assume that Y is observed, that is, $Y \in \mathbf{Z}$. Assume that the algorithm first encounters Y via the direct edge $Y \rightarrow X$. Any extension of this trail is blocked by Y , and hence the algorithm stops the traversal along this trail. However, the trail $X \leftarrow Z \rightarrow Y \leftarrow W$ is not blocked by Y . Thus, when we encounter Y for the second time via the edge $Z \rightarrow Y$, we should not ignore it. Therefore, after the first visit to Y , we can mark it as visited for the purpose of trails coming in from children of Y , but not for the purpose of trails coming in from parents of Y . ■

In general, we see that, for each node Y , we must keep track separately of whether it has been visited from the top and whether it has been visited from the bottom. Only when both directions have been explored is the node no longer useful for discovering new active trails.

Based on this intuition, we can now show that the algorithm achieves the desired result:

Theorem 3.6

The algorithm $\text{Reachable}(\mathcal{G}, X, \mathbf{Z})$ returns the set of all nodes reachable from X via trails that are active in $\mathcal{G}$ given $\mathbf{Z}$.

The proof is left as an exercise (exercise 3.14).

3.3.4 I-Equivalence

The notion of $\mathcal{I}(\mathcal{G})$ specifies a set of conditional independence assertions that are associated with a graph. This notion allows us to abstract away the details of the graph structure, viewing it purely as a specification of independence properties. In particular, one important implication of this perspective is the observation that very different BN structures can actually be equivalent, in that they encode precisely the same set of conditional independence assertions. Consider, for example, the three networks in figure 3.5a(b),(c). All three of them encode precisely the same independence assumptions: $(X \perp Y \mid Z)$.

Definition 3.9

I-equivalence

Two graph structures $\mathcal{K}_1$ and $\mathcal{K}_2$ over $\mathcal{X}$ are I-equivalent if $\mathcal{I}(\mathcal{K}_1) = \mathcal{I}(\mathcal{K}_2)$. The set of all graphs over $\mathcal{X}$ is partitioned into a set of mutually exclusive and exhaustive I-equivalence classes, which are the set of equivalence classes induced by the I-equivalence relation. ■

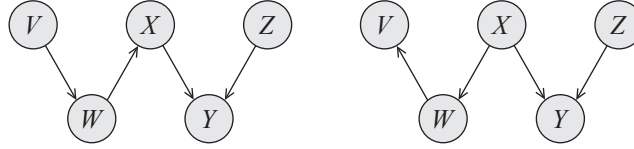


Figure 3.7 Skeletons and v-structures in a network. The two networks shown have the same skeleton and v-structures ($X \rightarrow Y \leftarrow Z$).

Note that the v-structure network in figure 3.5d induces a very different set of d-separation assertions, and hence it does not fall into the same I-equivalence class as the first three. Its I-equivalence class contains only that single network.



I-equivalence of two graphs immediately implies that any distribution P that can be factorized over one of these graphs can be factorized over the other. Furthermore, **there is no intrinsic property of P that would allow us to associate it with one graph rather than an equivalent one. This observation has important implications with respect to our ability to determine the directionality of influence.** In particular, although we can determine, for a distribution $P(X, Y)$, whether X and Y are correlated, there is nothing in the distribution that can help us determine whether the correct structure is $X \rightarrow Y$ or $Y \rightarrow X$. We return to this point when we discuss the causal interpretation of Bayesian networks in chapter 21.

The d-separation criterion allows us to test for I-equivalence using a very simple graph-based algorithm. We start by considering the trails in the networks.

Definition 3.10
skeleton

The skeleton of a Bayesian network graph $\mathcal{G}$ over $\mathcal{X}$ is an undirected graph over $\mathcal{X}$ that contains an edge $\{X, Y\}$ for every edge (X, Y) in $\mathcal{G}$. ■

In the networks of figure 3.7, the networks (a) and (b) have the same skeleton.

If two networks have a common skeleton, then the set of trails between two variables X and Y is same in both networks. If they do not have a common skeleton, we can find a trail in one network that does not exist in the other and use this trail to find a counterexample for the equivalence of the two networks.

Ensuring that the two networks have the same trails is clearly not enough. For example, the networks in figure 3.5 all have the same skeleton. Yet, as the preceding discussion shows, the network of figure 3.5d is not equivalent to the networks of figure 3.5a–(c). The difference, is of course, the v-structure in figure 3.5d. Thus, it seems that if the two networks have the same skeleton and exactly the same set of v-structures, they are equivalent. Indeed, this property provides a sufficient condition for I-equivalence:

Theorem 3.7

Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be two graphs over $\mathcal{X}$. If $\mathcal{G}_1$ and $\mathcal{G}_2$ have the same skeleton and the same set of v-structures then they are I-equivalent.

The proof is left as an exercise (see exercise 3.16).

Unfortunately, this characterization is not an equivalence: there are graphs that are I-equivalent but do not have the same set of v-structures. As a counterexample, consider *complete* graphs over a set of variables. Recall that a complete graph is one to which we cannot add

additional arcs without causing cycles. Such graphs encode the empty set of conditional independence assertions. Thus, any two complete graphs are I-equivalent. Although they have the same skeleton, they invariably have different v-structures. Thus, by using the criterion on theorem 3.7, we can conclude (in certain cases) only that two networks are I-equivalent, but we cannot use it to guarantee that they are not.

We can provide a stronger condition that does correspond exactly to I-equivalence. Intuitively, the unique independence pattern that we want to associate with a v-structure $X \rightarrow Z \leftarrow Y$ is that X and Y are independent (conditionally on their parents), but dependent given Z . If there is a direct edge between X and Y , as there was in our example of the complete graph, the first part of this pattern is eliminated.

Definition 3.11

immorality

A v-structure $X \rightarrow Z \leftarrow Y$ is an immorality if there is no direct edge between X and Y . If there is such an edge, it is called a covering edge for the v-structure. ■

covering edge

Note that not every v-structure is an immorality, so that two networks with the same immoralities do not necessarily have the same v-structures. For example, two different complete directed graphs always have the same immoralities (none) but different v-structures.

Theorem 3.8

Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be two graphs over $\mathcal{X}$. Then $\mathcal{G}_1$ and $\mathcal{G}_2$ have the same skeleton and the same set of immoralities if and only if they are I-equivalent.

The proof of this (more difficult) result is also left as an exercise (see exercise 3.17).

We conclude with a final characterization of I-equivalence in terms of local operations on the graph structure.

Definition 3.12

covered edge

An edge $X \rightarrow Y$ in a graph $\mathcal{G}$ is said to be covered if $\text{Pa}_Y^{\mathcal{G}} = \text{Pa}_X^{\mathcal{G}} \cup \{X\}$. ■

Theorem 3.9

Two graphs $\mathcal{G}$ and $\mathcal{G}'$ are I-equivalent if and only if there exists a sequence of networks $\mathcal{G} = \mathcal{G}_1, \dots, \mathcal{G}_k = \mathcal{G}'$ that are all I-equivalent to $\mathcal{G}$ such that the only difference between $\mathcal{G}_i$ and $\mathcal{G}_{i+1}$ is a single reversal of a covered edge.

The proof of this theorem is left as an exercise (exercise 3.18).

3.4 From Distributions to Graphs

In the previous sections, we showed that, if P factorizes over $\mathcal{G}$, we can derive a rich set of independence assertions that hold for P by simply examining $\mathcal{G}$. This result immediately leads to the idea that we can use a graph as a way of revealing the structure in a distribution. In particular, we can test for independencies in P by constructing a graph $\mathcal{G}$ that represents P and testing d-separation in $\mathcal{G}$. As we will see, having a graph that reveals the structure in P has other important consequences, in terms of reducing the number of parameters required to specify or learn the distribution, and in terms of the complexity of performing inference on the network.

In this section, we examine the following question: Given a distribution P , to what extent can we construct a graph $\mathcal{G}$ whose independencies are a reasonable surrogate for the independencies

in P ? It is important to emphasize that we will never actually take a fully specified distribution P and construct a graph $\mathcal{G}$ for it: As we discussed, a full joint distribution is much too large to represent explicitly. However, answering this question is an important conceptual exercise, which will help us later on when we try to understand the process of constructing a Bayesian network that represents our model of the world, whether manually or by learning from data.

3.4.1 Minimal I-Maps

One approach to finding a graph that represents a distribution P is simply to take any graph that is an I-map for P . The problem with this naive approach is clear: As we saw in example 3.3, the complete graph is an I-map for any distribution, yet it does not reveal any of the independence structure in the distribution. However, examples such as this one are not very interesting. The graph that we used as an I-map is clearly and trivially unrepresentative of the distribution, in that there are edges that are obviously redundant. This intuition leads to the following definition, which we also define more broadly:

Definition 3.13
minimal I-map

A graph $\mathcal{K}$ is a minimal I-map for a set of independencies $\mathcal{I}$ if it is an I-map for $\mathcal{I}$, and if the removal of even a single edge from $\mathcal{K}$ renders it not an I-map. ■

This notion of an I-map applies to multiple types of graphs, both Bayesian networks and other types of graphs that we will encounter later on. Moreover, because it refers to a set of independencies $\mathcal{I}$, it can be used to define an I-map for a distribution P , by taking $\mathcal{I} = \mathcal{I}(P)$, or to another graph $\mathcal{K}'$, by taking $\mathcal{I} = \mathcal{I}(\mathcal{K}')$.

Recall that definition 3.5 defines a Bayesian network to be a distribution P that factorizes over $\mathcal{G}$, thereby implying that $\mathcal{G}$ is an I-map for P . It is standard to restrict the definition even further, by requiring that $\mathcal{G}$ be a minimal I-map for P .

variable ordering

How do we obtain a minimal I-map for the set of independencies induced by a given distribution P ? The proof of the factorization theorem (theorem 3.1) gives us a procedure, which is shown in algorithm 3.2. We assume we are given a predetermined *variable ordering*, say, $\{X_1, \dots, X_n\}$. We now examine each variable X_i , $i = 1, \dots, n$ in turn. For each X_i , we pick some minimal subset $\mathbf{U}$ of $\{X_1, \dots, X_{i-1}\}$ to be X_i 's parents in $\mathcal{G}$. More precisely, we require that $\mathbf{U}$ satisfy $(X_i \perp \{X_1, \dots, X_{i-1}\} - \mathbf{U} \mid \mathbf{U})$, and that no node can be removed from $\mathbf{U}$ without violating this property. We then set $\mathbf{U}$ to be the parents of X_i .

The proof of theorem 3.1 tells us that, if each node X_i is independent of $X_1, \dots, X_{i-1}$ given its parents in $\mathcal{G}$, then P factorizes over $\mathcal{G}$. We can then conclude from theorem 3.2 that $\mathcal{G}$ is an I-map for P . By construction, $\mathcal{G}$ is minimal, so that $\mathcal{G}$ is a minimal I-map for P .

Note that our choice of $\mathbf{U}$ may not be unique. Consider, for example, a case where two variables A and B are logically equivalent, that is, our distribution P only gives positive probability to instantiations where A and B have the same value. Now, consider a node C that is correlated with A . Clearly, we can choose either A or B to be a parent of C , but having chosen the one, we cannot choose the other without violating minimality. Hence, the minimal parent set $\mathbf{U}$ in our construction is not necessarily unique. However, one can show that, if the distribution is positive (see definition 2.5), that is, if for any instantiation ξ to all the network variables $\mathcal{X}$ we have that $P(\xi) > 0$, then the choice of parent set, given an ordering, is unique. Under this assumption, algorithm 3.2 can produce all minimal I-maps for P : Let $\mathcal{G}$ be any

Algorithm 3.2 Procedure to build a minimal I-map given an ordering

```

Procedure Build-Minimal-I-Map (
     $X_1, \dots, X_n$  // an ordering of random variables in  $\mathcal{X}$ 
     $\mathcal{I}$  // Set of independencies
)
1  Set  $\mathcal{G}$  to an empty graph over  $\mathcal{X}$ 
2  for  $i = 1, \dots, n$ 
3       $\mathbf{U} \leftarrow \{X_1, \dots, X_{i-1}\}$  //  $\mathbf{U}$  is the current candidate for parents of  $X_i$ 
4      for  $\mathbf{U}' \subseteq \{X_1, \dots, X_{i-1}\}$ 
5          if  $\mathbf{U}' \subset \mathbf{U}$  and  $(X_i \perp \{X_1, \dots, X_{i-1}\} - \mathbf{U}' \mid \mathbf{U}') \in \mathcal{I}$  then
6               $\mathbf{U} \leftarrow \mathbf{U}'$ 
7          // At this stage  $\mathbf{U}$  is a minimal set satisfying  $(X_i \perp \{X_1, \dots, X_{i-1}\} - \mathbf{U} \mid \mathbf{U})$ 
8          // Now set  $\mathbf{U}$  to be the parents of  $X_i$ 
9      for  $X_j \in \mathbf{U}$ 
10         Add  $X_j \rightarrow X_i$  to  $\mathcal{G}$ 
11  return  $\mathcal{G}$ 

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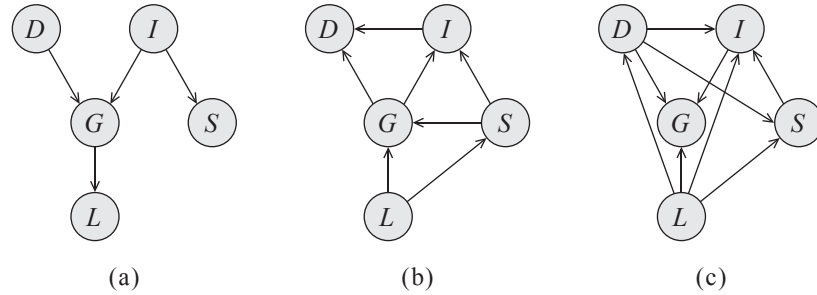


Figure 3.8 Three minimal I-maps for $P_{\mathcal{B}\text{student}}$, induced by different orderings: (a) D, I, S, G, L ; (b) L, S, G, I, D ; (c) L, D, S, I, G .

minimal I-map for P . If we give call Build-Minimal-I-Map with an ordering $\prec$ that is topological for $\mathcal{G}$, then, due to the uniqueness argument, the algorithm must return $\mathcal{G}$.

At first glance, the minimal I-map seems to be a reasonable candidate for capturing the structure in the distribution: It seems that if $\mathcal{G}$ is a minimal I-map for a distribution P , then we should be able to “read off” all of the independencies in P directly from $\mathcal{G}$. Unfortunately, this intuition is false.

Example 3.5

Consider the distribution $P_{\mathcal{B}\text{student}}$, as defined in figure 3.4, and let us go through the process of constructing a minimal I-map for $P_{\mathcal{B}\text{student}}$. We note that the graph G_{student} precisely reflects the independencies in this distribution $P_{\mathcal{B}\text{student}}$ (that is, $\mathcal{I}(P_{\mathcal{B}\text{student}}) = \mathcal{I}(G_{\text{student}})$), so that we can use G_{student} to determine which independencies hold in $P_{\mathcal{B}\text{student}}$.

Our construction process starts with an arbitrary ordering on the nodes; we will go through this

process for three different orderings. Throughout this process, it is important to remember that we are testing independencies relative to the distribution $P_{\mathcal{B}^{\text{student}}}$. We can use G_{student} (figure 3.4) to guide our intuition about which independencies hold in $P_{\mathcal{B}^{\text{student}}}$, but we can always resort to testing these independencies in the joint distribution $P_{\mathcal{B}^{\text{student}}}$.

The first ordering is a very natural one: D, I, S, G, L . We add one node at a time and see which of the possible edges from the preceding nodes are redundant. We start by adding D , then I . We can now remove the edge from D to I because this particular distribution satisfies $(I \perp D)$, so I is independent of D given its other parents (the empty set). Continuing on, we add S , but we can remove the edge from D to S because our distribution satisfies $(S \perp D \mid I)$. We then add G , but we can remove the edge from S to G , because the distribution satisfies $(G \perp S \mid I, D)$. Finally, we add L , but we can remove all edges from D, I, S . Thus, our final output is the graph in figure 3.8a, which is precisely our original network for this distribution.

Now, consider a somewhat less natural ordering: L, S, G, I, D . In this case, the resulting I-map is not quite as natural or as sparse. To see this, let us consider the sequence of steps. We start by adding L to the graph. Since it is the first variable in the ordering, it must be a root. Next, we consider S . The decision is whether to have L as a parent of S . Clearly, we need an edge from L to S , because the quality of the student's letter is correlated with his SAT score in this distribution, and S has no other parents that help render it independent of L . Formally, we have that $(S \perp L)$ does not hold in the distribution. In the next iteration of the algorithm, we introduce G . Now, all possible subsets of $\{L, S\}$ are potential parents set for G . Clearly, G is dependent on L . Moreover, although G is independent of S given I , it is not independent of S given L . Hence, we must add the edge between S and G . Carrying out the procedure, we end up with the graph shown in figure 3.8b.

Finally, consider the ordering: L, D, S, I, G . In this case, a similar analysis results in the graph shown in figure 3.8c, which is almost a complete graph, missing only the edge from S to G , which we can remove because G is independent of S given I . ■

Note that the graphs in figure 3.8b,c really are minimal I-maps for this distribution. However, they fail to capture some or all of the independencies that hold in the distribution. Thus, they show that the fact that $\mathcal{G}$ is a minimal I-map for P is far from a guarantee that $\mathcal{G}$ captures the independence structure in P .

3.4.2 Perfect Maps

We aim to find a graph $\mathcal{G}$ that precisely captures the independencies in a given distribution P .

Definition 3.14
perfect map

We say that a graph $\mathcal{K}$ is a perfect map (P-map) for a set of independencies $\mathcal{I}$ if we have that $\mathcal{I}(\mathcal{K}) = \mathcal{I}$. We say that $\mathcal{K}$ is a perfect map for P if $\mathcal{I}(\mathcal{K}) = \mathcal{I}(P)$. ■

If we obtain a graph $\mathcal{G}$ that is a P-map for a distribution P , then we can (by definition) read the independencies in P directly from $\mathcal{G}$. By construction, our original graph G_{student} is a P-map for $P_{\mathcal{B}^{\text{student}}}$.

If our goal is to find a perfect map for a distribution, an immediate question is whether every distribution has a perfect map. Unfortunately, the answer is no, and for several reasons. The first type of counterexample involves regularity in the parameterization of the distribution that cannot be captured in the graph structure.

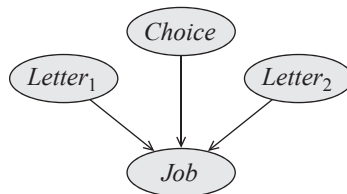


Figure 3.9 Network for the OneLetter example

Example 3.6

Consider a joint distribution P over 3 random variables X, Y, Z such that:

$$P(x, y, z) = \begin{cases} 1/12 & x \oplus y \oplus z = \text{false} \\ 1/6 & x \oplus y \oplus z = \text{true} \end{cases}$$

where $\oplus$ is the XOR (exclusive OR) function. A simple calculation shows that $(X \perp Y) \in \mathcal{I}(P)$, and that Z is not independent of X given Y or of Y given X . Hence, one minimal I-map for this distribution is the network $X \rightarrow Z \leftarrow Y$, using a deterministic XOR for the CPD of Z . However, this network is not a perfect map; a precisely analogous calculation shows that $(X \perp Z) \in \mathcal{I}(P)$, but this conclusion is not supported by a d -separation analysis. ■

Thus, we see that deterministic relationships can lead to distributions that do not have a P-map. Additional examples arise as a consequence of other regularities in the CPD.

Example 3.7

Consider a slight elaboration of our Student example. During his academic career, our student George has taken both Econ101 and CS102. The professors of both classes have written him letters, but the recruiter at Acme Consulting asks for only a single recommendation. George's chance of getting the job depends on the quality of the letter he gives the recruiter. We thus have four random variables: $L1$ and $L2$, corresponding to the quality of the recommendation letters for Econ101 and CS102 respectively; C , whose value represents George's choice of which letter to use; and J , representing the event that George is hired by Acme Consulting.

The obvious minimal I-map for this distribution is shown in figure 3.9. Is this a perfect map? Clearly, it does not reflect independencies that are not at the variable level. In particular, we have that $(L1 \perp J \mid C = 2)$. However, this limitation is not surprising; by definition, a BN structure makes independence assertions only at the level of variables. (We return to this issue in section 5.2.2.) However, our problems are not limited to these finer-grained independencies. Some thought reveals that, in our target distribution, we also have that $(L1 \perp L2 \mid C, J)$! This independence is not implied by d -separation, because the v -structure $L1 \rightarrow J \leftarrow L2$ is enabled. However, we can convince ourselves that the independence holds using reasoning by cases. If $C = 1$, then there is no dependence of J on $L2$. Intuitively, the edge from $L2$ to J disappears, eliminating the trail between $L1$ and $L2$, so that $L1$ and $L2$ are independent in this case. A symmetric analysis applies in the case that $C = 2$. Thus, in both cases, we have that $L1$ and $L2$ are independent. This independence assertion is not captured by our minimal I-map, which is therefore not a P-map. ■

A different class of examples is not based on structure within a CPD, but rather on symmetric variable-level independencies that are not naturally expressed within a Bayesian network.

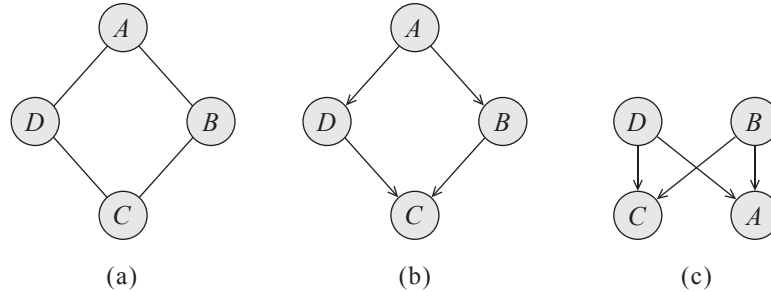


Figure 3.10 Attempted Bayesian network models for the Misconception example: (a) Study pairs over four students. (b) First attempt at a Bayesian network model. (c) Second attempt at a Bayesian network model.

A second class of distributions that do not have a perfect map are those for which the independence assumptions imposed by the structure of Bayesian networks is simply not appropriate.

Example 3.8

Consider a scenario where we have four students who get together in pairs to work on the homework for a class. For various reasons, only the following pairs meet: Alice and Bob; Bob and Charles; Charles and Debbie; and Debbie and Alice. (Alice and Charles just can't stand each other, and Bob and Debbie had a relationship that ended badly.) The study pairs are shown in figure 3.10a.

In this example, the professor accidentally misspoke in class, giving rise to a possible misconception among the students in the class. Each of the students in the class may subsequently have figured out the problem, perhaps by thinking about the issue or reading the textbook. In subsequent study pairs, he or she may transmit this newfound understanding to his or her study partners. We therefore have four binary random variables, representing whether the student has the misconception or not. We assume that for each $X \in \{A, B, C, D\}$, x^1 denotes the case where the student has the misconception, and x^0 denotes the case where he or she does not.

Because Alice and Charles never speak to each other directly, we have that A and C are conditionally independent given B and D . Similarly, B and D are conditionally independent given A and C . Can we represent this distribution (with these independence properties) using a BN? One attempt is shown in figure 3.10b. Indeed, it encodes the independence assumption that $(A \perp C \mid \{B, D\})$. However, it also implies that B and D are independent given only A , but dependent given both A and C . Hence, it fails to provide a perfect map for our target distribution. A second attempt, shown in figure 3.10c, is equally unsuccessful. It also implies that $(A \perp C \mid \{B, D\})$, but it also implies that B and D are marginally independent. It is clear that all other candidate BN structures are also flawed, so that this distribution does not have a perfect map. ■

3.4.3 Finding Perfect Maps ★

Earlier we discussed an algorithm for finding minimal I-maps. We now consider an algorithm for finding a perfect map (P-map) of a distribution. Because the requirements from a P-map are stronger than the ones we require from an I-map, the algorithm will be more involved.

Throughout the discussion in this section, we assume that P has a P-map. In other words, there is an unknown DAG $\mathcal{G}^*$ that is P-map of P . Since $\mathcal{G}^*$ is a P-map, we will interchangeably refer to independencies in P and in $\mathcal{G}^*$ (since these are the same). We note that the algorithms we describe do fail when they are given a distribution that does not have a P-map. We discuss this issue in more detail later.

Thus, our goal is to identify $\mathcal{G}^*$ from P . One obvious difficulty that arises when we consider this goal is that $\mathcal{G}^*$ is, in general, not uniquely identifiable from P . A P-map of a distribution, if one exists, is generally not unique: As we saw, for example, in figure 3.5, multiple graphs can encode precisely the same independence assumptions. However, the P-map of a distribution is unique up to I-equivalence between networks. That is, a distribution P can have many P-maps, but all of them are I-equivalent.

If we require that a P-map construction algorithm return a single network, the output we get may be some arbitrary member of the I-equivalence class of $\mathcal{G}^*$. A more correct answer would be to return the entire equivalence class, thus avoiding an arbitrary commitment to a possibly incorrect structure. Of course, we do not want our algorithm to return a (possibly very large) set of distinct networks as output. Thus, one of our tasks in this section is to develop a compact representation of an entire equivalence class of DAGs. As we will see later in the book, this representation plays a useful role in other contexts as well.

This formulation of the problem points us toward a solution. Recall that, according to theorem 3.8, two DAGs are I-equivalent if they share the same skeleton and the same set of immoralities. Thus, we can construct the I-equivalence class for $\mathcal{G}^*$ by determining its skeleton and its immoralities from the independence properties of the given distribution P . We then use both of these components to build a representation of the equivalence class.

3.4.3.1 Identifying the Undirected Skeleton

At this stage we want to construct an undirected graph S that contains an edge $X—Y$ if X and Y are adjacent in $\mathcal{G}^*$; that is, if either $X \rightarrow Y$ or $Y \rightarrow X$ is an edge in $\mathcal{G}^*$.

The basic idea is to use independence queries of the form $(X \perp Y \mid U)$ for different sets of variables U . This idea is based on the observation that if X and Y are adjacent in $\mathcal{G}^*$, we cannot separate them with any set of variables.

Lemma 3.1

Let $\mathcal{G}^$ be a P-map of a distribution $\mathcal{P}$, and let X and Y be two variables such that $X \rightarrow Y$ is in $\mathcal{G}^*$. Then, $P \not\models (X \perp Y \mid U)$ for any set U that does not include X and Y .*

PROOF Assume that that $X \rightarrow Y \in \mathcal{G}^*$, and let U be a set of variables. According to d-separation the trail $X \rightarrow Y$ cannot be blocked by the evidence set U . Thus, X and Y are not d-separated by U . Since $\mathcal{G}^*$ is a P-map of P , we have that $P \not\models (X \perp Y \mid U)$. ■

This lemma implies that if X and Y are adjacent in $\mathcal{G}^*$, all conditional independence queries that involve both of them would fail. Conversely, if X and Y are not adjacent in $\mathcal{G}^*$, we would hope to be able to find a set of variables that makes these two variables conditionally independent. Indeed, as we now show, we can provide a precise characterization of such a set:

Lemma 3.2

Let $\mathcal{G}^$ be an I-map of a distribution P , and let X and Y be two variables that are not adjacent in $\mathcal{G}^*$. Then either $P \models (X \perp Y \mid \text{Pa}_X^{\mathcal{G}^*})$ or $P \models (X \perp Y \mid \text{Pa}_Y^{\mathcal{G}^*})$.*

The proof is left as an exercise (exercise 3.19).

witness

Thus, if X and Y are not adjacent in $\mathcal{G}^*$, then we can find a set $\mathbf{U}$ so that $\mathcal{P} \models (X \perp Y \mid \mathbf{U})$. We call this set $\mathbf{U}$ a *witness* of their independence. Moreover, the lemma shows that we can find a witness of bounded size. Thus, if we assume that $\mathcal{G}^*$ has bounded indegree, say less than or equal to d , then we do not need to consider witness sets larger than d .

Algorithm 3.3 Recovering the undirected skeleton for a distribution P that has a P-map

```

Procedure Build-PMap-Skeleton (
   $\mathcal{X} = \{X_1, \dots, X_n\}$ ,    // Set of random variables
   $P$ ,    // Distribution over  $\mathcal{X}$ 
   $d$     // Bound on witness set
)
1   Let  $\mathcal{H}$  be the complete undirected graph over  $\mathcal{X}$ 
2   for  $X_i, X_j$  in  $\mathcal{X}$ 
3      $\mathbf{U}_{X_i, X_j} \leftarrow \emptyset$ 
4     for  $\mathbf{U} \in \text{Witnesses}(X_i, X_j, \mathcal{H}, d)$ 
5       // Consider  $\mathbf{U}$  as a witness set for  $X_i, X_j$ 
6       if  $P \models (X_i \perp X_j \mid \mathbf{U})$  then
7          $\mathbf{U}_{X_i, X_j} \leftarrow \mathbf{U}$ 
8         Remove  $X_i - X_j$  from  $\mathcal{H}$ 
9         break
10  return  $(\mathcal{H}, \{\mathbf{U}_{X_i, X_j} : i, j \in \{1, \dots, n\}\})$ 

```

With these tools in hand, we can now construct an algorithm for building a skeleton of $\mathcal{G}^*$, shown in algorithm 3.3. For each pair of variables, we consider all potential witness sets and test for independence. If we find a witness that separates the two variables, we record it (we will soon see why) and move on to the next pair of variables. If we do not find a witness, then we conclude that the two variables are adjacent in $\mathcal{G}^*$ and add them to the skeleton. The list $\text{Witnesses}(X_i, X_j, \mathcal{H}, d)$ in line 4 specifies the set of possible witness sets that we consider for separating X_i and X_j . From our earlier discussion, if we assume a bound d on the indegree, then we can restrict attention to sets $\mathbf{U}$ of size at most d . Moreover, using the same analysis, we saw that we have a witness that consists either of the parents of X_i or of the parents of X_j . In the first case, we can restrict attention to sets $\mathbf{U} \subseteq \text{Nb}_{X_i}^{\mathcal{H}} - \{X_j\}$, where $\text{Nb}_{X_i}^{\mathcal{H}}$ are the neighbors of X_i in the current graph $\mathcal{H}$; in the second, we can similarly restrict attention to sets $\mathbf{U} \subseteq \text{Nb}_{X_j}^{\mathcal{H}} - \{X_i\}$. Finally, we note that if $\mathbf{U}$ separates X_i and X_j , then also many of $\mathbf{U}$'s supersets will separate X_i and X_j . Thus, we search the set of possible witnesses in order of increasing size.

This algorithm will recover the correct skeleton given that $\mathcal{G}^*$ is a P-map of P and has bounded indegree d . If P does not have a P-map, then the algorithm can fail; see exercise 3.22. This algorithm has complexity of $O(n^{d+2})$ since we consider $O(n^2)$ pairs, and for each we perform $O((n-2)^d)$ independence tests. We greatly reduce the number of independence tests by ordering potential witnesses accordingly, and by aborting the inner loop once we find a witness for a pair (after line 9). However, for pairs of variables that are directly connected in the skeleton, we still need to evaluate all potential witnesses.

Algorithm 3.4 Marking immoralities in the construction of a perfect map

```

Procedure Mark-Immoralities (
     $\mathcal{X} = \{X_1, \dots, X_n\}$ ,
     $S$  // Skeleton
     $\{U_{X_i, X_j} : 1 \leq i, j \leq n\}$  // Witnesses found by Build-PMMap-Skeleton
)
1   $\mathcal{K} \leftarrow S$ 
2  for  $X_i, X_j, X_k$  such that  $X_i - X_j - X_k \in S$  and  $X_i - X_k \notin S$ 
3      //  $X_i - X_j - X_k$  is a potential immorality
4      if  $X_j \notin U_{X_i, X_k}$  then
5          Add the orientations  $X_i \rightarrow X_j$  and  $X_j \leftarrow X_k$  to  $\mathcal{K}$ 
6  return  $\mathcal{K}$ 

```

3.4.3.2 Identifying Immoralities

At this stage we have reconstructed the undirected skeleton S using Build-PMMap-Skeleton. Now, we want to reconstruct edge direction. The main cue for learning about edge directions in $\mathcal{G}^*$ are immoralities. As shown in theorem 3.8, all DAGs in the equivalence class of $\mathcal{G}^*$ share the same set of immoralities. Thus, our goal is to consider *potential immoralities* in the skeleton and for each one determine whether it is indeed an immorality. A triplet of variables X, Z, Y is a *potential immorality* if the skeleton contains $X - Z - Y$ but does not contain an edge between X and Y . If such a triplet is indeed an immorality in $\mathcal{G}^*$, then X and Y cannot be independent given Z . Nor will they be independent given a set U that contains Z . More precisely,

potential
immorality

Proposition 3.1

Let $\mathcal{G}^*$ be a P-map of a distribution P , and let X, Y and Z be variables that form an immorality $X \rightarrow Z \leftarrow Y$. Then, $P \not\models (X \perp Y \mid U)$ for any set U that contains Z .

PROOF Let U be a set of variables that contains Z . Since Z is observed, the trail $X \rightarrow Z \leftarrow Y$ is active, and so X and Y are not d-separated in $\mathcal{G}^*$. Since $\mathcal{G}^*$ is a P-map of P , we have that $P^* \not\models (X \perp Y \mid U)$. ■

What happens in the complementary situation? Suppose $X - Z - Y$ in the skeleton, but is not an immorality. This means that one of the following three cases is in $\mathcal{G}^*$: $X \rightarrow Z \rightarrow Y$, $Y \rightarrow Z \rightarrow X$, or $X \leftarrow Z \rightarrow Y$. In all three cases, X and Y are d-separated only if Z is observed.

Proposition 3.2

Let $\mathcal{G}^*$ be a P-map of a distribution P , and let the triplet X, Y, Z be a potential immorality in the skeleton of $\mathcal{G}^*$, such that $X \rightarrow Z \leftarrow Y$ is not in $\mathcal{G}^*$. If U is such that $P \models (X \perp Y \mid U)$, then $Z \in U$.

PROOF Consider all three configurations of the trail $X \rightleftharpoons Z \rightleftharpoons Y$. In all three, Z must be observed in order to block the trail. Since $\mathcal{G}^*$ is a P-map of P , we have that if $P \models (X \perp Y \mid U)$, then $Z \in U$. ■

Combining these two results, we see that a potential immorality $X - Z - Y$ is an immorality if and only if Z is not in the witness set(s) for X and Y . That is, if $X - Z - Y$ is an immorality,

then proposition 3.1 shows that Z is not in any witness set U ; conversely, if $X-Z-Y$ is not an immorality, the Z must be in every witness set U . Thus, we can use the specific witness set $U_{X,Y}$ that we recorded for X, Y in order to determine whether this triplet is an immorality or not: we simply check whether $Z \in U_{X,Y}$. If $Z \notin U_{X,Y}$, then we declare the triplet an immorality. Otherwise, we declare that it is not an immorality. The Mark-Immoralities procedure shown in algorithm 3.4 summarizes this process.

3.4.3.3 Representing Equivalence Classes

Once we have the skeleton and identified the immoralities, we have a specification of the equivalence class of $\mathcal{G}^*$. For example, to test if $\mathcal{G}$ is equivalent to $\mathcal{G}^*$ we can check whether it has the same skeleton as $\mathcal{G}^*$ and whether it agrees on the location of the immoralities.

The description of an equivalence class using only the skeleton and the set of immoralities is somewhat unsatisfying. For example, we might want to know whether the fact that our network is in the equivalence class implies that there is an arc $X \rightarrow Y$. Although the definition does tell us whether there is some edge between X and Y , it leaves the direction unresolved. In other cases, however, the direction of an edge is fully determined, for example, by the presence of an immorality. To encode both of these cases, we use a graph that allows both directed and undirected edges, as defined in section 2.2. Indeed, as we show, the chain graph, or PDAG, representation (definition 2.21) provides precisely the right framework.

Definition 3.15

class PDAG

Let $\mathcal{G}$ be a DAG. A chain graph $\mathcal{K}$ is a class PDAG of the equivalence class of $\mathcal{G}$ if shares the same skeleton as $\mathcal{G}$, and contains a directed edge $X \rightarrow Y$ if and only if all $\mathcal{G}'$ that are I-equivalent to $\mathcal{G}$ contain the edge $X \rightarrow Y$.² ■

In other words, a class PDAG represents potential edge orientations in the equivalence classes. If the edge is directed, then all the members of the equivalence class agree on the orientation of the edge. If the edge is undirected, there are two DAGs in the equivalence class that disagree on the orientation of the edge.

For example, the networks in figure 3.5a–(c) are I-equivalent. The class PDAG of this equivalence class is the graph $X-Z-Y$, since both edges can be oriented in either direction in some member of the equivalence class. Note that, although both edges in this PDAG are undirected, not all joint orientations of these edges are in the equivalence class. As discussed earlier, setting the orientations $X \rightarrow Z \leftarrow Y$ results in the network of figure 3.5d, which does not belong this equivalence class. More generally, if the class PDAG has k undirected edges, the equivalence class can contain at most 2^k networks, but the actual number can be much smaller.

Can we effectively construct the class PDAG $\mathcal{K}$ for $\mathcal{G}^*$ from the reconstructed skeleton and immoralities? Clearly, edges involved in immoralities must be directed in $\mathcal{K}$. The obvious question is whether $\mathcal{K}$ can contain directed edges that are not involved in immoralities. In other words, can there be additional edges whose direction is necessarily the same in every member of the equivalence class? To understand this issue better, consider the following example:

Example 3.9

Consider the DAG of figure 3.11a. This DAG has a single immorality $A \rightarrow C \leftarrow B$. This immorality implies that the class PDAG of this DAG must have the arcs $A \rightarrow C$ and $B \rightarrow C$ directed, as

2. For consistency with standard terminology, we use the PDAG terminology when referring to the chain graph representing an I-equivalence class.

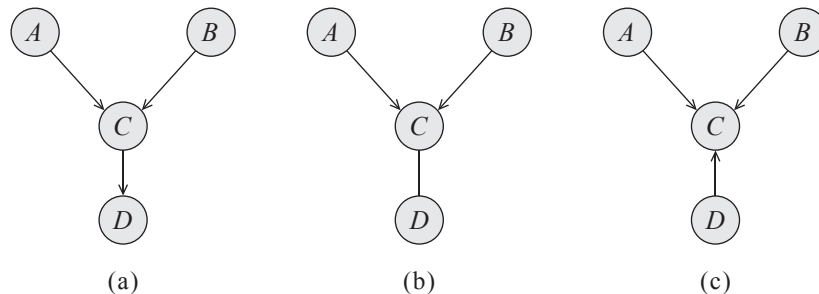


Figure 3.11 Simple example of compelled edges in the representation of an equivalence class. (a) Original DAG $\mathcal{G}^*$. (b) Skeleton of $\mathcal{G}^*$ annotated with immoralities. (c) a DAG that is not equivalent to $\mathcal{G}^*$.

shown in figure 3.11b. This PDAG representation suggests that the edge $C—D$ can assume either orientation. Note, however, that the DAG of figure 3.11c, where we orient the edge between C and D as $D \rightarrow C$, contains additional immoralities (that is, $A \rightarrow C \leftarrow D$ and $B \rightarrow C \leftarrow D$). Thus, this DAG is not equivalent to our original DAG.

In this example, there is only one possible orientation of $C—D$ that is consistent with the finding that $A—C—D$ is not an immorality. Thus, we conclude that the class PDAG for the DAG of figure 3.11a is simply the DAG itself. In other words, the equivalence class of this DAG is a singleton. ■

As this example shows, a negative result in an immorality test also provides information about edge orientation. In particular, in any case where the PDAG $\mathcal{K}$ contains a structure $X \rightarrow Y—Z$ and there is no edge from X to Z , then we must orient the edge $Y \rightarrow Z$, for otherwise we would create an immorality $X \rightarrow Y \leftarrow Z$.

Some thought reveals that there are other local configurations of edges where some ways of orienting edges are inconsistent, forcing a particular direction for an edge. Each such configuration can be viewed as a local constraint on edge orientation, give rise to a rule that can be used to orient more edges in the PDAG. Three such rules are shown in figure 3.12.

Let us understand the intuition behind these rules. Rule R1 is precisely the one we discussed earlier. Rule R2 is derived from the standard acyclicity constraint: If we have the directed path $X \rightarrow Y \rightarrow Z$, and an undirected edge $X—Z$, we cannot direct the edge $X \leftarrow Z$ without creating a cycle. Hence, we can conclude that the edge must be directed $X \rightarrow Z$. The third rule seems a little more complex, but it is also easily motivated. Assume, by contradiction, that we direct the edge $Z \rightarrow X$. In this case, we cannot direct the edge $X—Y_1$ as $X \rightarrow Y_1$ without creating a cycle; thus, we must have $Y_1 \rightarrow X$. Similarly, we must have $Y_2 \rightarrow X$. But, in this case, $Y_1 \rightarrow X \leftarrow Y_2$ forms an immorality (as there is no edge between Y_1 and Y_2), which contradicts the fact that the edges $X—Y_1$ and $X—Y_2$ are undirected in the original PDAG.

These three rules can be applied constructively in an obvious way: A rule applies to a PDAG whenever the induced subgraph on a subset of variables exactly matches the graph on the left-hand side of the rule. In that case, we modify this subgraph to match the subgraph on the right-hand side of the rule. Note that, by applying one rule and orienting a previously undirected edge, we create a new graph. This might create a subgraph that matches the antecedent of a rule, enforcing the orientation of additional edges. This process, however, must terminate at

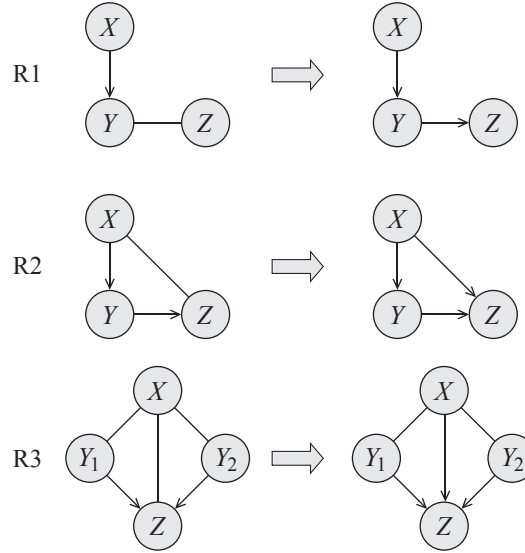


Figure 3.12 Rules for orienting edges in PDAG. Each rule lists a configuration of edges before and after an application of the rule.

some point (since we are only adding orientations at each step, and the number of edges is finite). This implies that iterated application of this local constraint to the graph (a process known as *constraint propagation*) is guaranteed to converge.

constraint
propagation

Algorithm 3.5 Finding the class PDAG characterizing the P-map of a distribution P

```

Procedure Build-PDAG (
   $\mathcal{X} = \{X_1, \dots, X_n\}$  // A specification of the random variables
   $P$  // Distribution of interest
)
1   $S, \{U_{X_i, X_j}\} \leftarrow \text{Build-PMap-Skeleton}(\mathcal{X}, P)$ 
2   $\mathcal{K} \leftarrow \text{Find-Immoralities}(\mathcal{X}, S, \{U_{X_i, X_j}\})$ 
3  while not converged
4    Find a subgraph in  $\mathcal{K}$  matching the left-hand side of a rule R1–R3
5    Replace the subgraph with the right-hand side of the rule
6  return  $\mathcal{K}$ 

```

Algorithm 3.5 implements this process. It builds an initial graph using Build-PMap-Skeleton and Mark-Immoralities, and then iteratively applies the three rules until convergence, that is, until we cannot find a subgraph that matches a left-hand side of any of the rules.

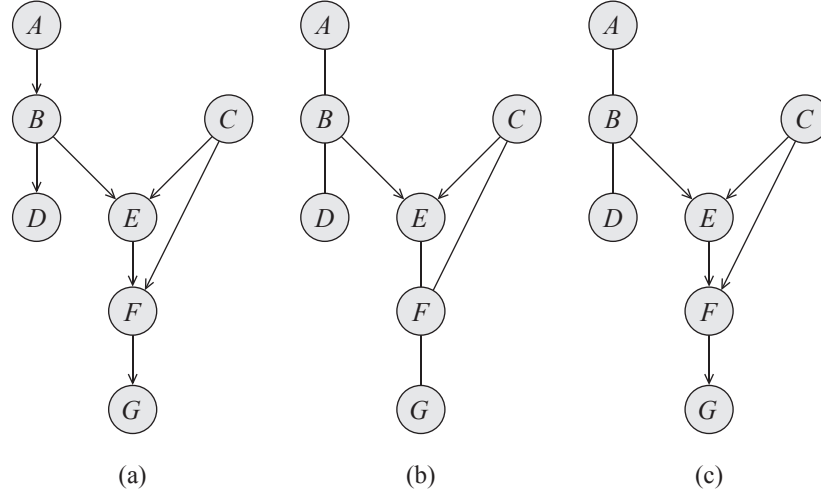


Figure 3.13 More complex example of compelled edges in the representation of an equivalence class. (a) Original DAG $\mathcal{G}^*$. (b) Skeleton of $\mathcal{G}^*$ annotated with immoralities. (c) Complete PDAG representation of the equivalence class of $\mathcal{G}^*$.

Example 3.10

Consider the DAG shown in figure 3.13a. After checking for immoralities, we find the graph shown in figure 3.13b. Now, we can start applying the preceding rules. For example, consider the variables B , E , and F . They induce a subgraph that matches the left-hand side of rule R1. Thus, we orient the edge between E and F to $E \rightarrow F$. Now, consider the variables C , E , and F . Their induced subgraph matches the left-hand side of rule R2, so we now orient the edge between C and F to $C \rightarrow F$. At this stage, if we consider the variables E , F , G , we can apply the rule R1, and orient the edge $F \rightarrow G$. (Alternatively, we could have arrived at the same orientation using C , F , and G .) The resulting PDAG is shown in figure 3.13c. ■

It seems fairly obvious that this algorithm is guaranteed to be sound: Any edge that is oriented by this procedure is, indeed, directed in exactly the same way in all of the members of the equivalence class. Much more surprising is the fact that it is also complete: Repeated application of these three local rules is guaranteed to capture all edge orientations in the equivalence class, without the need for additional global constraints. More precisely, we can prove that this algorithm produces the correct class PDAG for the distribution P :

Theorem 3.10

Let P be a distribution that has a P-map $\mathcal{G}^*$, and let $\mathcal{K}$ be the PDAG returned by $\text{Build-PDAG}(\mathcal{X}, P)$. Then, $\mathcal{K}$ is a class PDAG of $\mathcal{G}^*$.

The proof of this theorem can be decomposed into several aspects of correctness. We have already established the correctness of the skeleton found by $\text{Build-PMap-Skeleton}$. Thus, it remains to show that the directionality of the edges is correct. Specifically, we need to establish three basic facts:

- **Acyclicity:** The graph returned by $\text{Build-PDAG}(\mathcal{X}, P)$ is acyclic.

- **Soundness:** If $X \rightarrow Y \in \mathcal{K}$, then $X \rightarrow Y$ appears in all DAGs in $\mathcal{G}^*$'s I-equivalence class.
- **Completeness:** If $X \text{---} Y \in \mathcal{K}$, then we can find a DAG $\mathcal{G}$ that is I-equivalent to $\mathcal{G}^*$ such that $X \rightarrow Y \in \mathcal{G}$.

The last condition establishes completeness, since there is no constraint on the direction of the arc. In other words, the same condition can be used to prove the existence of a graph with $X \rightarrow Y$ and of a graph with $Y \rightarrow X$. Hence, it shows that either direction is possible within the equivalence class.

We begin with the soundness of the procedure.

Proposition 3.3 *Let P be a distribution that has a P -map $\mathcal{G}^*$, and let $\mathcal{K}$ be the graph returned by $\text{Build-PDAG}(\mathcal{X}, P)$. Then, if $X \rightarrow Y \in \mathcal{K}$, then $X \rightarrow Y$ appears in all DAGs in the I-equivalence class of $\mathcal{G}^*$.*

The proof is left as an exercise (exercise 3.23).

Next, we consider the acyclicity of the graph. We start by proving a property of graphs returned by the procedure. (Note that, once we prove that the graph returned by the procedure is the correct PDAG, it will follow that this property also holds for class PDAGs in general.)

Proposition 3.4 *Let $\mathcal{K}$ be the graph returned by Build-PDAG . Then, if $X \rightarrow Y \in \mathcal{K}$ and $Y \text{---} Z \in \mathcal{K}$, then $X \rightarrow Z \in \mathcal{K}$.*

The proof is left as an exercise (exercise 3.24).

Proposition 3.5 *Let $\mathcal{K}$ be the chain graph returned by Build-PDAG . Then $\mathcal{K}$ is acyclic.*

PROOF Suppose, by way of contradiction, that $\mathcal{K}$ contains a cycle. That is, there is a (partially) directed path $X_1 \rightleftharpoons X_2 \rightleftharpoons \dots \rightleftharpoons X_n \rightleftharpoons X_1$. Without loss of generality, assume that this path is the shortest cycle in $\mathcal{K}$. We claim that the path cannot contain an undirected edge. To see that, suppose that the path contains the triplet $X_i \rightarrow X_{i+1} \text{---} X_{i+2}$. Then, invoking proposition 3.4, we have that $X_i \rightarrow X_{i+2} \in \mathcal{K}$, and thus, we can construct a shorter path without X_{i+1} that contains the edge $X_i \rightarrow X_{i+2}$. At this stage, we have a directed cycle $X_1 \rightarrow X_2 \rightarrow \dots \rightarrow X_n \rightarrow X_1$. Using proposition 3.3, we conclude that this cycle appears in any DAG in the I-equivalence class, and in particular in $\mathcal{G}^*$. This conclusion contradicts the assumption that $\mathcal{G}^*$ is acyclic. It follows that $\mathcal{K}$ is acyclic. ■

The final step is the completeness proof. Again, we start by examining a property of the graph $\mathcal{K}$.

Proposition 3.6 *The PDAG $\mathcal{K}$ returned by Build-PDAG is necessarily chordal.*

The proof is left as an exercise (exercise 3.25).

This property allows us to characterize the structure of the PDAG $\mathcal{K}$ returned by Build-PDAG . Recall that, since $\mathcal{K}$ is an undirected chain graph, we can partition $\mathcal{X}$ into chain components $\mathbf{K}_1, \dots, \mathbf{K}_\ell$, where each chain component contains variables that are connected by undirected edges (see definition 2.21). It turns out that, in an undirected chordal graph, we can orient any edge in any direction without creating an immorality.

Proposition 3.7 *Let $\mathcal{K}$ be a undirected chordal graph over $\mathcal{X}$, and let $X, Y \in \mathcal{X}$. Then, there is a DAG $\mathcal{G}$ such that*

- (a) *The skeleton of $\mathcal{G}$ is $\mathcal{K}$.*
- (b) *$\mathcal{G}$ does not contain immoralities.*
- (c) *$X \rightarrow Y \in \mathcal{G}$.*

The proof of this proposition requires some additional machinery that we introduce in chapter 4, so we defer the proof to that chapter.

Using this proposition, we see that we can orient edges in the chain component K_j without introducing immoralities within the component. We still need to ensure that orienting an edge $X-Y$ within a component cannot introduce an immorality involving edges from outside the component. To see why this situation cannot occur, suppose we orient the edge $X \rightarrow Y$, and suppose that $Z \rightarrow Y \in \mathcal{K}$. This seems like a potential immorality. However, applying proposition 3.4, we see that since $Z \rightarrow Y$ and $Y-X$ are in $\mathcal{K}$, then so must be $Z \rightarrow X$. Since Z is a parent of both X and Y , we have that $X \rightarrow Y \leftarrow Z$ is not an immorality. This argument applies to any edge we orient within an undirected component, and thus no new immoralities are introduced.

With these tools, we can complete the completeness proof of Build-PDAG.

Proposition 3.8 *Let P be a distribution that has a P-map $\mathcal{G}^*$, and let $\mathcal{K}$ be the graph returned by Build-PDAG($\mathcal{X}, P$). If $X-Y \in \mathcal{K}$, then we can find a DAG $\mathcal{G}$ that is I-equivalent to $\mathcal{G}^*$ such that $X \rightarrow Y \in \mathcal{G}$.*

PROOF Suppose we have an undirected edge $X-Y \in \mathcal{K}$. We want to show that there is a DAG $\mathcal{G}$ that has the same skeleton and immoralities as $\mathcal{K}$ such that $X \rightarrow Y \in \mathcal{G}$. If can build such a graph $\mathcal{G}$, then clearly it is in the I-equivalence class of $\mathcal{G}^*$.

The construction is simple. We start with the chain component that contains $X-Y$, and use proposition 3.7 to orient the edges in the component so that $X \rightarrow Y$ is in the resulting DAG. Then, we use the same construction to orient all other chain components. Since the chain components are ordered and acyclic, and our orientation of each chain component is acyclic, the resulting directed graph is acyclic. Moreover, as shown, the new orientation in each component does not introduce immoralities. Thus, the resulting DAG has exactly the same skeleton and immoralities as $\mathcal{K}$. ■

3.5 Summary

In this chapter, we discussed the issue of specifying a high-dimensional joint distribution compactly by exploiting its independence properties. We provided two complementary definitions of a Bayesian network. The first is as a directed graph $\mathcal{G}$, annotated with a set of conditional probability distributions $P(X_i \mid \text{Pa}_{X_i})$. The network together with the CPDs define a distribution via the chain rule for Bayesian networks. In this case, we say that P factorizes over $\mathcal{G}$. We also defined the independence assumptions associated with the graph: the local independencies, the set of basic independence assumptions induced by the network structure; and the larger set of global independencies that are derived from the d-separation criterion. We showed the

4

Undirected Graphical Models

So far, we have dealt only with directed graphical models, or Bayesian networks. These models are useful because both the structure and the parameters provide a natural representation for many types of real-world domains. In this chapter, we turn our attention to another important class of graphical models, defined on the basis of undirected graphs.

As we will see, these models are useful in modeling a variety of phenomena where one cannot naturally ascribe a directionality to the interaction between variables. Furthermore, the undirected models also offer a different and often simpler perspective on directed models, in terms of both the independence structure and the inference task. We also introduce a combined framework that allows both directed and undirected edges. We note that, unlike our results in the previous chapter, some of the results in this chapter require that we restrict attention to distributions over discrete state spaces.

4.1 The Misconception Example

To motivate our discussion of an alternative graphical representation, let us reexamine the Misconception example of section 3.4.2 (example 3.8). In this example, we have four students who get together in pairs to work on their homework for a class. The pairs that meet are shown via the edges in the undirected graph of figure 3.10a.

As we discussed, we intuitively want to model a distribution that satisfies $(A \perp C \mid \{B, D\})$ and $(B \perp D \mid \{A, C\})$, but no other independencies. As we showed, these independencies cannot be naturally captured in a Bayesian network: any Bayesian network I-map of such a distribution would necessarily have extraneous edges, and it would not capture at least one of the desired independence statements. More broadly, a Bayesian network requires that we ascribe a directionality to each influence. In this case, the interactions between the variables seem symmetrical, and we would like a model that allows us to represent these correlations without forcing a specific direction to the influence.

A representation that implements this intuition is an undirected graph. As in a Bayesian network, the nodes in the graph of a *Markov network* represent the variables, and the edges correspond to a notion of direct probabilistic interaction between the neighboring variables — an interaction that is not mediated by any other variable in the network. In this case, the graph of figure 3.10, which captures the interacting pairs, is precisely the Markov network structure that captures our intuitions for this example. As we will see, this similarity is not an accident.

$\phi_1(A, B)$			$\phi_2(B, C)$			$\phi_3(C, D)$			$\phi_4(D, A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	b^1	c^1	100	c^1	d^1	1	d^1	a^1	100
(a)			(b)			(c)			(d)		

Figure 4.1 Factors for the Misconception example

The remaining question is how to parameterize this undirected graph. Because the interaction is not directed, there is no reason to use a standard CPD, where we represent the distribution over one node given others. Rather, we need a more symmetric parameterization. Intuitively, what we want to capture is the *affinities* between related variables. For example, we might want to represent the fact that Alice and Bob are more likely to agree than to disagree. We associate with A, B a general-purpose function, also called a *factor*:

Definition 4.1

factor

scope

Let $\mathbf{D}$ be a set of random variables. We define a factor ϕ to be a function from $\text{Val}(\mathbf{D})$ to $\mathbb{R}$. A factor is nonnegative if all its entries are nonnegative. The set of variables $\mathbf{D}$ is called the scope of the factor and denoted $\text{Scope}[\phi]$. ■

Unless stated otherwise, we restrict attention to nonnegative factors.

In our example, we have a factor $\phi_1(A, B) : \text{Val}(A, B) \mapsto \mathbb{R}^+$. The value associated with a particular assignment a, b denotes the affinity between these two values: the higher the value $\phi_1(a, b)$, the more compatible these two values are.

Figure 4.1a shows one possible compatibility factor for these variables. Note that this factor is not normalized; indeed, the entries are not even in $[0, 1]$. Roughly speaking, $\phi_1(A, B)$ asserts that it is more likely that Alice and Bob agree. It also adds more weight for the case where they are both right than for the case where they are both wrong. This factor function also has the property that $\phi_1(a^1, b^0) < \phi_1(a^0, b^1)$. Thus, if they disagree, there is less weight for the case where Alice has the misconception but Bob does not than for the converse case.

In a similar way, we define a compatibility factor for each other interacting pair: $\{B, C\}$, $\{C, D\}$, and $\{A, D\}$. Figure 4.1 shows one possible choice of factors for all four pairs. For example, the factor over C, D represents the compatibility of Charles and Debbie. It indicates that Charles and Debbie argue all the time, so that the most likely instantiations are those where they end up disagreeing.

As in a Bayesian network, the parameterization of the Markov network defines the local interactions between directly related variables. To define a global model, we need to combine these interactions. As in Bayesian networks, we combine the local models by multiplying them. Thus, we want $P(a, b, c, d)$ to be $\phi_1(a, b) \cdot \phi_2(b, c) \cdot \phi_3(c, d) \cdot \phi_4(d, a)$. In this case, however, we have no guarantees that the result of this process is a normalized joint distribution. Indeed, in this example, it definitely is not. Thus, we define the distribution by taking the product of

Assignment				Unnormalized	Normalized
a^0	b^0	c^0	d^0	300,000	0.04
a^0	b^0	c^0	d^1	300,000	0.04
a^0	b^0	c^1	d^0	300,000	0.04
a^0	b^0	c^1	d^1	30	$4.1 \cdot 10^{-6}$
a^0	b^1	c^0	d^0	500	$6.9 \cdot 10^{-5}$
a^0	b^1	c^0	d^1	500	$6.9 \cdot 10^{-5}$
a^0	b^1	c^1	d^0	5,000,000	0.69
a^0	b^1	c^1	d^1	500	$6.9 \cdot 10^{-5}$
a^1	b^0	c^0	d^0	100	$1.4 \cdot 10^{-5}$
a^1	b^0	c^0	d^1	1,000,000	0.14
a^1	b^0	c^1	d^0	100	$1.4 \cdot 10^{-5}$
a^1	b^0	c^1	d^1	100	$1.4 \cdot 10^{-5}$
a^1	b^1	c^0	d^0	10	$1.4 \cdot 10^{-6}$
a^1	b^1	c^0	d^1	100,000	0.014
a^1	b^1	c^1	d^0	100,000	0.014
a^1	b^1	c^1	d^1	100,000	0.014

Figure 4.2 Joint distribution for the Misconception example. The unnormalized measure and the normalized joint distribution over A, B, C, D , obtained from the parameterization of figure 4.1. The value of the partition function in this example is 7,201,840.

the local factors, and then normalizing it to define a legal distribution. Specifically, we define

$$P(a, b, c, d) = \frac{1}{Z} \phi_1(a, b) \cdot \phi_2(b, c) \cdot \phi_3(c, d) \cdot \phi_4(d, a),$$

where

$$Z = \sum_{a,b,c,d} \phi_1(a, b) \cdot \phi_2(b, c) \cdot \phi_3(c, d) \cdot \phi_4(d, a)$$

partition function
Markov random
field

is a normalizing constant known as the *partition function*. The term “partition” originates from the early history of Markov networks, which originated from the concept of *Markov random field* (or *MRF*) in statistical physics (see box 4.C); the “function” is because the value of Z is a function of the parameters, a dependence that will play a significant role in our discussion of learning.

In our example, the unnormalized measure (the simple product of the four factors) is shown in the next-to-last column in figure 4.2. For example, the entry corresponding to a^1, b^1, c^0, d^1 is obtained by multiplying:

$$\phi_1(a^1, b^1) \cdot \phi_2(b^1, c^0) \cdot \phi_3(c^0, d^1) \cdot \phi_4(d^1, a^1) = 10 \cdot 1 \cdot 100 \cdot 100 = 100,000.$$

The last column shows the normalized distribution.

We can use this joint distribution to answer queries, as usual. For example, by summing out A, C , and D , we obtain $P(b^1) \approx 0.732$ and $P(b^0) \approx 0.268$; that is, Bob is 26 percent likely to have the misconception. On the other hand, if we now observe that Charles does not have the misconception (c^0), we obtain $P(b^1 \mid c^0) \approx 0.06$.

The benefit of this representation is that it allows us great flexibility in representing interactions between variables. For example, if we want to change the nature of the interaction between A and B , we can simply modify the entries in that factor, without having to deal with normalization constraints and the interaction with other factors. The flip side of this flexibility, as we will see later, is that the effects of these changes are not always intuitively understandable.

As in Bayesian networks, there is a tight connection between the factorization of the distribution and its independence properties. The key result here is stated in exercise 2.5: $P \models (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$ if and only if we can write P in the form $P(\mathcal{X}) = \phi_1(\mathbf{X}, \mathbf{Z})\phi_2(\mathbf{Y}, \mathbf{Z})$. In our example, the structure of the factors allows us to decompose the distribution in several ways; for example:

$$P(A, B, C, D) = \left[\frac{1}{Z} \phi_1(A, B) \phi_2(B, C) \right] \phi_3(C, D) \phi_4(A, D).$$

From this decomposition, we can infer that $P \models (B \perp D \mid A, C)$. We can similarly infer that $P \models (A \perp C \mid B, D)$. These are precisely the two independencies that we tried, unsuccessfully, to achieve using a Bayesian network, in example 3.8. Moreover, these properties correspond to our intuition of “paths of influence” in the graph, where we have that B and D are separated given A, C , and that A and C are separated given B, D . Indeed, as in a Bayesian network, independence properties of the distribution P correspond directly to separation properties in the graph over which P factorizes.

4.2 Parameterization

We begin our formal discussion by describing the parameterization used in the class of undirected graphical models that are the focus of this chapter. In the next section, we make the connection to the graph structure and demonstrate how it captures the independence properties of the distribution.

To represent a distribution, we need to associate the graph structure with a set of parameters, in the same way that CPDs were used to parameterize the directed graph structure. However, the parameterization of Markov networks is not as intuitive as that of Bayesian networks, since the factors do not correspond either to probabilities or to conditional probabilities. As a consequence, the parameters are not intuitively understandable, making them hard to elicit from people. As we will see in chapter 20, they are also significantly harder to estimate from data.

4.2.1 Factors

A key issue in parameterizing a Markov network is that the representation is undirected, so that the parameterization cannot be directed in nature. We therefore use factors, as defined in definition 4.1. Note that a factor subsumes both the notion of a joint distribution and the notion of a CPD. A joint distribution over $\mathbf{D}$ is a factor over $\mathbf{D}$: it specifies a real number for every assignment of values of $\mathbf{D}$. A conditional distribution $P(X \mid \mathbf{U})$ is a factor over $\{X\} \cup \mathbf{U}$. However, both CPDs and joint distributions must satisfy certain normalization constraints (for example, in a joint distribution the numbers must sum to 1), whereas there are no constraints on the parameters in a factor.

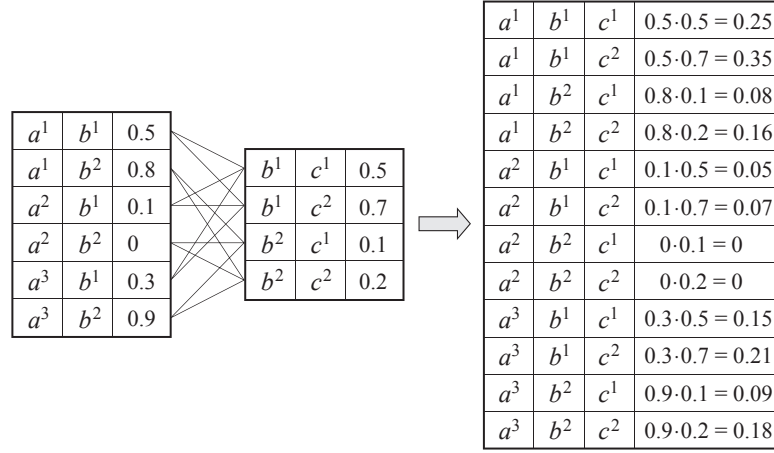


Figure 4.3 An example of factor product

As we discussed, we can view a factor as roughly describing the “compatibilities” between different values of the variables in its scope. We can now parameterize the graph by associating a set of factors with it. One obvious idea might be to associate parameters directly with the edges in the graph. However, a simple calculation will convince us that this approach is insufficient to parameterize a full distribution.

Example 4.1

Consider a fully connected graph over $\mathcal{X}$; in this case, the graph specifies no conditional independence assumptions, so that we should be able to specify an arbitrary joint distribution over $\mathcal{X}$. If all of the variables are binary, each factor over an edge would have 4 parameters, and the total number of parameters in the graph would be $4\binom{n}{2}$. However, the number of parameters required to specify a joint distribution over n binary variables is $2^n - 1$. Thus, pairwise factors simply do not have enough parameters to encompass the space of joint distributions. More intuitively, such factors capture only the pairwise interactions, and not interactions that involve combinations of values of larger subsets of variables. ■

A more general representation can be obtained by allowing factors over arbitrary subsets of variables. To provide a formal definition, we first introduce the following important operation on factors.

Definition 4.2

factor product

Let $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$ be three disjoint sets of variables, and let $\phi_1(\mathbf{X}, \mathbf{Y})$ and $\phi_2(\mathbf{Y}, \mathbf{Z})$ be two factors. We define the factor product $\phi_1 \times \phi_2$ to be a factor $\psi : \text{Val}(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) \mapsto \mathbb{R}$ as follows:

$$\psi(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) = \phi_1(\mathbf{X}, \mathbf{Y}) \cdot \phi_2(\mathbf{Y}, \mathbf{Z}).$$

■

The key aspect to note about this definition is the fact that the two factors ϕ_1 and ϕ_2 are multiplied in a way that “matches up” the common part $\mathbf{Y}$. Figure 4.3 shows an example of the product of two factors. We have deliberately chosen factors that do not correspond either to probabilities or to conditional probabilities, in order to emphasize the generality of this operation.

As we have already observed, both CPDs and joint distributions are factors. Indeed, the chain rule for Bayesian networks defines the joint distribution factor as the product of the CPD factors. For example, when computing $P(A, B) = P(A)P(B | A)$, we always multiply entries in the $P(A)$ and $P(B | A)$ tables that have the same value for A . Thus, letting $\phi_{X_i}(X_i, \text{Pa}_{X_i})$ represent $P(X_i | \text{Pa}_{X_i})$, we have that

$$P(X_1, \dots, X_n) = \prod_i \phi_{X_i}.$$

4.2.2 Gibbs Distributions and Markov Networks

We can now use the more general notion of factor product to define an undirected parameterization of a distribution.

Definition 4.3

Gibbs
distribution

A distribution P_Φ is a Gibbs distribution parameterized by a set of factors $\Phi = \{\phi_1(\mathbf{D}_1), \dots, \phi_K(\mathbf{D}_K)\}$ if it is defined as follows:

$$P_\Phi(X_1, \dots, X_n) = \frac{1}{Z} \tilde{P}_\Phi(X_1, \dots, X_n),$$

where

$$\tilde{P}_\Phi(X_1, \dots, X_n) = \phi_1(\mathbf{D}_1) \times \phi_2(\mathbf{D}_2) \times \dots \times \phi_m(\mathbf{D}_m)$$

is an unnormalized measure and

$$Z = \sum_{X_1, \dots, X_n} \tilde{P}_\Phi(X_1, \dots, X_n)$$

partition function

is a normalizing constant called the partition function. ■

It is tempting to think of the factors as representing the marginal probabilities of the variables in their scope. Thus, looking at any individual factor, we might be led to believe that the behavior of the distribution defined by the Markov network as a whole corresponds to the behavior defined by the factor. However, this intuition is overly simplistic. **A factor is only one contribution to the overall joint distribution. The distribution as a whole has to take into consideration the contributions from all of the factors involved.**



Example 4.2

Consider the distribution of figure 4.2. The marginal distribution over A, B , is

a^0	b^0	0.13
a^0	b^1	0.69
a^1	b^0	0.14
a^1	b^1	0.04

The most likely configuration is the one where Alice and Bob disagree. By contrast, the highest entry in the factor $\phi_1(A, B)$ in figure 4.1 corresponds to the assignment a^0, b^0 . The reason for the discrepancy is the influence of the other factors on the distribution. In particular, $\phi_3(C, D)$ asserts that Charles and Debbie disagree, whereas $\phi_2(B, C)$ and $\phi_4(D, A)$ assert that Bob and Charles agree and that Debbie and Alice agree. Taking just these factors into consideration, we would conclude that Alice and Bob are likely to disagree. In this case, the “strength” of these other factors is much stronger than that of the $\phi_1(A, B)$ factor, so that the influence of the latter is overwhelmed. ■

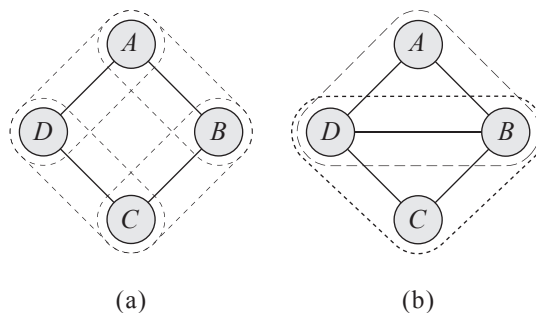


Figure 4.4 The cliques in two simple Markov networks. In (a), the cliques are the pairs $\{A, B\}$, $\{B, C\}$, $\{C, D\}$, and $\{D, A\}$. In (b), the cliques are $\{A, B, D\}$ and $\{B, C, D\}$.

We now want to relate the parameterization of a Gibbs distribution to a graph structure. If our parameterization contains a factor whose scope contains both X and Y , we are introducing a direct interaction between them. Intuitively, we would like these direct interactions to be represented in the graph structure. Thus, if our parameterization contains such a factor, we would like the associated Markov network structure $\mathcal{H}$ to contain an edge between X and Y .

Definition 4.4

Markov network
factorization

clique potentials

We say that a distribution P_Φ with $\Phi = \{\phi_1(\mathbf{D}_1), \dots, \phi_K(\mathbf{D}_K)\}$ factorizes over a Markov network $\mathcal{H}$ if each $\mathbf{D}_k$ ($k = 1, \dots, K$) is a complete subgraph of $\mathcal{H}$. ■

The factors that parameterize a Markov network are often called *clique potentials*.

As we will see, if we associate factors only with complete subgraphs, as in this definition, we are not violating the independence assumptions induced by the network structure, as defined later in this chapter.

Note that, because every complete subgraph is a subset of some (maximal) clique, we can reduce the number of factors in our parameterization by allowing factors only for maximal cliques. More precisely, let $\mathbf{C}_1, \dots, \mathbf{C}_l$ be the cliques in $\mathcal{H}$. We can parameterize P using a set of factors $\phi_1(\mathbf{C}_1), \dots, \phi_l(\mathbf{C}_l)$. Any factorization in terms of complete subgraphs can be converted into this form simply by assigning each factor to a clique that encompasses its scope and multiplying all of the factors assigned to each clique to produce a clique potential. In our Misconception example, we have four cliques: $\{A, B\}$, $\{B, C\}$, $\{C, D\}$, and $\{A, D\}$. Each of these cliques can have its own clique potential. One possible setting of the parameters in these clique potential is shown in figure 4.1. Figure 4.4 shows two examples of a Markov network and the (maximal) cliques in that network.



Although it can be used without loss of generality, the parameterization using maximal clique potentials generally obscures structure that is present in the original set of factors. For example, consider the Gibbs distribution described in example 4.1. Here, we have a potential for every pair of variables, so the Markov network associated with this distribution is a single large clique containing all variables. If we associate a factor with this single clique, it would be exponentially large in the number of variables, whereas the original parameterization in terms of edges requires only a quadratic number of parameters. See section 4.4.1.1 for further discussion.

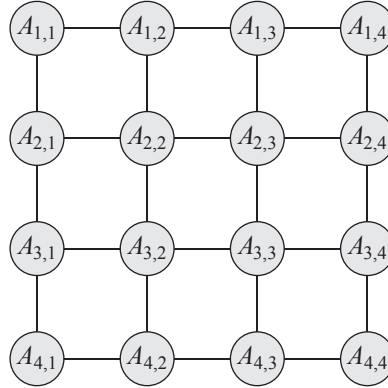


Figure 4.A.1 — A pairwise Markov network (MRF) structured as a grid.

pairwise Markov
network
node potential
edge potential

Box 4.A — Concept: Pairwise Markov Networks. A subclass of Markov networks that arises in many contexts is that of pairwise Markov networks, representing distributions where all of the factors are over single variables or pairs of variables. More precisely, a pairwise Markov network over a graph $\mathcal{H}$ is associated with a set of node potentials $\{\phi(X_i) : i = 1, \dots, n\}$ and a set of edge potentials $\{\phi(X_i, X_j) : (X_i, X_j) \in \mathcal{H}\}$. The overall distribution is (as always) the normalized product of all of the potentials (both node and edge). Pairwise MRFs are attractive because of their simplicity, and because interactions on edges are an important special case that often arises in practice (see, for example, box 4.C and box 4.B).

A class of pairwise Markov networks that often arises, and that is commonly used as a benchmark for inference, is the class of networks structured in the form of a grid, as shown in figure 4.A.1. As we discuss in the inference chapters of this book, although these networks have a simple and compact representation, they pose a significant challenge for inference algorithms.

4.2.3 Reduced Markov Networks

We end this section with one final concept that will prove very useful in later sections. Consider the process of conditioning a distribution on some assignment $\mathbf{u}$ to some subset of variables $\mathbf{U}$. Conditioning a distribution corresponds to eliminating all entries in the joint distribution that are inconsistent with the event $\mathbf{U} = \mathbf{u}$, and renormalizing the remaining entries to sum to 1. Now, consider the case where our distribution has the form P_Φ for some set of factors Φ . Each entry in the unnormalized measure $\tilde{P}_\Phi$ is a product of entries from the factors Φ , one entry from each factor. If, in some factor, we have an entry that is inconsistent with $\mathbf{U} = \mathbf{u}$, it will only contribute to entries in $\tilde{P}_\Phi$ that are also inconsistent with this event. Thus, we can eliminate all such entries from every factor in Φ .

More generally, we can define:

a^1	b^1	c^1	0.25
a^1	b^2	c^1	0.08
a^2	b^1	c^1	0.05
a^2	b^2	c^1	0
a^3	b^1	c^1	0.15
a^3	b^2	c^1	0.09

Figure 4.5 Factor reduction: The factor computed in figure 4.3, reduced to the context $C = c^1$.

Definition 4.5
factor reduction

Let $\phi(Y)$ be a factor, and $U = \mathbf{u}$ an assignment for $U \subseteq Y$. We define the reduction of the factor ϕ to the context $U = \mathbf{u}$, denoted $\phi[U = \mathbf{u}]$ (and abbreviated $\phi[\mathbf{u}]$), to be a factor over scope $Y' = Y - U$, such that

$$\phi[\mathbf{u}](\mathbf{y}') = \phi(\mathbf{y}', \mathbf{u}).$$

For $U \not\subseteq Y$, we define $\phi[\mathbf{u}]$ to be $\phi[U' = \mathbf{u}']$, where $U' = U \cap Y$, and $\mathbf{u}' = \mathbf{u}(U')$, where $\mathbf{u}(U')$ denotes the assignment in $\mathbf{u}$ to the variables in U' . ■

Figure 4.5 illustrates this operation, reducing the of figure 4.3 to the context $C = c^1$.

Now, consider a product of factors. An entry in the product is consistent with $\mathbf{u}$ if and only if it is a product of entries that are all consistent with $\mathbf{u}$. We can therefore define:

Definition 4.6
reduced Gibbs distribution

Let P_Φ be a Gibbs distribution parameterized by $\Phi = \{\phi_1, \dots, \phi_K\}$ and let $\mathbf{u}$ be a context. The reduced Gibbs distribution $P_\Phi[\mathbf{u}]$ is the Gibbs distribution defined by the set of factors $\Phi[\mathbf{u}] = \{\phi_1[\mathbf{u}], \dots, \phi_K[\mathbf{u}]\}$. ■

Reducing the set of factors defining P_Φ to some context $\mathbf{u}$ corresponds directly to the operation of conditioning P_Φ on the observation $\mathbf{u}$. More formally:

Proposition 4.1

Let $P_\Phi(X)$ be a Gibbs distribution. Then $P_\Phi[\mathbf{u}] = P_\Phi(W \mid \mathbf{u})$ where $W = X - U$.

Thus, to condition a Gibbs distribution on a context $\mathbf{u}$, we simply reduce every one of its factors to that context. Intuitively, the renormalization step needed to account for $\mathbf{u}$ is simply folded into the standard renormalization of any Gibbs distribution. This result immediately provides us with a construction for the Markov network that we obtain when we condition the associated distribution on some observation $\mathbf{u}$.

Definition 4.7
reduced Markov network

Let $\mathcal{H}$ be a Markov network over X and $U = \mathbf{u}$ a context. The reduced Markov network $\mathcal{H}[\mathbf{u}]$ is a Markov network over the nodes $W = X - U$, where we have an edge $X - Y$ if there is an edge $X - Y$ in $\mathcal{H}$. ■

Proposition 4.2

Let $P_\Phi(X)$ be a Gibbs distribution that factorizes over $\mathcal{H}$, and $U = \mathbf{u}$ a context. Then $P_\Phi[\mathbf{u}]$ factorizes over $\mathcal{H}[\mathbf{u}]$.

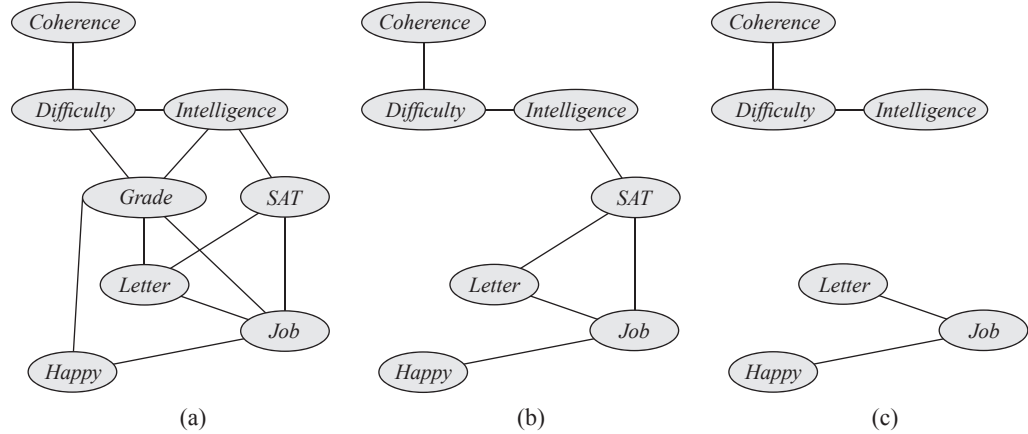


Figure 4.6 Markov networks for the factors in an extended Student example: (a) The initial set of factors; (b) Reduced to the context $G = g$; (c) Reduced to the context $G = g, S = s$.

Note the contrast to the effect of conditioning in a Bayesian network: Here, conditioning on a context $\mathbf{u}$ only eliminates edges from the graph; in a Bayesian network, conditioning on evidence can activate a v-structure, creating new dependencies. We return to this issue in section 4.5.1.1.

Example 4.3

Consider, for example, the Markov network shown in figure 4.6a; as we will see, this network is the Markov network required to capture the distribution encoded by an extended version of our Student Bayesian network (see figure 9.8). Figure 4.6b shows the same Markov network reduced over a context of the form $G = g$, and (c) shows the network reduced over a context of the form $G = g, S = s$. As we can see, the network structures are considerably simplified. ■

Box 4.B — Case Study: Markov Networks for Computer Vision. One important application area for Markov networks is computer vision. Markov networks, typically called MRFs in this vision community, have been used for a wide variety of visual processing tasks, such as image segmentation, removal of blur or noise, stereo reconstruction, object recognition, and many more.

In most of these applications, the network takes the structure of a pairwise MRF, where the variables correspond to pixels and the edges (factors) to interactions between adjacent pixels in the grid that represents the image; thus, each (interior) pixel has exactly four neighbors. The value space of the variables and the exact form of factors depend on the task. These models are usually formulated in terms of energies (negative log-potentials), so that values represent “penalties,” and a lower value corresponds to a higher-probability configuration.

image denoising

In image denoising, for example, the task is to restore the “true” value of all of the pixels given possibly noisy pixel values. Here, we have a node potential for each pixel X_i that penalizes large discrepancies from the observed pixel value y_i . The edge potential encodes a preference for continuity between adjacent pixel values, penalizing cases where the inferred value for X_i is too

far from the inferred pixel value for one of its neighbors X_j . However, it is important not to overpenalize true disparities (such as edges between objects or regions), leading to oversmoothing of the image. Thus, we bound the penalty, using, for example, some truncated norm, as described in box 4.D: $\epsilon(x_i, x_j) = \min(c\|x_i - x_j\|_p, \text{dist}_{\max})$ (for $p \in \{1, 2\}$).

stereo
reconstruction

Slight variants of the same model are used in many other applications. For example, in stereo reconstruction, the goal is to reconstruct the depth disparity of each pixel in the image. Here, the values of the variables represent some discretized version of the depth dimension (usually more finely discretized for distances close to the camera and more coarsely discretized as the distance from the camera increases). The individual node potential for each pixel X_i uses standard techniques from computer vision to estimate, from a pair of stereo images, the individual depth disparity of this pixel. The edge potentials, precisely as before, often use a truncated metric to enforce continuity of the depth estimates, with the truncation avoiding an overpenalization of true depth disparities (for example, when one object is partially in front of the other). Here, it is also quite common to make the penalty inversely proportional to the image gradient between the two pixels, allowing a smaller penalty to be applied in cases where a large image gradient suggests an edge between the pixels, possibly corresponding to an occlusion boundary.

image
segmentation

In image segmentation, the task is to partition the image pixels into regions corresponding to distinct parts of the scene. There are different variants of the segmentation task, many of which can be formulated as a Markov network. In one formulation, known as multiclass segmentation, each variable X_i has a domain $\{1, \dots, K\}$, where the value of X_i represents a region assignment for pixel i (for example, grass, water, sky, car). Since classifying every pixel can be computationally expensive, some state-of-the-art methods for image segmentation and other tasks first oversegment the image into superpixels (or small coherent regions) and classify each region — all pixels within a region are assigned the same label. The oversegmented image induces a graph in which there is one node for each superpixel and an edge between two nodes if the superpixels are adjacent (share a boundary) in the underlying image. We can now define our distribution in terms of this graph.

Features are extracted from the image for each pixel or superpixel. The appearance features depend on the specific task. In image segmentation, for example, features typically include statistics over color, texture, and location. Often the features are clustered or provided as input to local classifiers to reduce dimensionality. The features used in the model are then the soft cluster assignments or local classifier outputs for each superpixel. The node potential for a pixel or superpixel is then a function of these features. We note that the factors used in defining this model depend on the specific values of the pixels in the image, so that each image defines a different probability distribution over the segment labels for the pixels or superpixels. In effect, the model used here is a conditional random field, a concept that we define more formally in section 4.6.1.

conditional
random field

The model contains an edge potential between every pair of neighboring superpixels X_i, X_j . Most simply, this potential encodes a contiguity preference, with a penalty of λ whenever $X_i \neq X_j$. Again, we can improve the model by making the penalty depend on the presence of an image gradient between the two pixels. An even better model does more than penalize discontinuities. We can have nondefault values for other class pairs, allowing us to encode the fact that we more often find tigers adjacent to vegetation than adjacent to water; we can even make the model depend on the relative pixel location, allowing us to encode the fact that we usually find water below vegetation, cars over roads, and sky above everything.

Figure 4.B.1 shows segmentation results in a model containing only potentials on single pixels (thereby labeling each of them independently) versus results obtained from a model also containing

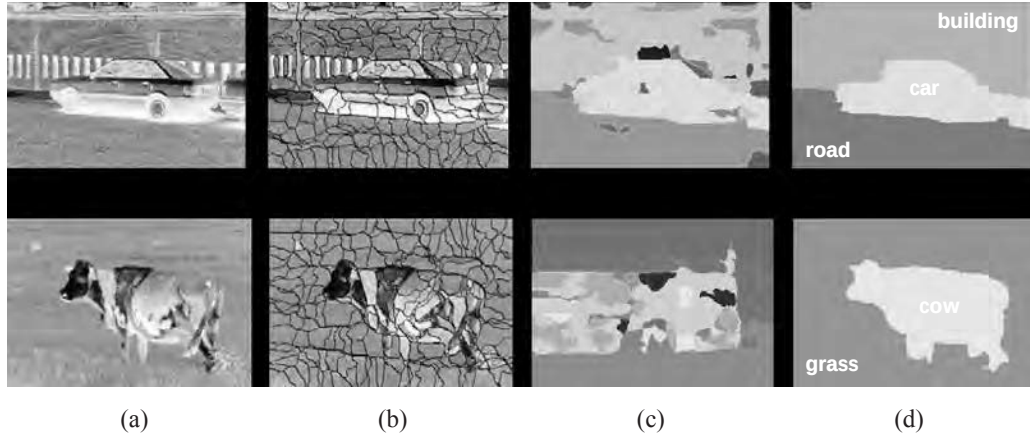


Figure 4.B.1 — Two examples of image segmentation results (a) The original image. (b) An oversegmentation known as superpixels; each superpixel is associated with a random variable that designates its segment assignment. The use of superpixels reduces the size of the problems. (c) Result of segmentation using node potentials alone, so that each superpixel is classified independently. (d) Result of segmentation using a pairwise Markov network encoding interactions between adjacent superpixels.

pairwise potentials. The difference in the quality of the results clearly illustrates the importance of modeling the correlations between the superpixels.

4.3 Markov Network Independencies

In section 4.1, we gave an intuitive justification of why an undirected graph seemed to capture the types of interactions in the Misconception example. We now provide a formal presentation of the undirected graph as a representation of independence assertions.

4.3.1 Basic Independencies

As in the case of Bayesian networks, the graph structure in a Markov network can be viewed as encoding a set of independence assumptions. Intuitively, in Markov networks, probabilistic influence “flows” along the undirected paths in the graph, but it is blocked if we condition on the intervening nodes.

Definition 4.8

observed variable

active path

Let $\mathcal{H}$ be a Markov network structure, and let $X_1 - \dots - X_k$ be a path in $\mathcal{H}$. Let $\mathcal{Z} \subseteq \mathcal{X}$ be a set of observed variables. The path $X_1 - \dots - X_k$ is active given $\mathcal{Z}$ if none of the X_i 's, $i = 1, \dots, k$, is in $\mathcal{Z}$. ■

Using this notion, we can define a notion of *separation* in the graph.

Definition 4.9

separation

global
independencies

We say that a set of nodes $\mathbf{Z}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{H}$, denoted $\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})$, if there is no active path between any node $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ given $\mathbf{Z}$. We define the global independencies associated with $\mathcal{H}$ to be:

$$\mathcal{I}(\mathcal{H}) = \{(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) : \text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})\}. \quad \blacksquare$$

As we will discuss, the independencies in $\mathcal{I}(\mathcal{H})$ are precisely those that are guaranteed to hold for every distribution P over $\mathcal{H}$. In other words, the separation criterion is sound for detecting independence properties in distributions over $\mathcal{H}$.

Note that the definition of separation is monotonic in $\mathbf{Z}$, that is, if $\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})$, then $\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z}')$ for any $\mathbf{Z}' \supset \mathbf{Z}$. Thus, if we take separation as our definition of the independencies induced by the network structure, we are effectively restricting our ability to encode nonmonotonic independence relations. Recall that in the context of intercausal reasoning in Bayesian networks, nonmonotonic reasoning patterns are quite useful in many situations — for example, when two diseases are independent, but dependent given some common symptom. The nature of the separation property implies that such independence patterns cannot be expressed in the structure of a Markov network. We return to this issue in section 4.5.

As for Bayesian networks, we can show a connection between the independence properties implied by the Markov network structure, and the possibility of factorizing a distribution over the graph. As before, we can now state the analogue to both of our representation theorems for Bayesian networks, which assert the equivalence between the Gibbs factorization of a distribution P over a graph $\mathcal{H}$ and the assertion that $\mathcal{H}$ is an I-map for P , that is, that P satisfies the Markov assumptions $\mathcal{I}(\mathcal{H})$.

4.3.1.1 Soundness

soundness

We first consider the analogue to theorem 3.2, which asserts that a Gibbs distribution satisfies the independencies associated with the graph. In other words, this result states the *soundness* of the separation criterion.

Theorem 4.1

Let P be a distribution over $\mathcal{X}$, and $\mathcal{H}$ a Markov network structure over $\mathcal{X}$. If P is a Gibbs distribution that factorizes over $\mathcal{H}$, then $\mathcal{H}$ is an I-map for P .

PROOF Let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ be any three disjoint subsets in $\mathcal{X}$ such that $\mathbf{Z}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{H}$. We want to show that $P \models (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$.

We start by considering the case where $\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z} = \mathcal{X}$. As $\mathbf{Z}$ separates $\mathbf{X}$ from $\mathbf{Y}$, there are no direct edges between $\mathbf{X}$ and $\mathbf{Y}$. Hence, any clique in $\mathcal{H}$ is fully contained either in $\mathbf{X} \cup \mathbf{Z}$ or in $\mathbf{Y} \cup \mathbf{Z}$. Let $\mathcal{I}_{\mathbf{X}}$ be the indexes of the set of cliques that are contained in $\mathbf{X} \cup \mathbf{Z}$, and let $\mathcal{I}_{\mathbf{Y}}$ be the indexes of the remaining cliques. We know that

$$P(X_1, \dots, X_n) = \frac{1}{Z} \prod_{i \in \mathcal{I}_{\mathbf{X}}} \phi_i(\mathbf{D}_i) \cdot \prod_{i \in \mathcal{I}_{\mathbf{Y}}} \phi_i(\mathbf{D}_i).$$

As we discussed, none of the factors in the first product involve any variable in $\mathbf{Y}$, and none in the second product involve any variable in $\mathbf{X}$. Hence, we can rewrite this product in the form:

$$P(X_1, \dots, X_n) = \frac{1}{Z} f(\mathbf{X}, \mathbf{Z}) g(\mathbf{Y}, \mathbf{Z}).$$

From this decomposition, the desired independence follows immediately (exercise 2.5).

Now consider the case where $\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z} \subset \mathcal{X}$. Let $\mathbf{U} = \mathcal{X} - (\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z})$. We can partition $\mathbf{U}$ into two disjoint sets $\mathbf{U}_1$ and $\mathbf{U}_2$ such that $\mathbf{Z}$ separates $\mathbf{X} \cup \mathbf{U}_1$ from $\mathbf{Y} \cup \mathbf{U}_2$ in $\mathcal{H}$. Using the preceding argument, we conclude that $P \models (\mathbf{X}, \mathbf{U}_1 \perp \mathbf{Y}, \mathbf{U}_2 \mid \mathbf{Z})$. Using the decomposition property (equation (2.8)), we conclude that $P \models (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$. ■

Hammersley-
Clifford
theorem

The other direction (the analogue to theorem 3.1), which goes from the independence properties of a distribution to its factorization, is known as the *Hammersley-Clifford theorem*. Unlike for Bayesian networks, this direction does not hold in general. As we will show, it holds only under the additional assumption that P is a positive distribution (see definition 2.5).

Theorem 4.2

Let P be a positive distribution over $\mathcal{X}$, and $\mathcal{H}$ a Markov network graph over $\mathcal{X}$. If $\mathcal{H}$ is an I-map for P , then P is a Gibbs distribution that factorizes over $\mathcal{H}$.

To prove this result, we would need to use the independence assumptions to construct a set of factors over $\mathcal{H}$ that give rise to the distribution P . In the case of Bayesian networks, these factors were simply CPDs, which we could derive directly from P . As we have discussed, the correspondence between the factors in a Gibbs distribution and the distribution P is much more indirect. The construction required here is therefore significantly more subtle, and relies on concepts that we develop later in this chapter; hence, we defer the proof to section 4.4 (theorem 4.8).



This result shows that, **for positive distributions, the global independencies imply that the distribution factorizes according the network structure. Thus, for this class of distributions, we have that a distribution P factorizes over a Markov network $\mathcal{H}$ if and only if $\mathcal{H}$ is an I-map of P .** The positivity assumption is necessary for this result to hold:

Example 4.4

Consider a distribution P over four binary random variables X_1, X_2, X_3, X_4 , which gives probability $1/8$ to each of the following eight configurations, and probability zero to all others:

$(0,0,0,0)$	$(1,0,0,0)$	$(1,1,0,0)$	$(1,1,1,0)$
$(0,0,0,1)$	$(0,0,1,1)$	$(0,1,1,1)$	$(1,1,1,1)$

Let $\mathcal{H}$ be the graph $X_1 - X_2 - X_3 - X_4 - X_1$. Then P satisfies the global independencies with respect to $\mathcal{H}$. For example, consider the independence $(X_1 \perp X_3 \mid X_2, X_4)$. For the assignment $X_2 = x_2^1, X_4 = x_4^0$, we have that only assignments where $X_1 = x_1^1$ receive positive probability. Thus, $P(x_1^1 \mid x_2^1, x_4^0) = 1$, and X_1 is trivially independent of X_3 in this conditional distribution. A similar analysis applies to all other cases, so that the global independencies hold. However, the distribution P does not factorize according to $\mathcal{H}$. The proof of this fact is left as an exercise (see exercise 4.1). ■

4.3.1.2 Completeness

completeness

The preceding discussion shows the soundness of the separation condition as a criterion for detecting independencies in Markov networks: any distribution that factorizes over $\mathcal{G}$ satisfies the independence assertions implied by separation. The next obvious issue is the *completeness* of this criterion.

As for Bayesian networks, the strong version of completeness does not hold in this setting. In other words, it is not the case that every pair of nodes X and Y that are not separated in $\mathcal{H}$ are dependent in every distribution P which factorizes over $\mathcal{H}$. However, as in theorem 3.3, we can use a weaker definition of completeness that does hold:

Theorem 4.3

Let $\mathcal{H}$ be a Markov network structure. If X and Y are not separated given $\mathbf{Z}$ in $\mathcal{H}$, then X and Y are dependent given $\mathbf{Z}$ in some distribution P that factorizes over $\mathcal{H}$.

PROOF The proof is a constructive one: we construct a distribution P that factorizes over $\mathcal{H}$ where X and Y are dependent. We assume, without loss of generality, that all variables are binary-valued. If this is not the case, we can treat them as binary-valued by restricting attention to two distinguished values for each variable.

By assumption, X and Y are not separated given $\mathbf{Z}$ in $\mathcal{H}$; hence, they must be connected by some unblocked trail. Let $X = U_1 - U_2 - \dots - U_k = Y$ be some minimal trail in the graph such that, for all i , $U_i \notin \mathbf{Z}$, where we define a minimal trail in $\mathcal{H}$ to be a path with no shortcuts: thus, for any i and $j \neq i \pm 1$, there is no edge $U_i - U_j$. We can always find such a path: If we have a nonminimal path where we have $U_i - U_j$ for $j > i + 1$, we can always “shortcut” the original trail, converting it to one that goes directly from U_i to U_j .

For any $i = 1, \dots, k - 1$, as there is an edge $U_i - U_{i+1}$, it follows that U_i, U_{i+1} must both appear in some clique $\mathcal{C}_i$. We pick some very large weight W , and for each i we define the clique potential $\phi_i(\mathcal{C}_i)$ to assign weight W if $U_i = U_{i+1}$ and weight 1 otherwise, regardless of the values of the other variables in the clique. Note that the cliques $\mathcal{C}_i$ for U_i, U_{i+1} and $\mathcal{C}_j$ for U_j, U_{j+1} must be different cliques: If $\mathcal{C}_i = \mathcal{C}_j$, then U_j is in the same clique as U_i , and we have an edge $U_i - U_j$, contradicting the minimality of the trail. Hence, we can define the clique potential for each clique $\mathcal{C}_i$ separately. We define the clique potential for any other clique to be uniformly 1.

We now consider the distribution P resulting from multiplying all of these clique potentials. Intuitively, the distribution $P(U_1, \dots, U_k)$ is simply the distribution defined by multiplying the pairwise factors for the pairs U_i, U_{i+1} , regardless of the other variables (including the ones in $\mathbf{Z}$). One can verify that, in $P(U_1, \dots, U_k)$, we have that $X = U_1$ and $Y = U_k$ are dependent. We leave the conclusion of this argument as an exercise (exercise 4.5). ■

We can use the same argument as theorem 3.5 to conclude that, for almost all distributions P that factorize over $\mathcal{H}$ (that is, for all distributions except for a set of measure zero in the space of factor parameterizations) we have that $\mathcal{I}(P) = \mathcal{I}(\mathcal{H})$.

Once again, we can view this result as telling us that our definition of $\mathcal{I}(\mathcal{H})$ is the maximal one. For any independence assertion that is not a consequence of separation in $\mathcal{H}$, we can always find a counterexample distribution P that factorizes over $\mathcal{H}$.

4.3.2 Independencies Revisited

When characterizing the independencies in a Bayesian network, we provided two definitions: the local independencies (each node is independent of its nondescendants given its parents), and the global independencies induced by d-separation. As we showed, these two sets of independencies are equivalent, in that one implies the other.

So far, our discussion for Markov networks provides only a global criterion. While the global criterion characterizes the entire set of independencies induced by the network structure, a local criterion is also valuable, since it allows us to focus on a smaller set of properties when examining the distribution, significantly simplifying the process of finding an I-map for a distribution P .

Thus, it is natural to ask whether we can provide a local definition of the independencies induced by a Markov network, analogously to the local independencies of Bayesian networks. Surprisingly, as we now show, in the context of Markov networks, there are three different possible definitions of the independencies associated with the network structure — two local ones and the global one in definition 4.9. While these definitions are related, they are equivalent only for positive distributions. As we will see, nonpositive distributions allow for deterministic dependencies between the variables. Such deterministic interactions can “fool” local independence tests, allowing us to construct networks that are not I-maps of the distribution, yet the local independencies hold.

4.3.2.1 Local Markov Assumptions

The first, and weakest, definition is based on the following intuition: Whenever two variables are directly connected, they have the potential of being directly correlated in a way that is not mediated by other variables. Conversely, when two variables are not directly linked, there must be some way of rendering them conditionally independent. Specifically, we can require that X and Y be independent given all other nodes in the graph.

Definition 4.10

pairwise
independencies

Let $\mathcal{H}$ be a Markov network. We define the pairwise independencies associated with $\mathcal{H}$ to be:

$$\mathcal{I}_p(\mathcal{H}) = \{(X \perp Y \mid \mathcal{X} - \{X, Y\}) : X - Y \notin \mathcal{H}\}. \quad \blacksquare$$

Using this definition, we can easily represent the independencies in our Misconception example using a Markov network: We simply connect the nodes up in exactly the same way as the interaction structure between the students.

The second local definition is an undirected analogue to the local independencies associated with a Bayesian network. It is based on the intuition that we can block all influences on a node by conditioning on its immediate neighbors.

Definition 4.11

Markov blanket

local
independencies

For a given graph $\mathcal{H}$, we define the Markov blanket of X in $\mathcal{H}$, denoted $\text{MB}_{\mathcal{H}}(X)$, to be the neighbors of X in $\mathcal{H}$. We define the local independencies associated with $\mathcal{H}$ to be:

$$\mathcal{I}_\ell(\mathcal{H}) = \{(X \perp \mathcal{X} - \{X\} - \text{MB}_{\mathcal{H}}(X) \mid \text{MB}_{\mathcal{H}}(X)) : X \in \mathcal{X}\}. \quad \blacksquare$$

In other words, the local independencies state that X is independent of the rest of the nodes in the graph given its immediate neighbors. We will show that these local independence assumptions hold for any distribution that factorizes over $\mathcal{H}$, so that X 's Markov blanket in $\mathcal{H}$ truly does separate it from all other variables.

4.3.2.2 Relationships between Markov Properties

We have now presented three sets of independence assertions associated with a network structure $\mathcal{H}$. For general distributions, $\mathcal{I}_p(\mathcal{H})$ is strictly weaker than $\mathcal{I}_\ell(\mathcal{H})$, which in turn is strictly weaker than $\mathcal{I}(\mathcal{H})$. However, all three definitions are equivalent for positive distributions.

Proposition 4.3 *For any Markov network $\mathcal{H}$, and any distribution P , we have that if $P \models \mathcal{I}_\ell(\mathcal{H})$ then $P \models \mathcal{I}_p(\mathcal{H})$.*

The proof of this result is left as an exercise (exercise 4.8).

Proposition 4.4 *For any Markov network $\mathcal{H}$, and any distribution P , we have that if $P \models \mathcal{I}(\mathcal{H})$ then $P \models \mathcal{I}_\ell(\mathcal{H})$.*

The proof of this result follows directly from the fact that if X and Y are not connected by an edge, then they are necessarily separated by all of the remaining nodes in the graph.

The converse of these inclusion results holds only for positive distributions (see definition 2.5). More specifically, if we assume the intersection property (equation (2.11)), all three of the Markov conditions are equivalent.

Theorem 4.4 *Let P be a positive distribution. If P satisfies $\mathcal{I}_p(\mathcal{H})$, then P satisfies $\mathcal{I}(\mathcal{H})$.*

PROOF We want to prove that for all disjoint sets $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$:

$$\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z}) \implies P \models (\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}). \quad (4.1)$$

The proof proceeds by descending induction on the size of $\mathbf{Z}$.

The base case is $|\mathbf{Z}| = n - 2$; equation (4.1) follows immediately from the definition of $\mathcal{I}_p(\mathcal{H})$.

For the inductive step, assume that equation (4.1) holds for every $\mathbf{Z}'$ with size $|\mathbf{Z}'| = k$, and let $\mathbf{Z}$ be any set such that $|\mathbf{Z}| = k - 1$. We distinguish between two cases.

In the first case, $\mathbf{X} \cup \mathbf{Z} \cup \mathbf{Y} = \mathcal{X}$. As $|\mathbf{Z}| < n - 2$, we have that either $|\mathbf{X}| \geq 2$ or $|\mathbf{Y}| \geq 2$. Without loss of generality, assume that the latter holds; let $A \in \mathbf{Y}$ and $\mathbf{Y}' = \mathbf{Y} - \{A\}$. From the fact that $\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y} \mid \mathbf{Z})$, we also have that $\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y}' \mid \mathbf{Z})$ on one hand and $\text{sep}_{\mathcal{H}}(\mathbf{X}; A \mid \mathbf{Z})$ on the other hand. As separation is monotonic, we also have that $\text{sep}_{\mathcal{H}}(\mathbf{X}; \mathbf{Y}' \mid \mathbf{Z} \cup \{A\})$ and $\text{sep}_{\mathcal{H}}(\mathbf{X}; A \mid \mathbf{Z} \cup \mathbf{Y}')$. The separating sets $\mathbf{Z} \cup \{A\}$ and $\mathbf{Z} \cup \mathbf{Y}'$ are each at least size $|\mathbf{Z}| + 1 = k$ in size, so that equation (4.1) applies, and we can conclude that P satisfies:

$$(\mathbf{X} \perp \mathbf{Y}' \mid \mathbf{Z} \cup \{A\}) \quad \& \quad (\mathbf{X} \perp A \mid \mathbf{Z} \cup \mathbf{Y}').$$

Because P is positive, we can apply the intersection property (equation (2.11)) and conclude that $P \models (\mathbf{X} \perp \mathbf{Y}' \cup \{A\} \mid \mathbf{Z})$, that is, $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$.

The second case is where $\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z} \neq \mathcal{X}$. Here, we might have that both $\mathbf{X}$ and $\mathbf{Y}$ are singletons. This case requires a similar argument that uses the induction hypothesis and properties of independence. We leave it as an exercise (exercise 4.9). ■

Our previous results entail that, for positive distributions, the three conditions are equivalent.

Corollary 4.1 *The following three statements are equivalent for a positive distribution P :*

1. $P \models \mathcal{I}_\ell(\mathcal{H})$.
2. $P \models \mathcal{I}_p(\mathcal{H})$.
3. $P \models \mathcal{I}(\mathcal{H})$.

This equivalence relies on the positivity assumption. In particular, for nonpositive distributions, we can provide examples of a distribution P that satisfies one of these properties, but not the stronger one.

Example 4.5

Let P be any distribution over $\mathcal{X} = \{X_1, \dots, X_n\}$; let $\mathcal{X}' = \{X'_1, \dots, X'_n\}$. We now construct a distribution $P'(\mathcal{X}, \mathcal{X}')$ whose marginal over $X_1, \dots, X_n$ is the same as P , and where X'_i is deterministically equal to X_i . Let $\mathcal{H}$ be a Markov network over $\mathcal{X}, \mathcal{X}'$ that contains no edges other than $X_i - X'_i$. Then, in P' , X_i is independent of the rest of the variables in the network given its neighbor X'_i , and similarly for X'_i ; thus, $\mathcal{H}$ satisfies the local independencies for every node in the network. Yet clearly $\mathcal{H}$ is not an I-map for P' , since $\mathcal{H}$ makes many independence assertions regarding the X_i 's that do not hold in P (or in P'). ■

Thus, for nonpositive distributions, the local independencies do not imply the global ones.

A similar construction can be used to show that, for nonpositive distributions, the pairwise independencies do necessarily imply the local independencies.

Example 4.6

Let P be any distribution over $\mathcal{X} = \{X_1, \dots, X_n\}$, and now consider two auxiliary sets of variables $\mathcal{X}'$ and $\mathcal{X}''$, and define $\mathcal{X}^* = \mathcal{X} \cup \mathcal{X}' \cup \mathcal{X}''$. We now construct a distribution $P'(\mathcal{X}^*)$ whose marginal over $X_1, \dots, X_n$ is the same as P , and where X'_i and X''_i are both deterministically equal to X_i . Let $\mathcal{H}$ be the empty Markov network over $\mathcal{X}^*$. We argue that this empty network satisfies the pairwise assumptions for every pair of nodes in the network. For example, X_i and X'_i are rendered independent because $\mathcal{X}^* - \{X_i, X'_i\}$ contains X''_i . Similarly, X_i and X_j are independent given X'_i . Thus, $\mathcal{H}$ satisfies the pairwise independencies, but not the local or global independencies. ■

4.3.3 From Distributions to Graphs

Based on our deeper understanding of the independence properties associated with a Markov network, we can now turn to the question of encoding the independencies in a given distribution P using a graph structure. As for Bayesian networks, the notion of an I-map is not sufficient by itself: The complete graph implies no independence assumptions and is hence an I-map for any distribution. We therefore return to the notion of a minimal I-map, defined in definition 3.13, which was defined broadly enough to apply to Markov networks as well.

How can we construct a minimal I-map for a distribution P ? Our discussion in section 4.3.2 immediately suggests two approaches for constructing a minimal I-map: one based on the pairwise Markov independencies, and the other based on the local independencies.

In the first approach, we consider the pairwise independencies. They assert that, if the edge $\{X, Y\}$ is not in $\mathcal{H}$, then X and Y must be independent given all other nodes in the graph, regardless of which other edges the graph contains. Thus, at the very least, to guarantee that $\mathcal{H}$ is an I-map, we must add direct edges between all pairs of nodes X and Y such that

$$P \not\models (X \perp Y \mid \mathcal{X} - \{X, Y\}). \quad (4.2)$$

We can now define $\mathcal{H}$ to include an edge $X - Y$ for all X, Y for which equation (4.2) holds.

In the second approach, we use the local independencies and the notion of minimality. For each variable X , we define the neighbors of X to be a minimal set of nodes $\mathbf{Y}$ that render X independent of the rest of the nodes. More precisely, define:

Definition 4.12

Markov blanket

A set U is a Markov blanket of X in a distribution P if $X \notin U$ and if U is a minimal set of nodes such that

$$(X \perp \mathcal{X} - \{X\} - U \mid U) \in \mathcal{I}(P). \quad (4.3)$$

■

We then define a graph $\mathcal{H}$ by introducing an edge $\{X, Y\}$ for all X and all $Y \in \text{MB}_P(X)$. As defined, this construction is not unique, since there may be several sets U satisfying equation (4.3). However, theorem 4.6 will show that there is only one such minimal set. In fact, we now show that any positive distribution P has a unique minimal I-map, and that both of these constructions produce this I-map.

We begin with the proof for the pairwise definition:

Theorem 4.5

Let P be a positive distribution, and let $\mathcal{H}$ be defined by introducing an edge $\{X, Y\}$ for all X, Y for which equation (4.2) holds. Then the Markov network $\mathcal{H}$ is the unique minimal I-map for P .

PROOF The fact that $\mathcal{H}$ is an I-map for P follows immediately from fact that P , by construction, satisfies $\mathcal{I}_p(\mathcal{H})$, and, therefore, by corollary 4.1, also satisfies $\mathcal{I}(\mathcal{H})$. The fact that it is minimal follows from the fact that if we eliminate some edge $\{X, Y\}$ from $\mathcal{H}$, the graph would imply the pairwise independence $(X \perp Y \mid \mathcal{X} - \{X, Y\})$, which we know to be false for P (otherwise, the edge would have been omitted in the construction of $\mathcal{H}$). The uniqueness of the minimal I-map also follows trivially: By the same argument, any other I-map $\mathcal{H}'$ for P must contain at least the edges in $\mathcal{H}$ and is therefore either equal to $\mathcal{H}$ or contains additional edges and is therefore not minimal. ■

It remains to show that the second definition results in the same minimal I-map.

Theorem 4.6

Let P be a positive distribution. For each node X , let $\text{MB}_P(X)$ be a minimal set of nodes U satisfying equation (4.3). We define a graph $\mathcal{H}$ by introducing an edge $\{X, Y\}$ for all X and all $Y \in \text{MB}_P(X)$. Then the Markov network $\mathcal{H}$ is the unique minimal I-map for P .

The proof is left as an exercise (exercise 4.11).

Both of the techniques for constructing a minimal I-map make the assumption that the distribution P is positive. As we have shown, for nonpositive distributions, neither the pairwise independencies nor the local independencies imply the global one. Hence, for a nonpositive distribution P , constructing a graph $\mathcal{H}$ such that P satisfies the pairwise assumptions for $\mathcal{H}$ does not guarantee that $\mathcal{H}$ is an I-map for P . Indeed, we can easily demonstrate that both of these constructions break down for nonpositive distributions.

Example 4.7

Consider a nonpositive distribution P over four binary variables A, B, C, D that assigns nonzero probability only to cases where all four variables take on exactly the same value; for example, we might have $P(a^1, b^1, c^1, d^1) = 0.5$ and $P(a^0, b^0, c^0, d^0) = 0.5$. The graph $\mathcal{H}$ shown in figure 4.7 is one possible output of applying the local independence I-map construction algorithm to P : For example, $P \models (A \perp C, D \mid B)$, and hence $\{B\}$ is a legal choice for $\text{MB}_P(A)$. A similar analysis shows that this network satisfies the Markov blanket condition for all nodes. However, it is not an I-map for the distribution.

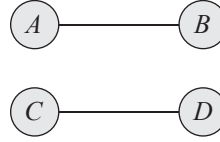


Figure 4.7 An attempt at an I-map for a nonpositive distribution P

If we use the pairwise independence I-map construction algorithm for this distribution, the network constructed is the empty network. For example, the algorithm would not place an edge between A and B , because $P \models (A \perp B \mid C, D)$. Exactly the same analysis shows that no edges will be placed into the graph. However, the resulting network is not an I-map for P . ■

Both these examples show that deterministic relations between variables can lead to failure in the construction based on local and pairwise independence. Suppose that A and B are two variables that are identical to each other and that both C and D are variables that correlated to both A and B so that $(C \perp D \mid A, B)$ holds. Since A is identical to B , we have that both $(A, D \perp C \mid B)$ and $(B, D \perp C \mid A)$ hold. In other words, it suffices to observe one of these two variables to capture the relevant information both have about C and separate C from D . In this case the Markov blanket of C is not uniquely defined. This ambiguity leads to the failure of both local and pairwise constructions. Clearly, identical variables are only one way of getting such ambiguities in local independencies. Once we allow nonpositive distribution, other distributions can have similar problems.

Having defined the notion of a minimal I-map for a distribution P , we can now ask to what extent it represents the independencies in P . More formally, we can ask whether every distribution has a perfect map. Clearly, the answer is no, even for positive distributions:

Example 4.8

Consider a distribution arising from a three-node Bayesian network with a v -structure, for example, the distribution induced in the Student example over the nodes Intelligence, Difficulty, and Grade (figure 3.3). In the Markov network for this distribution, we must clearly have an edge between I and G and between D and G . Can we omit the edge between I and D ? No, because we do not have that $(I \perp D \mid G)$ holds for the distribution; rather, we have the opposite: I and D are dependent given G . Therefore, the only minimal I-map for this P is the fully connected graph, which does not capture the marginal independence $(I \perp D)$ that holds in P . ■

This example provides another counterexample to the strong version of completeness mentioned earlier. The only distributions for which separation is a sound and complete criterion for determining conditional independence are those for which $\mathcal{H}$ is a perfect map.

4.4 Parameterization Revisited

Now that we understand the semantics and independence properties of Markov networks, we revisit some alternative representations for the parameterization of a Markov network.

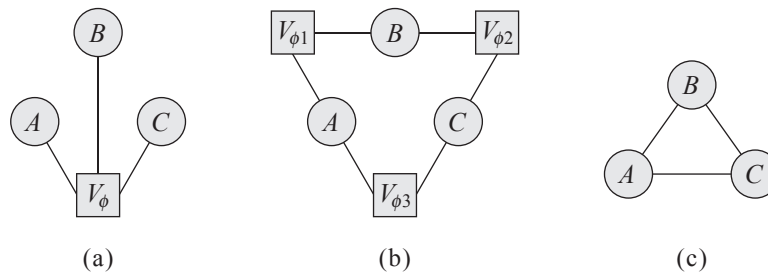


Figure 4.8 Different factor graphs for the same Markov network: (a) One factor graph over A, B, C , with a single factor over all three variables. (b) An alternative factor graph, with three pairwise factors. (c) The induced Markov network for both is a clique over A, B, C .

4.4.1 Finer-Grained Parameterization

4.4.1.1 Factor Graphs

A Markov network structure does not generally reveal all of the structure in a Gibbs parameterization. In particular, one cannot tell from the graph structure whether the factors in the parameterization involve maximal cliques or subsets thereof. Consider, for example, a Gibbs distribution P over a fully connected pairwise Markov network; that is, P is parameterized by a factor for each pair of variables $X, Y \in \mathcal{X}$. The clique potential parameterization would utilize a factor whose scope is the entire graph, and which therefore uses an exponential number of parameters. On the other hand, as we discussed in section 4.2.1, the number of parameters in the pairwise parameterization is quadratic in the number of variables. Note that the complete Markov network is not redundant in terms of conditional independencies — P does not factorize over any smaller network. Thus, although the finer-grained structure does not imply additional independencies in the distribution (see exercise 4.6), it is still very significant.

An alternative representation that makes this structure explicit is a *factor graph*. A factor graph is a graph containing two types of nodes: one type corresponds, as usual, to random variables; the other corresponds to factors over the variables. Formally:

Definition 4.13

factor graph

factorization

A factor graph $\mathcal{F}$ is an undirected graph containing two types of nodes: variable nodes (denoted as ovals) and factor nodes (denoted as squares). The graph only contains edges between variable nodes and factor nodes. A factor graph $\mathcal{F}$ is parameterized by a set of factors, where each factor node V_ϕ is associated with precisely one factor ϕ , whose scope is the set of variables that are neighbors of V_ϕ in the graph. A distribution P factorizes over $\mathcal{F}$ if it can be represented as a set of factors of this form. ■

Factor graphs make explicit the structure of the factors in the network. For example, in a fully connected pairwise Markov network, the factor graph would contain a factor node for each of the $\binom{n}{2}$ pairs of nodes; the factor node for a pair X_i, X_j would be connected to X_i and X_j ; by contrast, a factor graph for a distribution with a single factor over $X_1, \dots, X_n$ would have a single factor node connected to all of $X_1, \dots, X_n$ (see figure 4.8). Thus, although the Markov networks for these two distributions are identical, their factor graphs make explicit the

$\epsilon_1(A, B)$			$\epsilon_2(B, C)$			$\epsilon_3(C, D)$			$\epsilon_4(D, A)$		
a^0	b^0	-3.4	b^0	c^0	-4.61	c^0	d^0	0	d^0	a^0	-4.61
a^0	b^1	-1.61	b^0	c^1	0	c^0	d^1	-4.61	d^0	a^1	0
a^1	b^0	0	b^1	c^0	0	c^1	d^0	-4.61	d^1	a^0	0
a^1	b^1	-2.3	b^1	c^1	-4.61	c^1	d^1	0	d^1	a^1	-4.61
(a)			(b)			(c)			(d)		

Figure 4.9 Energy functions for the Misconception example

difference in their factorization.

4.4.1.2 Log-Linear Models

Although factor graphs make certain types of structure more explicit, they still encode factors as complete tables over the scope of the factor. As in Bayesian networks, factors can also exhibit a type of context-specific structure — patterns that involve particular values of the variables. These patterns are often more easily seen in terms of an alternative parameterization of the factors that converts them into log-space.

More precisely, we can rewrite a factor $\phi(\mathbf{D})$ as

$$\phi(\mathbf{D}) = \exp(-\epsilon(\mathbf{D})),$$

energy function

where $\epsilon(\mathbf{D}) = -\ln \phi(\mathbf{D})$ is often called an *energy function*. The use of the word “energy” derives from statistical physics, where the probability of a physical state (for example, a configuration of a set of electrons), depends inversely on its energy. In this logarithmic representation, we have

$$P(X_1, \dots, X_n) \propto \exp \left[- \sum_{i=1}^m \epsilon_i(\mathbf{D}_i) \right].$$

The logarithmic representation ensures that the probability distribution is positive. Moreover, the logarithmic parameters can take any value along the real line.

Any Markov network parameterized using positive factors can be converted to a logarithmic representation.

Example 4.9

Figure 4.9 shows the logarithmic representation of the clique potential parameters in figure 4.1. We can see that the “1” entries in the clique potentials translate into “0” entries in the energy function. ■

This representation makes certain types of structure in the potentials more apparent. For example, we can see that both $\epsilon_2(B, C)$ and $\epsilon_4(D, A)$ are constant multiples of an energy function that ascribes 1 to instantiations where the values of the two variables agree, and 0 to the instantiations where they do not.

We can provide a general framework for capturing such structure using the following notion:

Definition 4.14

feature

indicator feature

Let $\mathbf{D}$ be a subset of variables. We define a feature $f(\mathbf{D})$ to be a function from $\text{Val}(\mathbf{D})$ to $\mathbb{R}$. ■

A feature is simply a factor without the nonnegativity requirement. One type of feature of particular interest is the *indicator feature* that takes on value 1 for some values $\mathbf{y} \in \text{Val}(\mathbf{D})$ and 0 otherwise.

Features provide us with an easy mechanism for specifying certain types of interactions more compactly.

Example 4.10

Consider a situation where A_1 and A_2 each have ℓ values $a^1, \dots, a^\ell$. Assume that our distribution is such that we prefer situations where A_1 and A_2 take on the same value, but otherwise have no preference. Thus, our energy function might have the following form:

$$\epsilon(A_1, A_2) = \begin{cases} -3 & A_1 = A_2 \\ 0 & \text{otherwise} \end{cases}$$

Represented as a full factor, this clique potential requires ℓ^2 values. However, it can also be represented as a log-linear function in terms of a feature $f(A_1, A_2)$ that is an indicator function for the event $A_1 = A_2$. The energy function is then simply a constant multiple -3 of this feature. ■

Thus, we can provide a more general definition for our notion of log-linear models:

Definition 4.15

log-linear model

A distribution P is a log-linear model over a Markov network $\mathcal{H}$ if it is associated with:

- a set of features $\mathcal{F} = \{f_1(\mathbf{D}_1), \dots, f_k(\mathbf{D}_k)\}$, where each $\mathbf{D}_i$ is a complete subgraph in $\mathcal{H}$,
- a set of weights $w_1, \dots, w_k$,

such that

$$P(X_1, \dots, X_n) = \frac{1}{Z} \exp \left[- \sum_{i=1}^k w_i f_i(\mathbf{D}_i) \right].$$

■

Note that we can have several features over the same scope, so that we can, in fact, represent a standard set of table potentials. (See exercise 4.13.)

The log-linear model provides a much more compact representation for many distributions, especially in situations where variables have large domains such as text (such as box 4.E).

4.4.1.3 Discussion

We now have three representations of the parameterization of a Markov network. The Markov network denotes a product over potentials on cliques. A factor graph denotes a product of factors. And a set of features denotes a product over feature weights. Clearly, each representation is finer-grained than the previous one and as rich. A factor graph can describe the Gibbs distribution, and a set of features can describe all the entries in each of the factors of a factor graph.



Depending on the question of interest, different representations may be more appropriate. For example, a Markov network provides the right level of abstraction for discussing independence queries: The finer-grained representations of factor graphs or log-linear

models do not change the independence assertions made by the model. On the other hand, as we will see in later chapters, factor graphs are useful when we discuss inference, and features are useful when we discuss parameterizations, both for hand-coded models and for learning.

Ising model

Box 4.C — Concept: Ising Models and Boltzmann Machines. *One of the earliest types of Markov network models is the Ising model, which first arose in statistical physics as a model for the energy of a physical system involving a system of interacting atoms. In these systems, each atom is associated with a binary-valued random variable $X_i \in \{+1, -1\}$, whose value defines the direction of the atom's spin. The energy function associated with the edges is defined by a particularly simple parametric form:*

$$\epsilon_{i,j}(x_i, x_j) = w_{i,j}x_ix_j \quad (4.4)$$

This energy is symmetric in X_i, X_j ; it makes a contribution of $w_{i,j}$ to the energy function when $X_i = X_j$ (so both atoms have the same spin) and a contribution of $-w_{i,j}$ otherwise. Our model also contains a set of parameters u_i that encode individual node potentials; these bias individual variables to have one spin or another.

As usual, the energy function defines the following distribution:

$$P(\xi) = \frac{1}{Z} \exp \left(- \sum_{i < j} w_{i,j}x_ix_j - \sum_i u_ix_i \right).$$

As we can see, when $w_{i,j} > 0$ the model prefers to align the spins of the two atoms; in this case, the interaction is called ferromagnetic. When $w_{i,j} < 0$ the interaction is called antiferromagnetic. When $w_{i,j} = 0$ the atoms are non-interacting.

temperature
parameter

Much work has gone into studying particular types of Ising models, attempting to answer a variety of questions, usually as the number of atoms goes to infinity. For example, we might ask the probability of a configuration in which a majority of the spins are $+1$ or -1 , versus the probability of more mixed configurations. The answer to this question depends heavily on the strength of the interaction between the variables; so, we can consider adapting this strength (by multiplying all weights by a temperature parameter) and asking whether this change causes a phase transition in the probability of skewed versus mixed configurations. These questions, and many others, have been investigated extensively by physicists, and the answers are known (in some cases even analytically) for several cases.

Boltzmann
distribution

Related to the Ising model is the Boltzmann distribution; here, the variables are usually taken to have values $\{0, 1\}$, but still with the energy form of equation (4.4). Here, we get a nonzero contribution to the model from an edge (X_i, X_j) only when $X_i = X_j = 1$; however, the resulting energy can still be reformulated in terms of an Ising model (exercise 4.12).

The popularity of the Boltzmann machine was primarily driven by its similarity to an activation model for neurons. To understand the relationship, we note that the probability distribution over each variable X_i given an assignment to its neighbors is $\text{sigmoid}(z)$ where

$$z = - \left(\sum_j w_{i,j}x_j \right) - u_i.$$

This function is a sigmoid of a weighted combination of X_i 's neighbors, weighted by the strength and direction of the connections between them. This is the simplest but also most popular mathematical approximation of the function employed by a neuron in the brain. Thus, if we imagine a process by which the network continuously adapts its assignment by resampling the value of each variable as a stochastic function of its neighbors, then the “activation” probability of each variable resembles a neuron’s activity. This model is a very simple variant of a stochastic, recurrent neural network.

labeling MRF

Box 4.D — Concept: Metric MRFs. One important class of MRFs comprises those used for labeling. Here, we have a graph of nodes $X_1, \dots, X_n$ related by a set of edges $\mathcal{E}$, and we wish to assign to each X_i a label in the space $\mathcal{V} = \{v_1, \dots, v_K\}$. Each node, taken in isolation, has its preferences among the possible labels. However, we also want to impose a soft “smoothness” constraint over the graph, in that neighboring nodes should take “similar” values.

We encode the individual node preferences as node potentials in a pairwise MRF and the smoothness preferences as edge potentials. For reasons that will become clear, it is traditional to encode these models in negative log-space, using energy functions. As our objective in these models is inevitably the MAP objective, we can also ignore the partition function, and simply consider the energy function:

$$E(x_1, \dots, x_n) = \sum_i \epsilon_i(x_i) + \sum_{(i,j) \in \mathcal{E}} \epsilon_{i,j}(x_i, x_j). \quad (4.5)$$

Our goal is then to minimize the energy:

$$\arg \min_{x_1, \dots, x_n} E(x_1, \dots, x_n).$$

We now need to provide a formal definition for the intuition of “smoothness” described earlier. There are many different types of conditions that we can impose; different conditions allow different methods to be applied.

Ising model

One of the simplest in this class of models is a slight variant of the Ising model, where we have that, for any i, j :

$$\epsilon_{i,j}(x_i, x_j) = \begin{cases} 0 & x_i = x_j \\ \lambda_{i,j} & x_i \neq x_j, \end{cases} \quad (4.6)$$

for $\lambda_{i,j} \geq 0$. In this model, we obtain the lowest possible pairwise energy (0) when two neighboring nodes X_i, X_j take the same value, and a higher energy $\lambda_{i,j}$ when they do not.

Potts model

This simple model has been generalized in many ways. The Potts model extends it to the setting of more than two labels. An even broader class contains models where we have a distance function on the labels, and where we prefer neighboring nodes to have labels that are a smaller distance apart. More precisely, a function $\mu : \mathcal{V} \times \mathcal{V} \mapsto [0, \infty)$ is a metric if it satisfies:

metric function

- **Reflexivity:** $\mu(v_k, v_l) = 0$ if and only if $k = l$;
- **Symmetry:** $\mu(v_k, v_l) = \mu(v_l, v_k)$;

$\epsilon'_1(A, B)$			$\epsilon'_2(B, C)$		
a^0	b^0	-4.4	b^0	c^0	-3.61
a^0	b^1	-1.61	b^0	c^1	+1
a^1	b^0	-1	b^1	c^0	0
a^1	b^1	-2.3	b^1	c^1	-4.61
(a)			(b)		

Figure 4.10 Alternative but equivalent energy functions

- **Triangle Inequality:** $\mu(v_k, v_l) + \mu(v_l, v_m) \geq \mu(v_k, v_m)$.

semimetric

We say that μ is a semimetric if it satisfies reflexivity and symmetry. We can now define a metric MRF (or a semimetric MRF) by defining $\epsilon_{i,j}(v_k, v_l) = \mu(v_k, v_l)$ for all i, j , where μ is a metric (semimetric). We note that, as defined, this model assumes that the distance metric used is the same for all pairs of variables. This assumption is made because it simplifies notation, it often holds in practice, and it reduces the number of parameters that must be acquired. It is not required for the inference algorithms that we present in later chapters. Metric interactions arise in many applications, and play a particularly important role in computer vision (see box 4.B and box 13.B). For example, one common metric used is some form of truncated p -norm (usually $p = 1$ or $p = 2$):

truncated norm

$$\epsilon(x_i, x_j) = \min(c\|x_i - x_j\|_p, \text{dist}_{\max}). \quad (4.7)$$

4.4.2 Overparameterization

Even if we use finer-grained factors, and in some cases, even features, the Markov network parameterization is generally overparameterized. That is, for any given distribution, there are multiple choices of parameters to describe it in the model. Most obviously, if our graph is a single clique over n binary variables $X_1, \dots, X_n$, then the network is associated with a clique potential that has 2^n parameters, whereas the joint distribution only has $2^n - 1$ independent parameters.

A more subtle point arises in the context of a nontrivial clique structure. Consider a pair of cliques $\{A, B\}$ and $\{B, C\}$. The energy function $\epsilon_1(A, B)$ (or its corresponding clique potential) contains information not only about the interaction between A and B , but also about the distribution of the individual variables A and B . Similarly, $\epsilon_2(B, C)$ gives us information about the individual variables B and C . The information about B can be placed in either of the two cliques, or its contribution can be split between them in arbitrary ways, resulting in many different ways of specifying the same distribution.

Example 4.11

Consider the energy functions $\epsilon_1(A, B)$ and $\epsilon_2(B, C)$ in figure 4.9. The pair of energy functions shown in figure 4.10 result in an equivalent distribution: Here, we have simply subtracted 1 from $\epsilon_1(A, B)$ and added 1 to $\epsilon_2(B, C)$ for all instantiations where $B = b^0$. It is straightforward to

check that this results in an identical distribution to that of figure 4.9. In instances where $B \neq b^0$ the energy function returns exactly the same value as before. In cases where $B = b^0$, the actual values of the energy functions have changed. However, because the sum of the energy functions on each instance is identical to the original sum, the probability of the instance will not change. ■

Intuitively, the standard Markov network representation gives us too many places to account for the influence of variables in shared cliques. Thus, the same distribution can be represented as a Markov network (of a given structure) in infinitely many ways. It is often useful to pick one of this infinite set as our chosen parameterization for the distribution.

4.4.2.1 Canonical Parameterization

canonical
parameterization

The *canonical parameterization* provides one very natural approach to avoiding this ambiguity in the parameterization of a Gibbs distribution P . This canonical parameterization requires that the distribution P be positive. It is most convenient to describe this parameterization using energy functions rather than clique potentials. For this reason, it is also useful to consider a log-transform of P : For any assignment ξ to $\mathcal{X}$, we use $\ell(\xi)$ to denote $\ln P(\xi)$. This transformation is well defined because of our positivity assumption.

The canonical parameterization of a Gibbs distribution over $\mathcal{H}$ is defined via a set of energy functions over all cliques. Thus, for example, the Markov network in figure 4.4b would have energy functions for the two cliques $\{A, B, D\}$ and $\{B, C, D\}$, energy functions for all possible pairs of variables except the pair $\{A, C\}$ (a total of five pairs), energy functions for all four singleton sets, and a constant energy function for the empty clique.

At first glance, it appears that we have only increased the number of parameters in the specification. However, as we will see, this approach uniquely associates the interaction parameters for a subset of variables with that subset, avoiding the ambiguity described earlier. As a consequence, many of the parameters in this canonical parameterization are often zero.

The canonical parameterization is defined relative to a particular fixed assignment $\xi^* = (x_1^*, \dots, x_n^*)$ to the network variables $\mathcal{X}$. This assignment can be chosen arbitrarily. For any subset of variables $\mathbf{Z}$, and any assignment $\mathbf{x}$ to some subset of $\mathcal{X}$ that contains $\mathbf{Z}$, we define the assignment $\mathbf{x}_{\mathbf{Z}}$ to be $\mathbf{x}_{\langle \mathbf{Z} \rangle}$, that is, the assignment in $\mathbf{x}$ to the variables in $\mathbf{Z}$. Conversely, we define $\xi_{-\mathbf{Z}}^*$ to be $\xi^*_{\langle \mathcal{X} - \mathbf{Z} \rangle}$, that is, the assignment in ξ^* to the variables outside $\mathbf{Z}$. We can now construct an assignment $(\mathbf{x}_{\mathbf{Z}}, \xi_{-\mathbf{Z}}^*)$ that keeps the assignments to the variables in $\mathbf{Z}$ as specified in $\mathbf{x}$, and augments it using the default values in ξ^* .

canonical energy
function

The *canonical energy function* for a clique $\mathbf{D}$ is now defined as follows:

$$\epsilon_{\mathbf{D}}^*(d) = \sum_{\mathbf{Z} \subseteq \mathbf{D}} (-1)^{|\mathbf{D} - \mathbf{Z}|} \ell(d_{\mathbf{Z}}, \xi_{-\mathbf{Z}}^*), \quad (4.8)$$

where the sum is over all subsets of $\mathbf{D}$, including $\mathbf{D}$ itself and the empty set $\emptyset$. Note that all of the terms in the summation have a scope that is contained in $\mathbf{D}$, which in turn is part of a clique, so that these energy functions are legal relative to our Markov network structure.

This formula performs an inclusion-exclusion computation. For a set $\{A, B, C\}$, it first subtracts out the influence of all of the pairs: $\{A, B\}$, $\{B, C\}$, and $\{C, A\}$. However, this process oversubtracts the influence of the individual variables. Thus, their influence is added back in, to compensate. More generally, consider any subset of variables $\mathbf{Z} \subseteq \mathbf{D}$. Intuitively, it

$\epsilon_1^*(A, B)$			$\epsilon_2^*(B, C)$			$\epsilon_3^*(C, D)$			$\epsilon_4^*(D, A)$		
a^0	b^0	0	b^0	c^0	0	c^0	d^0	0	d^0	a^0	0
a^0	b^1	0	b^0	c^1	0	c^0	d^1	0	d^0	a^1	0
a^1	b^0	0	b^1	c^0	0	c^1	d^0	0	d^1	a^0	0
a^1	b^1	4.09	b^1	c^1	9.21	c^1	d^1	-9.21	d^1	a^1	9.21

$\epsilon_5^*(A)$		$\epsilon_6^*(B)$		$\epsilon_7^*(C)$		$\epsilon_8^*(D)$		$\epsilon_9^*(\emptyset)$
a^0	0	b^0	0	c^0	0	d^0	0	-3.18
a^1	-8.01	b^1	-6.4	c^1	0	d^1	0	

Figure 4.11 Canonical energy function for the Misconception example

makes a “contribution” once for every subset $U \supseteq Z$. Except for $U = D$, the number of times that Z appears is even — there is an even number of subsets $U \supseteq Z$ — and the number of times it appears with a positive sign is equal to the number of times it appears with a negative sign. Thus, we have effectively eliminated the net contribution of the subsets from the canonical energy function.

Let us consider the effect of the canonical transformation on our Misconception network.

Example 4.12

Let us choose (a^0, b^0, c^0, d^0) as our arbitrary assignment on which to base the canonical parameterization. The resulting energy functions are shown in figure 4.11. For example, the energy value $\epsilon_1^*(a^1, b^1)$ was computed as follows:

$$\begin{aligned} & \ell(a^1, b^1, c^0, d^0) - \ell(a^1, b^0, c^0, d^0) - \ell(a^0, b^1, c^0, d^0) + \ell(a^0, b^0, c^0, d^0) = \\ & -13.49 - -11.18 - -9.58 + -3.18 = 4.09 \end{aligned}$$

Note that many of the entries in the energy functions are zero. As discussed earlier, this phenomenon is fairly general, and occurs because we have accounted for the influence of small subsets of variables separately, leaving the larger factors to deal only with higher-order influences. We also note that these canonical parameters are not very intuitive, highlighting yet again the difficulties of constructing a reasonable parameterization of a Markov network by hand. ■

This canonical parameterization defines the same distribution as our original distribution P :

Theorem 4.7

Let P be a positive Gibbs distribution over $\mathcal{H}$, and let $\epsilon^*(D_i)$ for each clique D_i be defined as specified in equation (4.8). Then

$$P(\xi) = \exp \left[\sum_i \epsilon_{D_i}^*(\xi \langle D_i \rangle) \right].$$

The proof for the case where $\mathcal{H}$ consists of a single clique is fairly simple, and it is left as an exercise (exercise 4.4). The general case follows from results in the next section.

The canonical parameterization gives us the tools to prove the Hammersley-Clifford theorem, which we restate for convenience.

Theorem 4.8

Let P be a positive distribution over $\mathcal{X}$, and $\mathcal{H}$ a Markov network graph over $\mathcal{X}$. If $\mathcal{H}$ is an I-map for P , then P is a Gibbs distribution over $\mathcal{H}$.

PROOF To prove this result, we need to show the existence of a Gibbs parameterization for any distribution P that satisfies the Markov assumptions associated with $\mathcal{H}$. The proof is constructive, and simply uses the canonical parameterization shown earlier in this section. Given P , we define an energy function for all subsets D of nodes in the graph, regardless of whether they are cliques in the graph. This energy function is defined exactly as in equation (4.8), relative to some specific fixed assignment ξ^* used to define the canonical parameterization. The distribution defined using this set of energy functions is P : the argument is identical to the proof of theorem 4.7, for the case where the graph consists of a single clique (see exercise 4.4).

It remains only to show that the resulting distribution is a Gibbs distribution over $\mathcal{H}$. To show that, we need to show that the factors $\epsilon^*(D)$ are identically 0 whenever D is not a clique in the graph, that is, whenever the nodes in D do not form a fully connected subgraph. Assume that we have $X, Y \in D$ such that there is no edge between X and Y . For this proof, it helps to introduce the notation

$$\sigma_Z[x] = (x_Z, \xi_{-Z}^*).$$

Plugging this notation into equation (4.8), we have that:

$$\epsilon_D^*(d) = \sum_{Z \subseteq D} (-1)^{|D-Z|} \ell(\sigma_Z[d]).$$

We now rearrange the sum over subsets Z into a sum over groups of subsets. Let $W \subseteq D - \{X, Y\}$; then W , $W \cup \{X\}$, $W \cup \{Y\}$, and $W \cup \{X, Y\}$ are all subsets of Z . Hence, we can rewrite the summation over subsets of D as a summation over subsets of $D - \{X, Y\}$:

$$\begin{aligned} \epsilon_D^*(d) &= \sum_{W \subseteq D - \{X, Y\}} (-1)^{|D - \{X, Y\} - W|} \\ &\quad (\ell(\sigma_W[d]) - \ell(\sigma_{W \cup \{X\}}[d]) - \ell(\sigma_{W \cup \{Y\}}[d]) + \ell(\sigma_{W \cup \{X, Y\}}[d])). \end{aligned} \quad (4.9)$$

Now consider a specific subset W in this sum, and let u^* be $\xi^* \langle \mathcal{X} - D \rangle$ — the assignment to $\mathcal{X} - D$ in ξ . We now have that:

$$\begin{aligned} \ell(\sigma_{W \cup \{X, Y\}}[d]) - \ell(\sigma_{W \cup \{X\}}[d]) &= \ln \frac{P(x, y, w, u^*)}{P(x, y^*, w, u^*)} \\ &= \ln \frac{P(y \mid x, w, u^*) P(x, w, u^*)}{P(y^* \mid x, w, u^*) P(x, w, u^*)} \\ &= \ln \frac{P(y \mid x^*, w, u^*) P(x, w, u^*)}{P(y^* \mid x^*, w, u^*) P(x, w, u^*)} \\ &= \ln \frac{P(y \mid x^*, w, u^*) P(x^*, w, u^*)}{P(y^* \mid x^*, w, u^*) P(x^*, w, u^*)} \\ &= \ln \frac{P(x^*, y, w, u^*)}{P(x^*, y^*, w, u^*)} \\ &= \ell(\sigma_{W \cup \{Y\}}[d]) - \ell(\sigma_W[d]), \end{aligned}$$

where the third equality is a consequence of the fact that X and Y are not connected directly by an edge, and hence we have that $P \models (X \perp Y \mid \mathcal{X} - \{X, Y\})$. Thus, we have that each term in the outside summation in equation (4.9) adds to zero, and hence the summation as a whole is also zero, as required. ■



For positive distributions, we have already shown that all three sets of Markov assumptions are equivalent; putting these results together with theorem 4.1 and theorem 4.2, we obtain that, **for positive distributions, all four conditions — factorization and the three types of Markov assumptions — are all equivalent.**

4.4.2.2 Eliminating Redundancy

An alternative approach to the issue of overparameterization is to try to eliminate it entirely. We can do so in the context of a feature-based representation, which is sufficiently fine-grained to allow us to eliminate redundancies without losing expressive power. The tools for detecting and eliminating redundancies come from linear algebra.

linear
dependence

We say that a set of features $f_1, \dots, f_k$ is *linearly dependent* if there are constants $\alpha_0, \alpha_1, \dots, \alpha_k$, not all of which are 0, so that for all ξ

$$\alpha_0 + \sum_i \alpha_i f_i(\xi) = 0.$$

This is the usual definition of linear dependencies in linear algebra, where we view each feature as a vector whose entries are the value of the feature in each of the possible instantiations.

Example 4.13

Consider again the Misconception example. We can encode the log-factors in example 4.9 as a set of features by introducing indicator features of the form:

$$f_{a,b}(A, B) = \begin{cases} 1 & A = a, B = b \\ 0 & \text{otherwise.} \end{cases}$$

Thus, to represent $\epsilon_1(A, B)$, we introduce four features that correspond to the four entries in the energy function. Since A, B take on exactly one of these possible four values, we have that

$$f_{a^0,b^0}(A, B) + f_{a^0,b^1}(A, B) + f_{a^1,b^0}(A, B) + f_{a^1,b^1}(A, B) = 1.$$

Thus, this set of features is linearly dependent. ■

Example 4.14

Now consider also the features that capture $\epsilon_2(B, C)$ and their interplay with the features that capture $\epsilon_1(A, B)$. We start by noting that the sum $f_{a^0,b^0}(A, B) + f_{a^1,b^0}(A, B)$ is equal to 1 when $B = b^0$ and 0 otherwise. Similarly, $f_{b^0,c^0}(B, C) + f_{b^0,c^1}(B, C)$ is also an indicator for $B = b^0$. Thus we get that

$$f_{a^0,b^0}(A, B) + f_{a^1,b^0}(A, B) - f_{b^0,c^0}(B, C) - f_{b^0,c^1}(B, C) = 0.$$

And so these four features are linearly dependent. ■

As we now show, linear dependencies imply non-unique parameterization.

Proposition 4.5

Let $f_1, \dots, f_k$ be a set of features with weights $\mathbf{w} = \{w_1, \dots, w_k\}$ that form a log-linear representation of a distribution P . If there are coefficients $\alpha_0, \alpha_1, \dots, \alpha_k$ such that for all ξ

$$\alpha_0 + \sum_i \alpha_i f_i(\xi) = 0 \quad (4.10)$$

then the log-linear model with weights $\mathbf{w}' = \{w_1 + \alpha_1, \dots, w_k + \alpha_k\}$ also represents P .

PROOF Consider the distribution

$$P_{\mathbf{w}'}(\xi) \propto \exp \left\{ - \sum_i (w_i + \alpha_i) f_i(\xi) \right\}.$$

Using equation (4.10) we see that

$$- \sum_i (w_i + \alpha_i) f_i(\xi) = \alpha_0 - \sum_i w_i f_i(\xi).$$

Thus,

$$P_{\mathbf{w}'}(\xi) \propto e^{\alpha_0} \exp \left\{ - \sum_i w_i f_i(\xi) \right\} \propto P(\xi).$$

We conclude that $P_{\mathbf{w}'}(\xi) = P(\xi)$. ■

redundant

Motivated by this result, we say that a set of linearly dependent features is *redundant*. A *nonredundant* set of features is one where the features are not linearly dependent on each other. In fact, if the set of features is nonredundant, then each set of weights describes a unique distribution.

Proposition 4.6

Let $f_1, \dots, f_k$ be a set of nonredundant features, and let $\mathbf{w}, \mathbf{w}' \in \mathbb{R}^k$. If $\mathbf{w} \neq \mathbf{w}'$ then $P_{\mathbf{w}} \neq P_{\mathbf{w}'}$.

Example 4.15

Can we construct a nonredundant set of features for the Misconception example? We can determine the number of nonredundant features by building the 16×16 matrix of the values of the 16 features (four factors with four features each) in the 16 instances of the joint distribution. This matrix has rank of 9, which implies that a subset of 8 features will be a nonredundant subset. In fact, there are several such subsets. In particular, the canonical parameterization shown in figure 4.11 has nine features of nonzero weight, which form a nonredundant parameterization. The equivalence of the canonical parameterization (theorem 4.7) implies that this set of features has the same expressive power as the original set of features. To verify this, we can show that adding any other feature will lead to a linear dependency. Consider, for example, the feature f_{a^1, b^0} . We can verify that

$$f_{a^1, b^0} + f_{a^1, b^1} - f_{a^1} = 0.$$

Similarly, consider the feature f_{a^0, b^0} . Again we can find a linear dependency on other features:

$$f_{a^0, b^0} + f_{a^1} + f_{b^1} - f_{a^1, b^1} = 1.$$

Using similar arguments, we can show that adding any of the original features will lead to redundancy. Thus, this set of features can represent any parameterization in the original model. ■

4.5 Bayesian Networks and Markov Networks

We have now described two graphical representation languages: Bayesian networks and Markov networks. Example 3.8 and example 4.8 show that these two representations are incomparable as a language for representing independencies: each can represent independence constraints that the other cannot. In this section, we strive to provide more insight about the relationship between these two representations.

4.5.1 From Bayesian Networks to Markov Networks

Let us begin by examining how we might take a distribution represented using one of these frameworks, and represent it in the other. One can view this endeavor from two different perspectives: Given a Bayesian network $\mathcal{B}$, we can ask how to represent the distribution $P_{\mathcal{B}}$ as a parameterized Markov network; or, given a graph $\mathcal{G}$, we can ask how to represent the independencies in $\mathcal{G}$ using an undirected graph $\mathcal{H}$. In other words, we might be interested in finding a minimal I-map for a distribution $P_{\mathcal{B}}$, or a minimal I-map for the independencies $\mathcal{I}(\mathcal{G})$. We can see that these two questions are related, but each perspective offers its own insights.

Let us begin by considering a distribution $P_{\mathcal{B}}$, where $\mathcal{B}$ is a parameterized Bayesian network over a graph $\mathcal{G}$. Importantly, the parameterization of $\mathcal{B}$ can also be viewed as a parameterization for a Gibbs distribution: We simply take each CPD $P(X_i \mid \text{Pa}_{X_i})$ and view it as a factor of scope X_i, Pa_{X_i} . This factor satisfies additional normalization properties that are not generally true of all factors, but it is still a legal factor. This set of factors defines a Gibbs distribution, one whose partition function happens to be 1.

What is more important, **a Bayesian network conditioned on evidence $E = e$ also induces a Gibbs distribution: the one defined by the original factors *reduced* to the context $E = e$.**



Proposition 4.7

Let $\mathcal{B}$ be a Bayesian network over $\mathcal{X}$ and $E = e$ an observation. Let $\mathbf{W} = \mathcal{X} - E$. Then $P_{\mathcal{B}}(\mathbf{W} \mid e)$ is a Gibbs distribution defined by the factors $\Phi = \{\phi_{X_i}\}_{X_i \in \mathcal{X}}$, where

$$\phi_{X_i} = P_{\mathcal{B}}(X_i \mid \text{Pa}_{X_i})[E = e].$$

The partition function for this Gibbs distribution is $P(e)$.

The proof follows directly from the definitions. This result allows us to view any Bayesian network conditioned as evidence as a Gibbs distribution, and to bring to bear techniques developed for analysis of Markov networks.

What is the structure of the undirected graph that can serve as an I-map for a set of factors in a Bayesian network? In other words, what is the I-map for the Bayesian network structure $\mathcal{G}$? Going back to our construction, we see that we have created a factor for each family of X_i , containing all the variables in the family. Thus, in the undirected I-map, we need to have an edge between X_i and each of its parents, as well as between all of the parents of X_i . This observation motivates the following definition:

Definition 4.16

moralized graph

The moral graph $\mathcal{M}[\mathcal{G}]$ of a Bayesian network structure $\mathcal{G}$ over $\mathcal{X}$ is the undirected graph over $\mathcal{X}$ that contains an undirected edge between X and Y if: (a) there is a directed edge between them (in either direction), or (b) X and Y are both parents of the same node.¹ ■

1. The name *moralized graph* originated because of the supposed “morality” of marrying the parents of a node.

For example, figure 4.6a shows the moralized graph for the extended $\mathcal{B}^{student}$ network of figure 9.8.

The preceding discussion shows the following result:

Corollary 4.2

Let $\mathcal{G}$ be a Bayesian network structure. Then for any distribution $P_{\mathcal{B}}$ such that $\mathcal{B}$ is a parameterization of $\mathcal{G}$, we have that $\mathcal{M}[\mathcal{G}]$ is an I-map for $P_{\mathcal{B}}$.

One can also view the moralized graph construction purely from the perspective of the independencies encoded by a graph, avoiding completely the discussion of parameterizations of the network.

Proposition 4.8

Let $\mathcal{G}$ be any Bayesian network graph. The moralized graph $\mathcal{M}[\mathcal{G}]$ is a minimal I-map for $\mathcal{G}$.

Markov blanket

PROOF We want to build a Markov network $\mathcal{H}$ such that $\mathcal{I}(\mathcal{H}) \subseteq \mathcal{I}(\mathcal{G})$, that is, that $\mathcal{H}$ is an I-map for $\mathcal{G}$ (see definition 3.3). We use the algorithm for constructing minimal I-maps based on the Markov independencies. Consider a node X in $\mathcal{X}$: our task is to select as X 's neighbors the smallest set of nodes $\mathcal{U}$ that are needed to render X independent of all other nodes in the network. We define the *Markov blanket* of X in a Bayesian network $\mathcal{G}$, denoted $\text{MB}_{\mathcal{G}}(X)$, to be the nodes consisting of X 's parents, X 's children, and other parents of X 's children. We now need to show that $\text{MB}_{\mathcal{G}}(X)$ d-separates X from all other variables in $\mathcal{G}$; and that no subset of $\text{MB}_{\mathcal{G}}(X)$ has that property. The proof uses straightforward graph-theoretic properties of trails, and it is left as an exercise (exercise 4.14). ■



moral graph

Now, let us consider how “close” the moralized graph is to the original graph $\mathcal{G}$. Intuitively, **the addition of the moralizing edges to the Markov network $\mathcal{H}$ leads to the loss of independence information implied by the graph structure.** For example, if our Bayesian network $\mathcal{G}$ has the form $X \rightarrow Z \leftarrow Y$, with no edge between X and Y , the Markov network $\mathcal{M}[\mathcal{G}]$ loses the information that X and Y are marginally independent (not given Z). However, information is not always lost. Intuitively, moralization causes loss of information about independencies only when it introduces new edges into the graph. We say that a Bayesian network $\mathcal{G}$ is *moral* if it contains no immoralities (as in definition 3.11); that is, for any pair of variables X, Y that share a child, there is a covering edge between X and Y . It is not difficult to show that:

Proposition 4.9

If the directed graph $\mathcal{G}$ is moral, then its moralized graph $\mathcal{M}[\mathcal{G}]$ is a perfect map of $\mathcal{G}$.

PROOF Let $\mathcal{H} = \mathcal{M}[\mathcal{G}]$. We have already shown that $\mathcal{I}(\mathcal{H}) \subseteq \mathcal{I}(\mathcal{G})$, so it remains to show the opposite inclusion. Assume by contradiction that there is an independence $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) \in \mathcal{I}(\mathcal{G})$ which is not in $\mathcal{I}(\mathcal{H})$. Thus, there must exist some trail from $\mathbf{X}$ to $\mathbf{Y}$ in $\mathcal{H}$ which is active given $\mathbf{Z}$. Consider some such trail that is minimal, in the sense that it has no shortcuts. As $\mathcal{H}$ and $\mathcal{G}$ have precisely the same edges, the same trail must exist in $\mathcal{G}$. As, by assumption, it cannot be active in $\mathcal{G}$ given $\mathbf{Z}$, we conclude that it must contain a v-structure $X_1 \rightarrow X_2 \leftarrow X_3$. However, because $\mathcal{G}$ is moralized, we also have some edge between X_1 and X_3 , contradicting the assumption that the trail is minimal. ■

Thus, a moral graph $\mathcal{G}$ can be converted to a Markov network without losing independence assumptions. This conclusion is fairly intuitive, inasmuch as the only independencies in $\mathcal{G}$ that are not present in an undirected graph containing the same edges are those corresponding to

v-structures. But if any v-structure can be short-cut, it induces no independencies that are not represented in the undirected graph.

We note, however, that very few directed graphs are moral. For example, assume that we have a v-structure $X \rightarrow Y \leftarrow Z$, which is moral due to the existence of an arc $X \rightarrow Z$. If Z has another parent W , it also has a v-structure $X \rightarrow Z \leftarrow W$, which, to be moral, requires some edge between X and W . We return to this issue in section 4.5.3.

4.5.1.1 Soundness of d-Separation

The connection between Bayesian networks and Markov networks provides us with the tools for proving the soundness of the d-separation criterion in Bayesian networks.

The idea behind the proof is to leverage the soundness of separation in undirected graphs, a result which (as we showed) is much easier to prove. Thus, we want to construct an undirected graph $\mathcal{H}$ such that active paths in $\mathcal{H}$ correspond to active paths in $\mathcal{G}$. A moment of thought shows that the moralized graph is not the right construct, because there are paths in the undirected graph that correspond to v-structures in $\mathcal{G}$ that may or may not be active. For example, if our graph $\mathcal{G}$ is $X \rightarrow Z \leftarrow Y$ and Z is not observed, d-separation tells us that X and Y are independent; but the moralized graph for $\mathcal{G}$ is the complete undirected graph, which does not have the same independence.

barren node

Therefore, to show the result, we first want to eliminate v-structures that are not active, so as to remove such cases. To do so, we first construct a subgraph where remove all *barren nodes* from the graph, thereby also removing all v-structures that do not have an observed descendant. The elimination of the barren nodes does not change the independence properties of the distribution over the remaining variables, but does eliminate paths in the graph involving v-structures that are not active. If we now consider only the subgraph, we can reduce d-separation to separation and utilize the soundness of separation to show the desired result.

upward closure

We first use these intuitions to provide an alternative formulation for d-separation. Recall that in definition 2.14 we defined the *upward closure* of a set of nodes U in a graph to be $U \cup \text{Ancestors}_U$. Letting U^* be the closure of a set U , we can define the network induced over U^* ; importantly, as all parents of every node in U^* are also in U^* , we have all the variables mentioned in every CPD, so that the induced graph defines a coherent probability distribution. We let $\mathcal{G}^+[U]$ be the induced Bayesian network over U and its ancestors.

Proposition 4.10

Let X, Y, Z be three disjoint sets of nodes in a Bayesian network $\mathcal{G}$. Let $U = X \cup Y \cup Z$, and let $\mathcal{G}' = \mathcal{G}^+[U]$ be the induced Bayesian network over $U \cup \text{Ancestors}_U$. Let $\mathcal{H}$ be the moralized graph $\mathcal{M}[\mathcal{G}']$. Then $\text{d-sep}_{\mathcal{G}}(X; Y \mid Z)$ if and only if $\text{sep}_{\mathcal{H}}(X; Y \mid Z)$.

Example 4.16

To gain some intuition for this result, consider the Bayesian network $\mathcal{G}$ of figure 4.12a (which extends our Student network). Consider the d-separation query $\text{d-sep}_{\mathcal{G}}(D; I \mid L)$. In this case, $U = \{D, I, L\}$, and hence the moralized graph $\mathcal{M}[\mathcal{G}^+[U]]$ is the graph shown in figure 4.12b, where we have introduced an undirected moralizing edge between D and I . In the resulting graph, D and I are not separated given L , exactly as we would have concluded using the d-separation procedure on the original graph.

On the other hand, consider the d-separation query $\text{d-sep}_{\mathcal{G}}(D; I \mid S, A)$. In this case, $U = \{D, I, S, A\}$. Because D and I are not spouses in $\mathcal{G}^+[U]$, the moralization process does not add

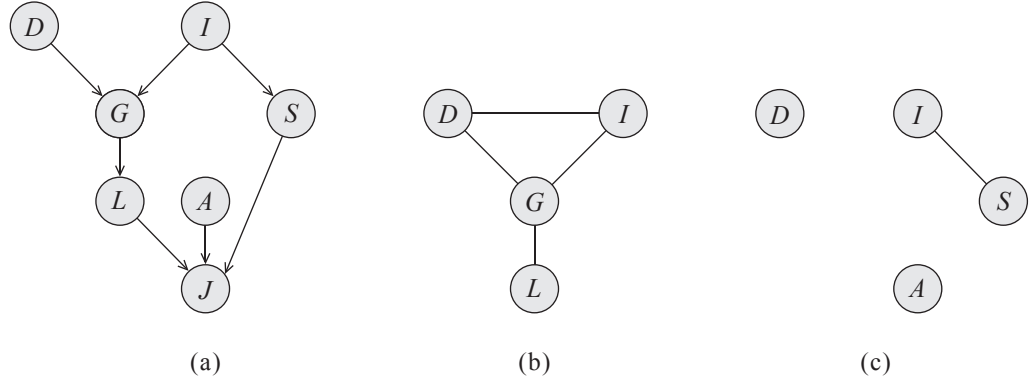


Figure 4.12 Example of alternative definition of d-separation based on Markov networks. (a) A Bayesian network $\mathcal{G}$. (b) The Markov network $\mathcal{M}[\mathcal{G}^+[D, I, L]]$. (c) The Markov network $\mathcal{M}[\mathcal{G}^+[D, I, A, S]]$.

an edge between them. The resulting moralized graph is shown in figure 4.12c. As we can see, we have that $\text{sep}_{\mathcal{M}[\mathcal{G}^+[U]]}(D; I \mid S, A)$, as desired. ■

The proof for the general case is similar and is left as an exercise (exercise 4.15).

With this result, the soundness of d-separation follows easily. We repeat the statement of theorem 3.3:

Theorem 4.9

If a distribution $P_{\mathcal{B}}$ factorizes according to $\mathcal{G}$, then $\mathcal{G}$ is an I-map for P .

PROOF As in proposition 4.10, let $U = X \cup Y \cup Z$, let $U^* = U \cup \text{Ancestors}_U$, let $\mathcal{G}_{U^*} = \mathcal{G}^+[U]$ be the induced graph over U^* , and let $\mathcal{H}$ be the moralized graph $\mathcal{M}[\mathcal{G}_{U^*}]$. Let P_{U^*} be the Bayesian network distribution defined over $\mathcal{G}_{U^*}$ in the obvious way: the CPD for any variable in U^* is the same as in $\mathcal{B}$. Because U^* is upwardly closed, all variables used in these CPDs are in U^* .

Now, consider an independence assertion $(X \perp Y \mid Z) \in \mathcal{I}(\mathcal{G})$; we want to prove that $P_{\mathcal{B}} \models (X \perp Y \mid Z)$. By definition 3.7, if $(X \perp Y \mid Z) \in \mathcal{I}(\mathcal{G})$, we have that $d\text{-sep}_{\mathcal{G}}(X; Y \mid Z)$. It follows that $\text{sep}_{\mathcal{H}}(X; Y \mid Z)$, and hence that $(X \perp Y \mid Z) \in \mathcal{I}(\mathcal{H})$. P_{U^*} is a Gibbs distribution over $\mathcal{H}$, and hence, from theorem 4.1, $P_{U^*} \models (X \perp Y \mid Z)$. Using exercise 3.8, the distribution $P_{U^*}(U^*)$ is the same as $P_{\mathcal{B}}(U^*)$. Hence, it follows also that $P_{\mathcal{B}} \models (X \perp Y \mid Z)$, proving the desired result. ■

4.5.2 From Markov Networks to Bayesian Networks

The previous section dealt with the conversion from a Bayesian network to a Markov network. We now consider the converse transformation: finding a Bayesian network that is a minimal I-map for a Markov network. It turns out that the transformation in this direction is significantly more difficult, both conceptually and computationally. Indeed, the Bayesian network that is a minimal I-map for a Markov network might be considerably larger than the Markov network.

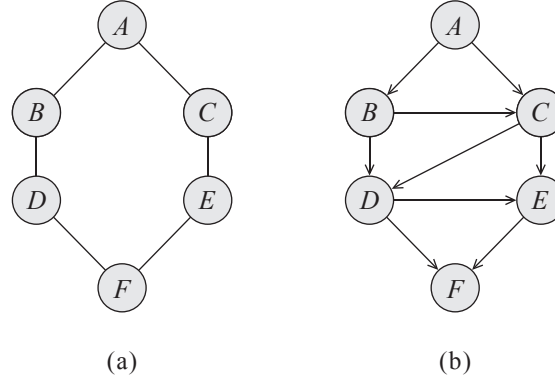


Figure 4.13 Minimal I-map Bayesian networks for a nonchordal Markov network. (a) A Markov network $\mathcal{H}_\ell$ with a loop. (b) A minimal I-map $\mathcal{G}_\ell$ Bayesian network for $\mathcal{H}$.

Example 4.17

Consider the Markov network structure $\mathcal{H}_\ell$ of figure 4.13a, and assume that we want to find a Bayesian network I-map for $\mathcal{H}_\ell$. As we discussed in section 3.4.1, we can find such an I-map by enumerating the nodes in $\mathcal{X}$ in some ordering, and define the parent set for each one in turn according to the independencies in the distribution. Assume we enumerate the nodes in the order A, B, C, D, E, F . The process for A and B is obvious. Consider what happens when we add C . We must, of course, introduce A as a parent for C . More interestingly, however, C is not independent of B given A ; hence, we must also add B as a parent for C . Now, consider the node D . One of its parents must be B . As D is not independent of C given B , we must add C as a parent for D . We do not need to add A , as D is independent of A given B and C . Similarly, E 's parents must be C and D . Overall, the minimal Bayesian network I-map according to this ordering has the structure $\mathcal{G}_\ell$ shown in figure 4.13b. ■

A quick examination of the structure $\mathcal{G}_\ell$ shows that we have added several edges to the graph, resulting in a set of triangles crisscrossing the loop. In fact, the graph $\mathcal{G}_\ell$ in figure 4.13b is chordal: all loops have been partitioned into triangles.

One might hope that a different ordering might lead to fewer edges being introduced. Unfortunately, this phenomenon is a general one: any Bayesian network I-map for this Markov network must add triangulating edges into the graph, so that the resulting graph is chordal (see definition 2.24). In fact, we can show the following property, which is even stronger:

Theorem 4.10

Let $\mathcal{H}$ be a Markov network structure, and let $\mathcal{G}$ be any Bayesian network minimal I-map for $\mathcal{H}$. Then $\mathcal{G}$ can have no immoralities (see definition 3.11).

PROOF Let $X_1, \dots, X_n$ be a topological ordering for $\mathcal{G}$. Assume, by contradiction, that there is some immorality $X_i \rightarrow X_j \leftarrow X_k$ in $\mathcal{G}$ such that there is no edge between X_i and X_k ; assume (without loss of generality) that $i < k < j$.

Owing to minimality of the I-map $\mathcal{G}$, if X_i is a parent of X_j , then X_i and X_j are not separated by X_j 's other parents. Thus, $\mathcal{H}$ necessarily contains one or more paths between X_i

and X_j that are not cut by X_k (or by X_j 's other parents). Similarly, $\mathcal{H}$ necessarily contains one or more paths between X_k and X_j that are not cut by X_i (or by X_j 's other parents).

Consider the parent set $\mathbf{U}$ that was chosen for X_k . By our previous argument, there are one or more paths in $\mathcal{H}$ between X_i and X_k via X_j . As $i < k$, and X_i is not a parent of X_k (by our assumption), we have that $\mathbf{U}$ must cut all of those paths. To do so, $\mathbf{U}$ must cut either all of the paths between X_i and X_j , or all of the paths between X_j and X_k : As long as there is at least one active path from X_i to X_j and one from X_j to X_k , there is an active path between X_i and X_k that is not cut by $\mathbf{U}$. Assume, without loss of generality, that $\mathbf{U}$ cuts all paths between X_j and X_k (the other case is symmetrical). Now, consider the choice of parent set for X_j , and recall that it is the (unique) minimal subset among $X_1, \dots, X_{j-1}$ that separates X_j from the others. In a Markov network, this set consists of all nodes in $X_1, \dots, X_{j-1}$ that are the first on some uncut path from X_j . As $\mathbf{U}$ separates X_k from X_j , it follows that X_k cannot be the first on any uncut path from X_j , and therefore X_k cannot be a parent of X_j . This result provides the desired contradiction. ■

Because any nontriangulated loop of length at least 4 in a Bayesian network graph necessarily contains an immorality, we conclude:

Corollary 4.3

Let $\mathcal{H}$ be a Markov network structure, and let $\mathcal{G}$ be any minimal I-map for $\mathcal{H}$. Then $\mathcal{G}$ is necessarily chordal.

triangulation

Thus, the process of turning a Markov network into a Bayesian network requires that we add enough edges to a graph to make it chordal. This process is called *triangulation*. As in the transformation from Bayesian networks to Markov networks, the addition of edges leads to the loss of independence information. For instance, in example 4.17, the Bayesian network $\mathcal{G}_\ell$ in figure 4.13b loses the information that C and D are independent given A and F . In the transformation from directed to undirected models, however, the edges added are only the ones that are, in some sense, implicitly there — the edges required by the fact that each factor in a Bayesian network involves an entire family (a node and its parents). By contrast, the transformation from Markov networks to Bayesian networks can lead to the introduction of a large number of edges, and, in many cases, to the creation of very large families (exercise 4.16).

4.5.3 Chordal Graphs

We have seen that the conversion in either direction between Bayesian networks to Markov networks can lead to the addition of edges to the graph and to the loss of independence information implied by the graph structure. It is interesting to ask when a set of independence assumptions can be represented perfectly by both a Bayesian network and a Markov network. It turns out that this class is precisely the class of undirected chordal graphs.

The proof of one direction is fairly straightforward, based on our earlier results.

Theorem 4.11

Let $\mathcal{H}$ be a nonchordal Markov network. Then there is no Bayesian network $\mathcal{G}$ which is a perfect map for $\mathcal{H}$ (that is, such that $\mathcal{I}(\mathcal{H}) = \mathcal{I}(\mathcal{G})$).

PROOF The proof follows from the fact that the minimal I-map for $\mathcal{G}$ must be chordal. Hence, any I-map $\mathcal{G}$ for $\mathcal{I}(\mathcal{H})$ must include edges that are not present in $\mathcal{H}$. Because any additional edge eliminates independence assumptions, it is not possible for any Bayesian network $\mathcal{G}$ to precisely encode $\mathcal{I}(\mathcal{H})$. ■

To prove the other direction of this equivalence, we first prove some important properties of chordal graphs. As we will see, chordal graphs and the properties we now show play a central role in the derivation of exact inference algorithms for graphical models. For the remainder of this discussion, we restrict attention to connected graphs; the extension to the general case is straightforward. The basic result we show is that we can decompose any connected chordal graph $\mathcal{H}$ into a *tree of cliques* — a tree whose nodes are the maximal cliques in $\mathcal{H}$ — so that the structure of the tree precisely encodes the independencies in $\mathcal{H}$. (In the case of disconnected graphs, we obtain a forest of cliques, rather than a tree.)

sepsset

We begin by introducing some notation. Let $\mathcal{H}$ be a connected undirected graph, and let $C_1, \dots, C_k$ be the set of maximal cliques in $\mathcal{H}$. Let $\mathcal{T}$ be any tree-structured graph whose nodes correspond to the maximal cliques $C_1, \dots, C_k$. Let C_i, C_j be two cliques in the tree that are directly connected by an edge; we define $S_{i,j} = C_i \cap C_j$ to be a *sepsset* between C_i and C_j . Let $W_{<(i,j)}$ ($W_{<(j,i)}$) be all of the variables that appear in any clique on the C_i (C_j) side of the edge. Thus, each edge decomposes $\mathcal{X}$ into three disjoint sets: $W_{<(i,j)} - S_{i,j}$, $W_{<(j,i)} - S_{i,j}$, and $S_{i,j}$.

Definition 4.17

clique tree

We say that a tree $\mathcal{T}$ is a *clique tree* for $\mathcal{H}$ if:

- each node corresponds to a clique in $\mathcal{H}$, and each maximal clique in $\mathcal{H}$ is a node in $\mathcal{T}$;
- each sepsset $S_{i,j}$ separates $W_{<(i,j)}$ and $W_{<(j,i)}$ in $\mathcal{H}$.

■

Note that this definition implies that each separator $S_{i,j}$ renders its two sides conditionally independent in $\mathcal{H}$.

Example 4.18

Consider the Bayesian network graph $\mathcal{G}_\ell$ in figure 4.13b. Since it contains no immoralities, its moralized graph $\mathcal{H}'_\ell$ is simply the same graph, but where all edges have been made undirected. As $\mathcal{G}_\ell$ is chordal, so is $\mathcal{H}'_\ell$. The clique tree for $\mathcal{H}'_\ell$ is simply a chain $\{A, B, C\} \rightarrow \{B, C, D\} \rightarrow \{C, D, E\} \rightarrow \{D, E, F\}$, which clearly satisfies the separation requirements of the clique tree definition. ■

Theorem 4.12

Every undirected chordal graph $\mathcal{H}$ has a clique tree $\mathcal{T}$.

PROOF We prove the theorem by induction on the number of nodes in the graph. The base case of a single node is trivial. Now, consider a chordal graph $\mathcal{H}$ of size > 1 . If $\mathcal{H}$ consists of a single clique, then the theorem holds trivially. Therefore, consider the case where we have at least two nodes X_1, X_2 that are not connected directly by an edge. Assume that X_1 and X_2

are connected, otherwise the inductive step holds trivially. Let S be a minimal subset of nodes that separates X_1 and X_2 .

The removal of the set S breaks up the graph into at least two disconnected components — one containing X_1 , another containing X_2 , and perhaps additional ones. Let W_1, W_2 be some partition of the variables in $\mathcal{X} - S$ into two disjoint components, such that W_i encompasses the connected component containing X_i . (The other connected components can be assigned to W_1 or W_2 arbitrarily.) We first show that S must be a complete subgraph. Let Z_1, Z_2 be any two variables in S . Due to the minimality of S , each Z_i must lie on a path between X_1 and X_2 that does not go through any other node in S . (Otherwise, we could eliminate Z_i from S while still maintaining separation.) We can therefore construct a minimal path from Z_1 to Z_2 that goes only through nodes in W_1 by constructing a path from Z_1 to X_1 to Z_2 that goes only through W_1 , and by eliminating any shortcuts. We can similarly construct a minimal path from Z_1 to Z_2 that goes only through nodes in W_2 . The two paths together form a cycle of length ≥ 4 . Because of chordality, the cycle must have a chord, which, by construction, must be the edge $Z_1 - Z_2$.

Now consider the induced graph $\mathcal{H}_1 = \mathcal{H}[W_1 \cup S]$. As $X_2 \notin \mathcal{H}_1$, this induced graph is smaller than $\mathcal{H}$. Moreover, $\mathcal{H}_1$ is chordal, so we can apply the inductive hypothesis. Let $\mathcal{T}_1$ be the clique tree for $\mathcal{H}_1$. Because S is a complete connected subgraph, it is either a maximal clique or a subset of some maximal clique in $\mathcal{H}_1$. Let C_1 be some clique in $\mathcal{T}_1$ containing S (there may be more than one such clique). We can similarly define $\mathcal{H}_2$ and C_2 for X_2 . If neither C_1 nor C_2 is equal to S , we construct a tree $\mathcal{T}$ that contains the union of the cliques in $\mathcal{T}_1$ and $\mathcal{T}_2$, and connects C_1 and C_2 by an edge. Otherwise, without loss of generality, let $C_1 = S$; we create $\mathcal{T}$ by merging $\mathcal{T}_1$ minus C_1 into $\mathcal{T}_2$, making all of C_1 's neighbors adjacent to C_2 instead.

It remains to show that the resulting structure is a clique tree for $\mathcal{H}$. First, we note that there is no clique in $\mathcal{H}$ that intersects both W_1 and W_2 ; hence, any maximal clique in $\mathcal{H}$ is a maximal clique in either $\mathcal{H}_1$ or $\mathcal{H}_2$ (or both in the possible case of S), so that all maximal cliques in $\mathcal{H}$ appear in $\mathcal{T}$. Thus, the nodes in $\mathcal{T}$ are precisely the maximal cliques in $\mathcal{H}$. Second, we need to show that any $S_{i,j}$ separates $W_{<(i,j)}$ and $W_{<(j,i)}$. Consider two variables $X \in W_{<(i,j)}$ and $Y \in W_{<(j,i)}$. First, assume that $X, Y \in \mathcal{H}_1$; as all the nodes in $\mathcal{H}_1$ are on the $\mathcal{T}_1$ side of the tree, we also have that $S_{i,j} \subset \mathcal{H}_1$. Any path between two nodes in $\mathcal{H}_1$ that goes through W_2 can be shortcut to go only through $\mathcal{H}_1$. Thus, if $S_{i,j}$ separates X, Y in $\mathcal{H}_1$, then it also separates them in $\mathcal{H}$. The same argument applies for $X, Y \in \mathcal{H}_2$. Now, consider $X \in W_1$ and $Y \in W_2$. If $S_{i,j} = S$, the result follows from the fact that S separates W_1 and W_2 . Otherwise, assume that $S_{i,j}$ is in $\mathcal{T}_1$, on the path from X to C_1 . In this case, we have that $S_{i,j}$ separates X from S , and S separates $S_{i,j}$ from Y . The conclusion now follows from the transitivity of graph separation.

We have therefore constructed a clique tree for $\mathcal{H}$, proving the inductive claim. $\blacksquare$

Using this result, we can show that the independencies in an undirected graph $\mathcal{H}$ can be captured perfectly in a Bayesian network if and only if $\mathcal{H}$ is chordal.

Theorem 4.13

Let $\mathcal{H}$ be a chordal Markov network. Then there is a Bayesian network $\mathcal{G}$ such that $\mathcal{I}(\mathcal{H}) = \mathcal{I}(\mathcal{G})$.

PROOF Let $\mathcal{T}$ be the clique tree for $\mathcal{H}$, whose existence is guaranteed by theorem 4.12. We can select an ordering over the nodes in the Bayesian network as follows. We select an arbitrary

clique C_1 to be the root of the clique tree, and then order the cliques $C_1, \dots, C_k$ using any topological ordering, that is, where cliques closer to the root are ordered first. We now order the nodes in the network in any ordering consistent with the clique ordering: if X_l first appears in C_i and X_m first appears in C_j , for $i < j$, then X_l must precede X_m in the ordering. We now construct a Bayesian network using the procedure Build-Minimal-I-Map of algorithm 3.2 applied to the resulting node ordering $X_1, \dots, X_n$ and to $\mathcal{I}(\mathcal{H})$.

Let $\mathcal{G}$ be the resulting network. We first show that, when X_i is added to the graph, then X_i 's parents are precisely $U_i = \text{Nb}_{X_i} \cap \{X_1, \dots, X_{i-1}\}$, where Nb_{X_i} is the set of neighbors of X_i in $\mathcal{H}$. In other words, we want to show that X_i is independent of $\{X_1, \dots, X_{i-1}\} - U_i$ given U_i . Let C_k be the first clique in the clique ordering to which X_i belongs. Then $U_i \subset C_k$. Let C_l be the parent of C_k in the rooted clique tree. According to our selected ordering, all of the variables in C_l are ordered before any variable in $C_k - C_l$. Thus, $S_{l,k} \subset \{X_1, \dots, X_{i-1}\}$. Moreover, from our choice of ordering, none of $\{X_1, \dots, X_{i-1}\} - U_i$ are in any descendants of C_k in the clique tree. Thus, they are all in $W_{<(l,k)}$. From theorem 4.12, it follows that $S_{l,k}$ separates X_i from all of $\{X_1, \dots, X_{i-1}\} - U_i$, and hence that X_i is independent of all of $\{X_1, \dots, X_{i-1}\} - U_i$ given U_i . It follows that $\mathcal{G}$ and $\mathcal{H}$ have the same set of edges. Moreover, we note that all of U_i are in C_k , and hence are connected in $\mathcal{G}$. Therefore, $\mathcal{G}$ is moralized. As $\mathcal{H}$ is the moralized undirected graph of $\mathcal{G}$, the result now follows from proposition 4.9. ■

For example, the graph $\mathcal{G}_\ell$ of figure 4.13b, and its moralized network $\mathcal{H}'_\ell$ encode precisely the same independencies. By contrast, as we discussed, there exists no Bayesian network that encodes precisely the independencies in the nonchordal network $\mathcal{H}_\ell$ of figure 4.13a.

Thus, we have shown that chordal graphs are precisely the intersection between Markov networks and Bayesian networks, in that the independencies in a graph can be represented exactly in both types of models if and only if the graph is chordal.

4.6 Partially Directed Models

So far, we have presented two distinct types of graphical models, based on directed and undirected graphs. We can unify both representations by allowing models that incorporate both directed and undirected dependencies. We begin by describing the notion of *conditional random field*, a Markov network with a directed dependency on some subset of variables. We then present a generalization of this framework to the class of *chain graphs*, an entire network in which undirected components depend on each other in a directed fashion.

4.6.1 Conditional Random Fields

So far, we have described the Markov network representation as encoding a joint distribution over $\mathcal{X}$. The same undirected graph representation and parameterization can also be used to encode a *conditional distribution* $P(\mathbf{Y} \mid \mathbf{X})$, where $\mathbf{Y}$ is a set of *target variables* and $\mathbf{X}$ is a (disjoint) set of *observed variables*. We will also see a directed analogue of this concept in section 5.6. In the case of Markov networks, this representation is generally called a *conditional random field* (CRF).

target variable

observed variable

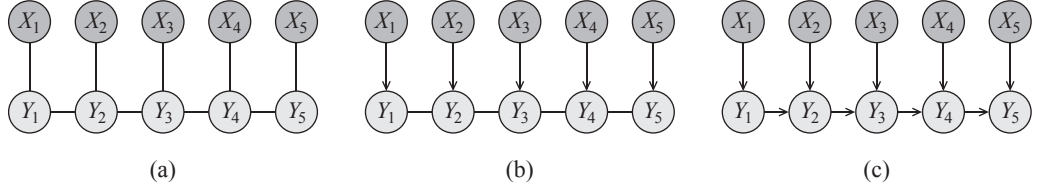


Figure 4.14 Different linear-chain graphical models: (a) a linear-chain-structured conditional random field, where the feature variables are denoted using grayed-out ovals; (b) a partially directed variant; (c) a fully directed, non-equivalent model. The X_i 's are assumed to be always observed when the network is used, and hence they are shown as darker gray.

4.6.1.1 CRF Representation and Semantics

More formally, a CRF is an undirected graph whose nodes correspond to $\mathbf{Y} \cup \mathbf{X}$. At a high level, this graph is parameterized in the same way as an ordinary Markov network, as a set of factors $\phi_1(\mathbf{D}_1), \dots, \phi_m(\mathbf{D}_m)$. (As before, these factors can also be encoded more compactly as a log-linear model; for uniformity of presentation, we view the log-linear model as encoding a set of factors.) However, rather than encoding the distribution $P(\mathbf{Y}, \mathbf{X})$, we view it as representing the conditional distribution $P(\mathbf{Y} \mid \mathbf{X})$. To have the network structure and parameterization correspond naturally to a conditional distribution, we want to avoid representing a probabilistic model over $\mathbf{X}$. We therefore disallow potentials that involve only variables in $\mathbf{X}$.

Definition 4.18

conditional
random field

A conditional random field is an undirected graph $\mathcal{H}$ whose nodes correspond to $\mathbf{X} \cup \mathbf{Y}$; the network is annotated with a set of factors $\phi_1(\mathbf{D}_1), \dots, \phi_m(\mathbf{D}_m)$ such that each $\mathbf{D}_i \not\subseteq \mathbf{X}$. The network encodes a conditional distribution as follows:

$$\begin{aligned}
 P(\mathbf{Y} \mid \mathbf{X}) &= \frac{1}{Z(\mathbf{X})} \tilde{P}(\mathbf{Y}, \mathbf{X}) \\
 \tilde{P}(\mathbf{Y}, \mathbf{X}) &= \prod_{i=1}^m \phi_i(\mathbf{D}_i) \\
 Z(\mathbf{X}) &= \sum_{\mathbf{Y}} \tilde{P}(\mathbf{Y}, \mathbf{X}).
 \end{aligned} \tag{4.11}$$

Two variables in $\mathcal{H}$ are connected by an (undirected) edge whenever they appear together in the scope of some factor. ■

The only difference between equation (4.11) and the (unconditional) Gibbs distribution of definition 4.3 is the different normalization used in the partition function $Z(\mathbf{X})$. The definition of a CRF induces a different value for the partition function for every assignment $\mathbf{x}$ to $\mathbf{X}$. This difference is denoted graphically by having the feature variables grayed out.

Example 4.19

Consider a CRF over $\mathbf{Y} = \{Y_1, \dots, Y_k\}$ and $\mathbf{X} = \{X_1, \dots, X_k\}$, with an edge $Y_i - Y_{i+1}$ ($i = 1, \dots, k-1$) and an edge $Y_i - X_i$ ($i = 1, \dots, k$), as shown in figure 4.14a. The distribution

represented by this network has the form:

$$\begin{aligned}
 P(\mathbf{Y} \mid \mathbf{X}) &= \frac{1}{Z(\mathbf{X})} \tilde{P}(\mathbf{Y}, \mathbf{X}) \\
 \tilde{P}(\mathbf{Y}, \mathbf{X}) &= \prod_{i=1}^{k-1} \phi(Y_i, Y_{i+1}) \prod_{i=1}^k \phi(Y_i, X_i) \\
 Z(\mathbf{X}) &= \sum_{\mathbf{Y}} \tilde{P}(\mathbf{Y}, \mathbf{X}).
 \end{aligned}$$

■

Note that, unlike the definition of a conditional Bayesian network, the structure of a CRF may still contain edges between variables in $\mathbf{X}$, which arise when two such variables appear together in a factor that also contains a target variable. However, these edges do not encode the structure of any distribution over $\mathbf{X}$, since the network explicitly does not encode any such distribution.



The fact that we avoid encoding the distribution over the variables in $\mathbf{X}$ is one of the main strengths of the CRF representation. This flexibility allows us to incorporate into the model a rich set of observed variables whose dependencies may be quite complex or even poorly understood. It also allows us to include continuous variables whose distribution may not have a simple parametric form. This flexibility allows us to use domain knowledge in order to define a rich set of features characterizing our domain, without worrying about modeling their joint distribution. For example, returning to the vision MRFs of box 4.B, rather than defining a joint distribution over pixel values and their region assignment, we can define a conditional distribution over segment assignments *given* the pixel values. The use of a conditional distribution here allows us to avoid making a parametric assumption over the (continuous) pixel values. Even more important, we can use image-processing routines to define rich features, such as the presence or direction of an image gradient at a pixel. Such features can be highly informative in determining the region assignment of a pixel. However, the definition of such features usually relies on multiple pixels, and defining a correct joint distribution or a set of independence assumptions over these features is far from trivial. The fact that we can condition on these features and avoid this whole issue allows us the flexibility to include them in the model. See box 4.E for another example.

4.6.1.2 Directed and Undirected Dependencies

A CRF defines a conditional distribution of $\mathbf{Y}$ on $\mathbf{X}$; thus, it can be viewed as a partially directed graph, where we have an undirected component over Y , which has the variables in $\mathbf{X}$ as parents.

Example 4.20

naive Markov

Consider a CRF over the binary-valued variables $\mathbf{X} = \{X_1, \dots, X_k\}$ and $\mathbf{Y} = \{Y\}$, and a pairwise potential between Y and each X_i ; this model is sometimes known as a naive Markov model, due to its similarity to the naive Bayes model. Assume that the pairwise potentials defined via the following log-linear model

$$\phi_i(X_i, Y) = \exp \{w_i \mathbf{I}\{X_i = 1, Y = 1\}\}.$$

We also introduce a single-node potential $\phi_0(Y) = \exp \{w_0 \mathbf{I}\{Y = 1\}\}$. Following equation (4.11),

we now have:

$$\begin{aligned}\tilde{P}(Y = 1 \mid x_1, \dots, x_k) &= \exp \left\{ w_0 + \sum_{i=1}^k w_i x_i \right\} \\ \tilde{P}(Y = 0 \mid x_1, \dots, x_k) &= \exp \{0\} = 1.\end{aligned}$$

In this case, we can show (exercise 5.16) that

$$P(Y = 1 \mid x_1, \dots, x_k) = \text{sigmoid} \left(w_0 + \sum_{i=1}^k w_i x_i \right),$$

where

$$\text{sigmoid}(z) = \frac{e^z}{1 + e^z}$$

sigmoid

logistic CPD

is the sigmoid function. This conditional distribution $P(Y \mid \mathbf{X})$ is of great practical interest: It defines a CPD that is not structured as a table, but that is induced by a small set of parameters $w_0, \dots, w_k$ — parameters whose number is linear, rather than exponential, in the number of parents. This type of CPD, often called a logistic CPD, is a natural model for many real-world applications, inasmuch as it naturally aggregates the influence of different parents. We discuss this CPD in greater detail in section 5.4.2 as part of our general presentation of structured CPDs. ■

The partially directed model for the CRF of example 4.19 is shown in figure 4.14b. We may be tempted to believe that we can construct an equivalent model that is fully directed, such as the one in figure 4.14c. In particular, conditioned on any assignment $\mathbf{x}$, the posterior distributions over $\mathbf{Y}$ in the two models satisfy the same independence assignments (the ones defined by the chain structure). However, the two models are not equivalent: In the Bayesian network, we have that Y_1 is independent of X_2 if we are not given Y_2 . By contrast, in the original CRF, the unnormalized marginal measure of $\mathbf{Y}$ depends on the entire parameterization of the chain, and specifically the values of all of the variables in $\mathbf{X}$. A sound conditional Bayesian network for this distribution would require edges from all of the variables in $\mathbf{X}$ to each of the variables Y_i , thereby losing much of the structure in the distribution. See also box 20.A for further discussion.

Box 4.E — Case Study: CRFs for Text Analysis. One important use for the CRF framework is in the domain of text analysis. Various models have been proposed for different tasks, including part-of-speech labeling, identifying named entities (people, places, organizations, and so forth), and extracting structured information from the text (for example, extracting from a reference list the publication titles, authors, journals, years, and the like). Most of these models share a similar structure: We have a target variable for each word (or perhaps short phrase) in the document, which encodes the possible labels for that word. Each target variable is connected to a set of feature variables that capture properties relevant to the target distinction. These methods are very popular in text analysis, both because the structure of the networks is a good fit for this domain, and because they produce state-of-the-art results for a broad range of natural-language processing problems.

As a concrete example, consider the named entity recognition task, as described by Sutton and McCallum (2004, 2007). Entities often span multiple words, and the type of an entity may not be

apparent from individual words; for example, “New York” is a location, but “New York Times” is an organization. The problem of extracting entities from a word sequence of length T can be cast as a graphical model by introducing for each word, $X_t, 1 \leq t \leq T$, a target variable, Y_t , which indicates the entity type of the word. The outcomes of Y_t include B-PERSON, I-PERSON, B-LOCATION, I-LOCATION, B-ORGANIZATION, I-ORGANIZATION, and OTHER. In this so-called “BIO notation,” OTHER indicates that the word is not part of an entity, the B- outcomes indicate the beginning of a named entity phrase, and the I- outcomes indicate the inside or end of the named entity phrase. Having a distinguishing label for the beginning versus inside of an entity phrase allows the model to segment adjacent entities of the same type.

linear-chain CRF

A common structure for this problem is a linear-chain CRF often having two factors for each word: one factor $\phi_t^1(Y_t, Y_{t+1})$ to represent the dependency between neighboring target variables, and another factor $\phi_t^2(Y_t, X_1, \dots, X_T)$ that represents the dependency between a target and its context in the word sequence. Note that the second factor can depend on arbitrary features of the entire input word sequence. We generally do not encode this model using table factors, but using a log-linear model. Thus, the factors are derived from a number of feature functions, such as $f_t(Y_t, X_t) = \mathbf{I}\{Y_t = \text{B-ORGANIZATION}, X_t = \text{“Times”}\}$. We note that, just as logistic CPDs are the conditional analog of the naive Bayes classifier (example 4.20), the linear-chain CRF is the conditional analog of the hidden Markov model (HMM) that we present in section 6.2.3.1.

hidden Markov
model

A large number of features of the word X_t and neighboring words are relevant to the named entity decision. These include features of the word itself: is it capitalized; does it appear in a list of common person names; does it appear in an atlas of location names; does it end with the character string “ton”; is it exactly the string “York”; is the following word “Times.” Also relevant are aggregate features of the entire word sequence, such as whether it contains more than two sports-related words, which might be an indicator that “New York” is an organization (sports team) rather than a location. In addition, including features that are conjunctions of all these features often increases accuracy. The total number of features can be quite large, often in the hundreds of thousands or more if conjunctions of word pairs are used as features. However, the features are sparse, meaning that most features are zero for most words.

Note that the same feature variable can be connected to multiple target variables, so that Y_t would typically be dependent on the identity of several words in a window around position t . These contextual features are often highly indicative: for example, “Mrs.” before a word and “spoke” after a word are both strong indicators that the word is a person. These context words would generally be used as a feature for multiple target variables. Thus, if we were using a simple naive-Bayes-style generative model, where each target variable is a parent of its associated feature, we either would have to deal with the fact that a context word has multiple parents or we would have to duplicate its occurrences (with one copy for each target variable for which it is in the context), and thereby overcount its contribution.

Linear-chain CRFs frequently provide per-token accuracies in the high 90 percent range on many natural data sets. Per-field precision and recall (where the entire phrase category and boundaries must be correct) are more often around 80–95 percent, depending on the data set.

Although the linear-chain model is often effective, additional information can be incorporated into the model by augmenting the graphical structure. For example, often when a word occurs multiple times in the same document, it often has the same label. This knowledge can be incorporated by including factors that connect identical words, resulting in a skip-chain CRF, as shown in figure 4.E.1a. The first occurrence of the word “Green” has neighboring words that provide strong

skip-chain CRF

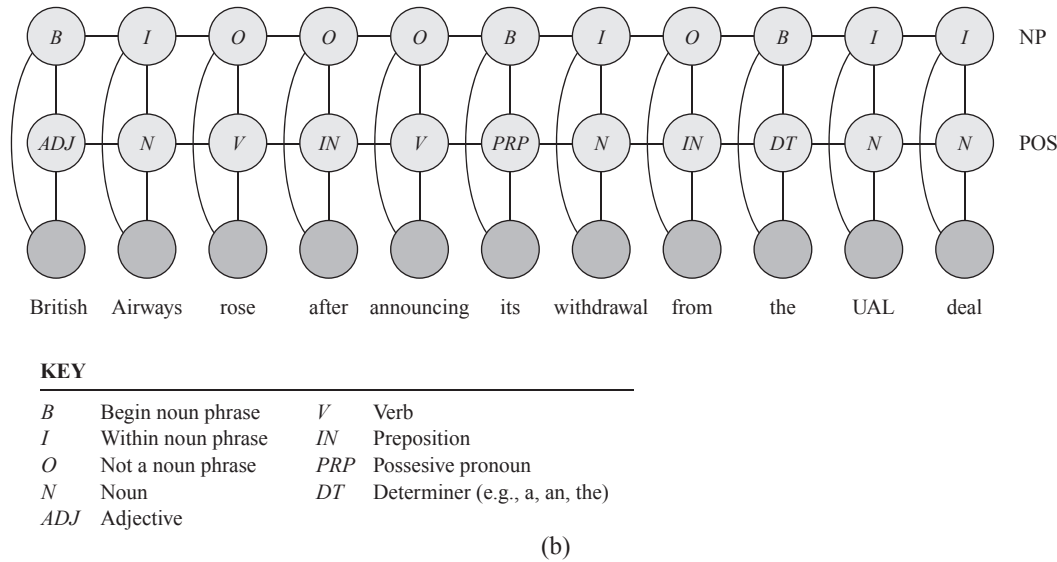
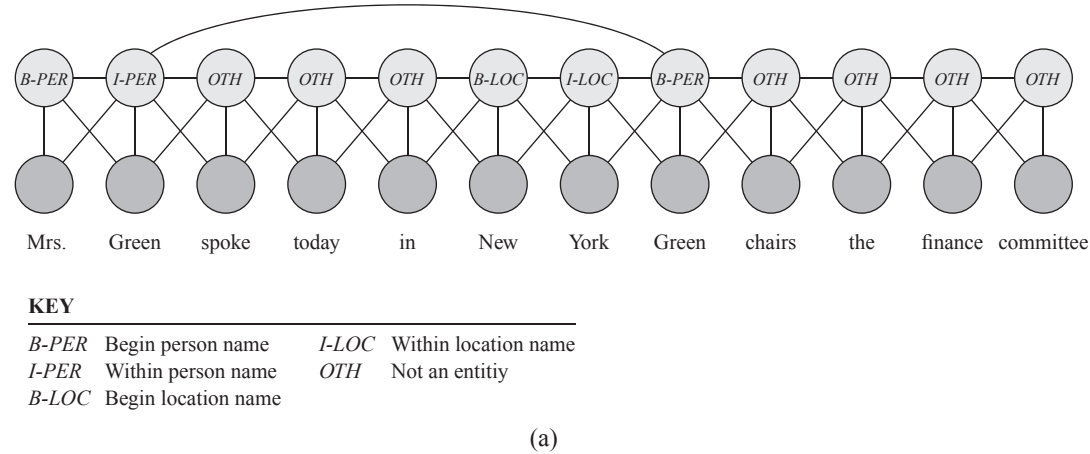


Figure 4.E.1 — Two models for text analysis based on a linear chain CRF Gray nodes indicate $\mathbf{X}$ and clear nodes $\mathbf{Y}$. The annotations inside the $\mathbf{Y}$ are the true labels. (a) A skip chain CRF for named entity recognition, with connections between adjacent words and long-range connections between multiple occurrences of the same word. (b) A pair of coupled linear-chain CRFs that performs joint part-of-speech labeling and noun-phrase segmentation. Here, B indicates the beginning of a noun phrase, I other words in the noun phrase, and O words not in a noun phrase. The labels for the second chain are parts of speech.

evidence that it is a PERSON's name; however, the second occurrence is much more ambiguous. By augmenting the original linear-chain CRF with an additional long-range factor that prefers its connected target variables to have the same value, the model is more likely to predict correctly that the second occurrence is also a PERSON. This example demonstrates another flexibility of conditional models, which is that the graphical structure over $\mathbf{Y}$ can easily depend on the value of the $\mathbf{X}$'s.

coupled HMM

CRFs having a wide variety of model structures have been successfully applied to many different tasks. Joint inference of both part-of-speech labels and noun-phrase segmentation has been performed with two connected linear chains (somewhat analogous to a coupled hidden Markov mode, shown in figure 6.3). This structure is illustrated in figure 4.E.1b.

4.6.2 Chain Graph Models ★

partially directed
acyclic graph

chain component

chain graph

We now present a more general framework that builds on the CRF representation and can be used to provide a general treatment of the independence assumptions made in these partially directed models. Recall from definition 2.21 that, in a *partially directed acyclic graph* (PDAG), the nodes can be disjointly partitioned into several *chain components*. An edge between two nodes in the same chain component must be undirected, while an edge between two nodes in different chain components must be directed. Thus, PDAGs are also called *chain graphs*.

4.6.2.1 Factorization

chain graph
model

As in our other graphical representations, the structure of a PDAG $\mathcal{K}$ can be used to define a factorization for a probability distribution over $\mathcal{K}$. Intuitively, the factorization for PDAGs represents the distribution as a product of each of the chain components given its parents. Thus, we call such a representation a *chain graph model*.

Intuitively, each chain component $\mathbf{K}_i$ in the chain graph model is associated with a CRF that defines $\mathcal{P}(\mathbf{K}_i \mid \text{Pa}_{\mathbf{K}_i})$ — the conditional distribution of $\mathbf{K}_i$ given its parents in the graph. More precisely, each is defined via a set of factors that involve the variables in $\mathbf{K}_i$ and their parents; the distribution $\mathcal{P}(\mathbf{K}_i \mid \text{Pa}_{\mathbf{K}_i})$ is defined by using the factors associated with $\mathbf{K}_i$ to define a CRF whose target variables are $\mathbf{K}_i$ and whose observable variables are $\text{Pa}_{\mathbf{K}_i}$.

To provide a formal definition, it helps to introduce the concept of a moralized PDAG.

Definition 4.19

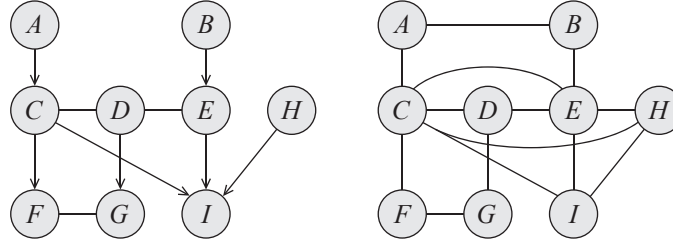
moralized graph

Let $\mathcal{K}$ be a PDAG and $\mathbf{K}_1, \dots, \mathbf{K}_\ell$ be its chain components. We define $\text{Pa}_{\mathbf{K}_i}$ to be the parents of nodes in $\mathbf{K}_i$. The moralized graph of $\mathcal{K}$ is an undirected graph $\mathcal{M}[\mathcal{K}]$ produced by first connecting, using undirected edges, any pair of nodes $X, Y \in \text{Pa}_{\mathbf{K}_i}$ for all $i = 1, \dots, \ell$, and then converting all directed edges into undirected edges. ■

This definition generalizes our earlier notion of a moralized directed graph. In the case of directed graphs, each node is its own chain component, and hence we are simply adding undirected edges between the parents of each node.

Example 4.21

Figure 4.15 shows a chain graph and its moral graph. We have added the edge between A and B , since they are both parents of the chain component $\{C, D, E\}$, and edges between C , E , and H , because they are parents of the chain component $\{I\}$. Note that we did not add an edge between

Figure 4.15 A chain graph $\mathcal{K}$ and its moralized version

D and H (even though D and C, E are in the same chain component), since D is not a parent of I . ■

We can now define the factorization of a chain graph:

Definition 4.20

chain graph
distribution

Let $\mathcal{K}$ be a PDAG, and $\mathbf{K}_1, \dots, \mathbf{K}_\ell$ be its chain components. A chain graph distribution is defined via a set of factors $\phi_i(\mathbf{D}_i)$ ($i = 1, \dots, m$), such that each $\mathbf{D}_i$ is a complete subgraph in the moralized graph $\mathcal{M}[\mathcal{K}^+[\mathbf{D}_i]]$. We associate each factor $\phi_i(\mathbf{D}_i)$ with a single chain component $\mathbf{K}_j$, such that $\mathbf{D}_i \subseteq \mathbf{K}_j \cup \text{Pa}_{\mathbf{K}_i}$ and define $P(\mathbf{K}_i \mid \text{Pa}_{\mathbf{K}_i})$ as a CRF with these factors, and with $\mathbf{Y}_i = \mathbf{K}_i$ and $\mathbf{X}_i = \text{Pa}_{\mathbf{K}_i}$. We now define

$$P(\mathcal{X}) = \prod_{i=1}^{\ell} P(\mathbf{K}_i \mid \text{Pa}_{\mathbf{K}_i}).$$

We say that a distribution P factorizes over $\mathcal{K}$ if it can be represented as a chain graph distribution over $\mathcal{K}$. ■

Example 4.22

In the chain graph model defined by the graph of figure 4.15, we require that the conditional distribution $P(C, D, E \mid A, B)$ factorize according to the graph of figure 4.16a. Specifically, we would have to define the conditional probability as a normalized product of factors:

$$\frac{1}{Z(A, B)} \phi_1(A, C) \phi_2(B, E) \phi_3(C, D) \phi_4(D, E).$$

A similar factorization applies to $P(F, G \mid C, D)$. ■

4.6.2.2 Independencies in Chain Graphs

boundary

As for undirected graphs, there are three distinct interpretations for the independence properties induced by a PDAG. Recall that in a PDAG, we have both the notion of parents of X (variables Y such that $Y \rightarrow X$ is in the graph) and neighbors of X (variables Y such that $Y - X$ is in the graph). Recall that the union of these two sets is the *boundary* of X , denoted Boundary_X . Also recall, from definition 2.15, that the descendants of X are those nodes Y that can be reached using any directed path, where a directed path can involve both directed and undirected edges but must contain at least one edge directed from X to Y , and no edges directed from Y to

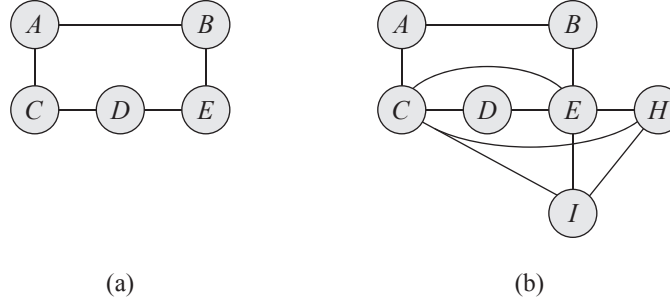


Figure 4.16 Example for definition of c-separation in a chain graph. (a) The Markov network $\mathcal{M}[\mathcal{K}^+[C, D, E]]$. (b) The Markov network $\mathcal{M}[\mathcal{K}^+[C, D, E, I]]$.

X . Thus, in the case of PDAGs, it follows that if Y is a descendant of X , then Y must be in a “lower” chain component.

Definition 4.21

pairwise
independencies

For a PDAG $\mathcal{K}$, we define the pairwise independencies associated with $\mathcal{K}$ to be:

$$\mathcal{I}_p(\mathcal{K}) = \{(X \perp Y \mid (\text{NonDescendants}_X - \{X, Y\})) : \\ X, Y \text{ non-adjacent}, Y \in \text{NonDescendants}_X\}.$$

This definition generalizes the pairwise independencies for undirected graphs: in an undirected graph, nodes have no descendants, so $\text{NonDescendants}_X = \mathcal{X}$. Similarly, it is not too hard to show that these independencies also hold in a directed graph.

Definition 4.22

local
independencies

For a PDAG $\mathcal{K}$, we define the local independencies associated with $\mathcal{K}$ to be:

$$\mathcal{I}_\ell(\mathcal{K}) = \{(X \perp \text{NonDescendants}_X - \text{Boundary}_X \mid \text{Boundary}_X) : X \in \mathcal{X}\}.$$

This definition generalizes the definition of local independencies for both directed and undirected graphs. For directed graphs, NonDescendants_X is precisely the set of nondescendants, whereas Boundary_X is the set of parents. For undirected graphs, NonDescendants_X is $\mathcal{X}$, whereas $\text{Boundary}_X = \text{Nb}_X$.

We define the global independencies in a PDAG using the definition of moral graph. Our definition follows the lines of proposition 4.10.

Definition 4.23

c-separation

Let $X, Y, Z \subset \mathcal{X}$ be three disjoint sets, and let $U = X \cup Y \cup Z$. We say that X is c-separated from Y given Z if X is separated from Y given Z in the undirected graph $\mathcal{M}[\mathcal{K}^+[X \cup Y \cup Z]]$. ■

Example 4.23

Consider again the PDAG of figure 4.15. Then C is c-separated from E given D, A , because C and E are separated given D, A in the undirected graph $\mathcal{M}[\mathcal{K}^+[\{C, D, E\}]]$, shown in figure 4.16a. However, C is not c-separated from E given only D , since there is a path between C and E via A, B . On the other hand, C is not separated from E given D, A, I . The graph $\mathcal{M}[\mathcal{K}^+[\{C, D, E, I\}]]$ is shown in figure 4.16b. As we can see, the introduction of I into the set U causes us to introduce a direct edge between C and E in order to moralize the graph. Thus, we cannot block the path between C and E using D, A, I . ■

This notion of c-separation clearly generalizes the notion of separation in undirected graphs, since the ancestors of a set U in an undirected graph are simply the entire set of nodes $\mathcal{X}$. It also generalizes the notion of d-separation in directed graphs, using the equivalent definition provided in proposition 4.10. Using the definition of c-separation, we can finally define the notion of global Markov independencies:

Definition 4.24

global
independencies

Let $\mathcal{K}$ be a PDAG. We define the global independencies associated with $\mathcal{K}$ to be:

$$\mathcal{I}(\mathcal{K}) = \{(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}) : \mathbf{X}, \mathbf{Y}, \mathbf{Z} \subset \mathcal{X}, \mathbf{X} \text{ is c-separated from } \mathbf{Y} \text{ given } \mathbf{Z}\}. \quad \blacksquare$$

As in the case of undirected models, these three criteria for independence are not equivalent for nonpositive distributions. The inclusions are the same: the global independencies imply the local independencies, which in turn imply the pairwise independencies. Because undirected models are a subclass of PDAGs, the same counterexamples used in section 4.3.3 show that the inclusions are strict for nonpositive distributions. For positive distributions, we again have that the three definitions are equivalent.

We note that, as in the case of Bayesian networks, the parents of a chain component are always fully connected in $\mathcal{M}[\mathcal{K}[\mathbf{K}_i \cup \text{Pa}_{\mathbf{K}_i}]]$. Thus, while the structure over the parents helps factorize the distribution over the chain components containing the parents, it does not give rise to independence assertions in the conditional distribution over the child chain component. Importantly, however, it does give rise to structure in the form of the parameterization of $P(\mathbf{K}_i \mid \text{Pa}_{\mathbf{K}_i})$, as we saw in example 4.20.

As in the case of directed and undirected models, we have an equivalence between the requirement of factorization of a distribution and the requirement that it satisfy the independencies associated with the graph. Not surprisingly, since PDAGs generalize undirected graphs, this equivalence only holds for positive distributions:

Theorem 4.14

A positive distribution P factorizes over a PDAG $\mathcal{K}$ if and only if $P \models \mathcal{I}(\mathcal{K})$.

We omit the proof.

4.7 Summary and Discussion

In this chapter, we introduced *Markov networks*, an alternative graphical modeling language for probability distributions, based on undirected graphs.

We showed that Markov networks, like Bayesian networks, can be viewed as defining a set of independence assumptions determined by the graph structure. In the case of undirected models, there are several possible definitions for the independence assumptions induced by the graph, which are equivalent for positive distributions. As in the case of Bayesian network, we also showed that the graph can be viewed as a data structure for specifying a probability distribution in a factored form. The factorization is defined as a product of factors (general nonnegative functions) over cliques in the graph. We showed that, for positive distributions, the two characterizations of undirected graphs — as specifying a set of independence assumptions and as defining a factorization — are equivalent.

Markov networks also provide useful insight on Bayesian networks. In particular, we showed how a Bayesian network can be viewed as a Gibbs distribution. More importantly, the unnormalized measure we obtain by introducing evidence into a Bayesian network is also a Gibbs

distribution, whose partition function is the probability of the evidence. This observation will play a critical role in providing a unified view of inference in graphical models.

We investigated the relationship between Bayesian networks and Markov networks and showed that the two represent different families of independence assumptions. The difference in these independence assumptions is a key factor in deciding which of the two representations to use in encoding a particular domain. There are domains where interactions have a natural directionality, often derived from causal intuitions. **In this case, the independencies derived from the network structure directly reflect patterns such as intercausal reasoning. Markov networks represent only monotonic independence patterns: observing a variable can only serve to remove dependencies, not to activate them.** Of course, we can encode a distribution with “causal” connections as a Gibbs distribution, and it will exhibit the same nonmonotonic independencies. However, these independencies will not be manifest in the network structure.



In other domains, the interactions are more symmetrical, and attempts to force a directionality give rise to models that are unintuitive and that often are incapable of capturing the independencies in the domain (see, for example, section 6.6). As a consequence, the use of undirected models has increased steadily, most notably in fields such as computer vision and natural language processing, where the acyclicity requirements of directed graphical models are often at odds with the nature of the model. The flexibility of the undirected model also allows the distribution to be decomposed into factors over multiple overlapping “features” without having to worry about defining a single normalized generating distribution for each variable. Conversely, **this very flexibility and the associated lack of clear semantics for the model parameters often make it difficult to elicit models from experts. Therefore, many recent applications use learning techniques to estimate parameters from data, avoiding the need to provide a precise semantic meaning for each of them.**



Finally, the question of which class of models better encodes the properties of the distribution is only one factor in the selection of a representation. There are other important distinctions between these two classes of models, especially when it comes to learning from data. We return to these topics later in the book (see, for example, box 20.A).

4.8 Relevant Literature

contingency table

The representation of a probability distribution as an undirected graph has its roots in the *contingency table* representation that is a staple in statistical modeling. The idea of representing probabilistic interactions in this representation dates back at least as early as the work of Bartlett (1935). This line of work is reviewed in detail by Whittaker (1990) and Lauritzen (1996), and we refer the reader to those sources and the references therein.

A parallel line of work involved the development of the Markov network (or Markov random field) representation. Here, the starting point was a graph object rather than a distribution object (such as a contingency table). Isham (1981) surveys some of the early work along these lines.

The connection between the undirected graph representation and the Gibbs factorization of the distribution was first made in the unpublished work of Hammersley and Clifford (1971). As a consequence, they also showed the equivalence of the different types of (local, pairwise, and global) independence properties for undirected graphs in the case of positive distributions.

Lauritzen (1982) made the connection between MRFs and contingency tables, and proved